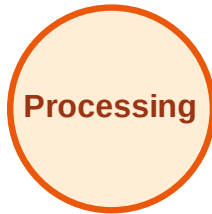


# BFS Traversal

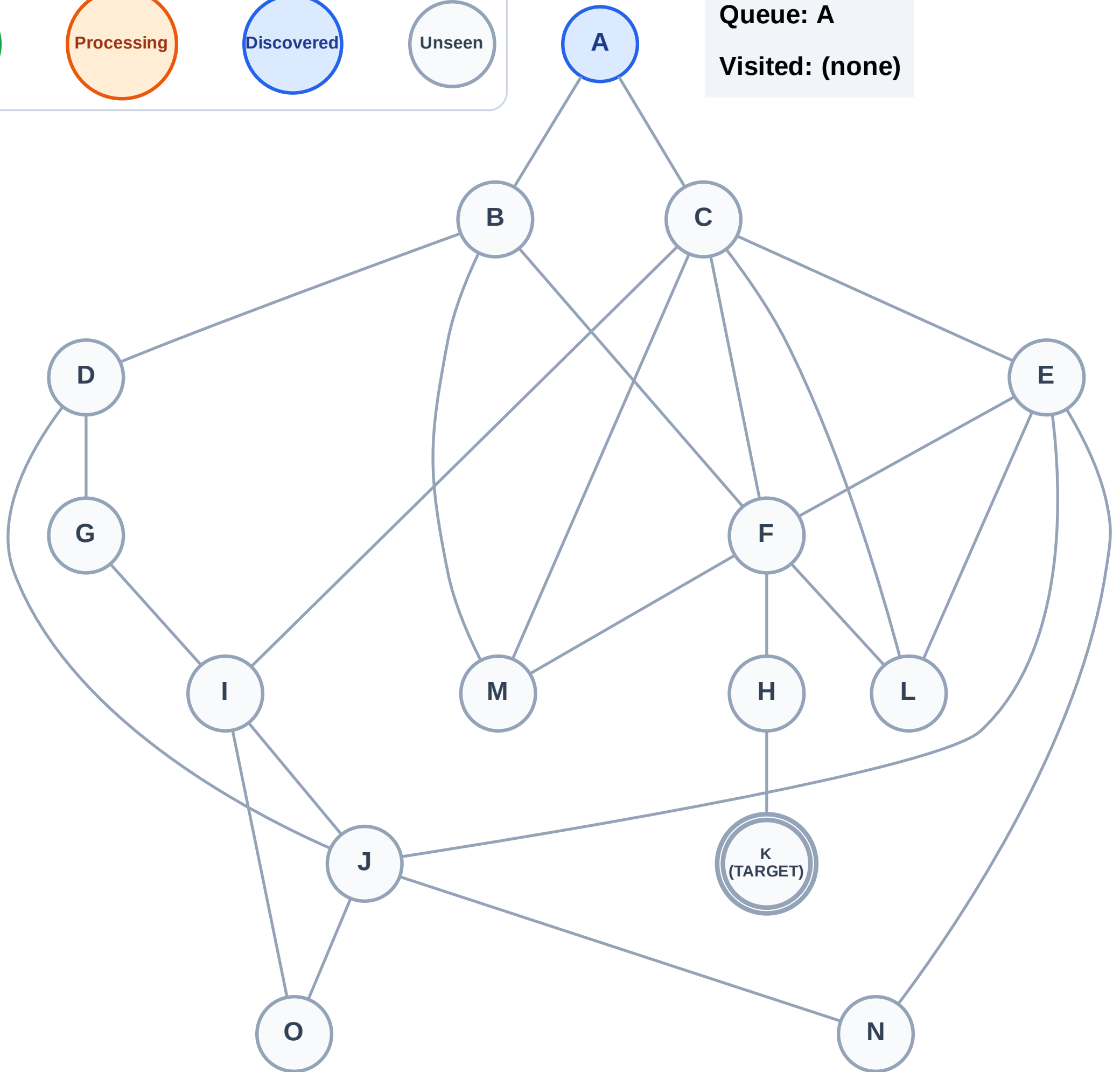
Step 1: Start at A (Seeking target node K)

## Legend



Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

Legend

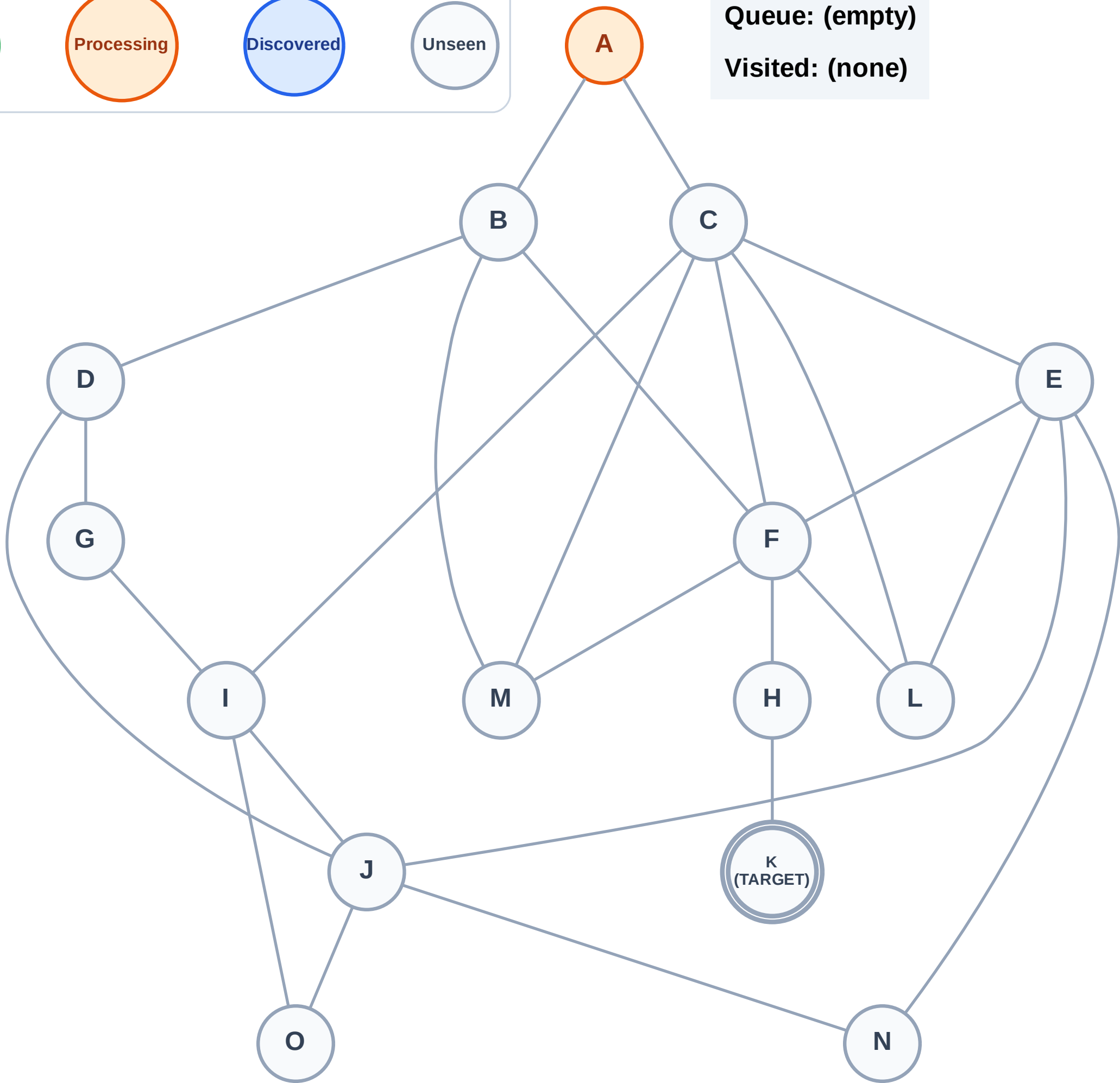
Complete

Processing

Discovered

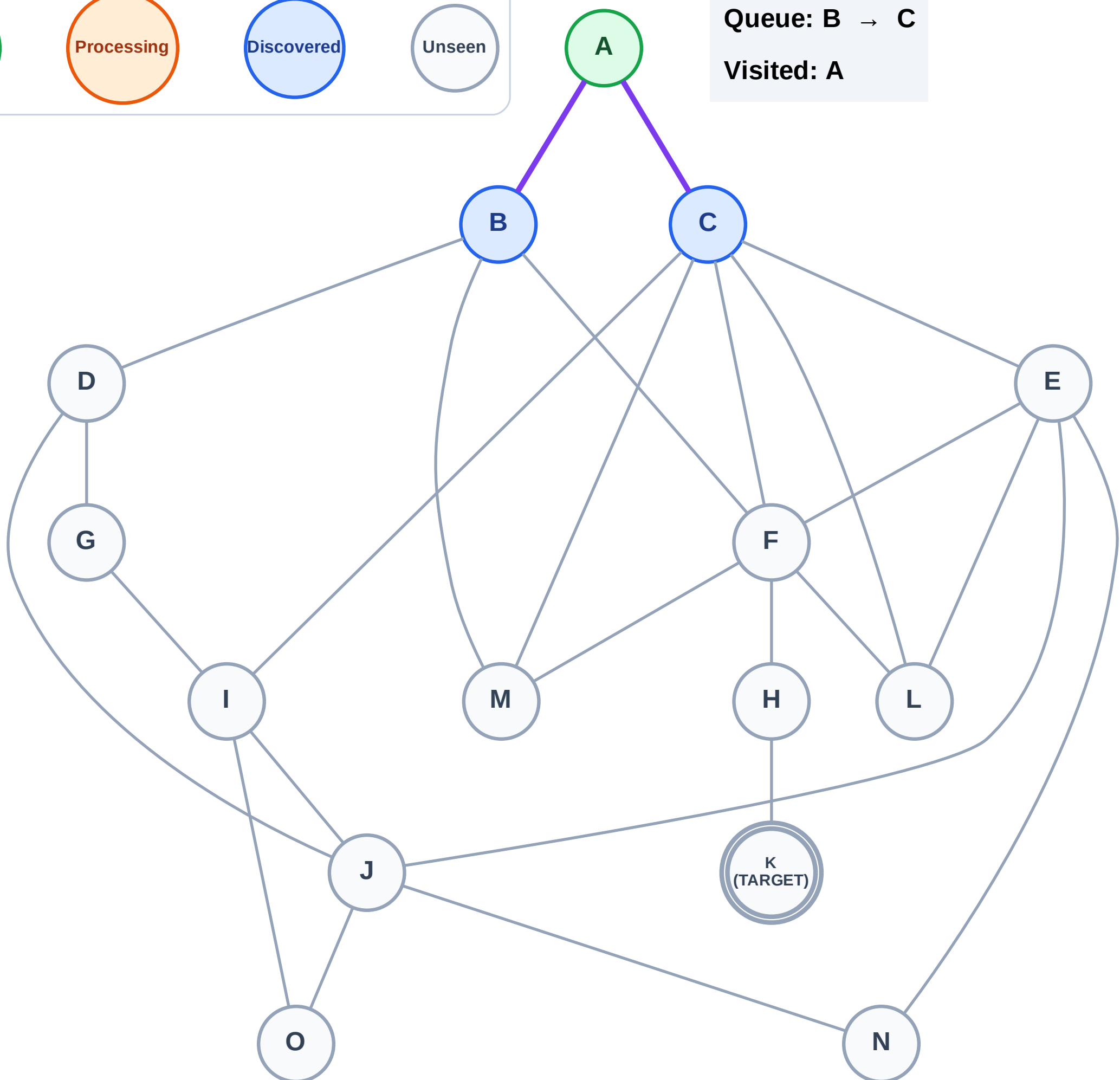
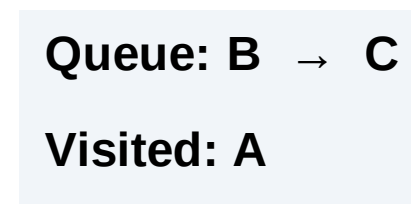
Unseen

Queue: (empty)  
Visited: (none)



### Step 3: Discover B, C from A

### Step 3: Discover B, C from A



# BFS Traversal

Step 4: Process node B

Legend

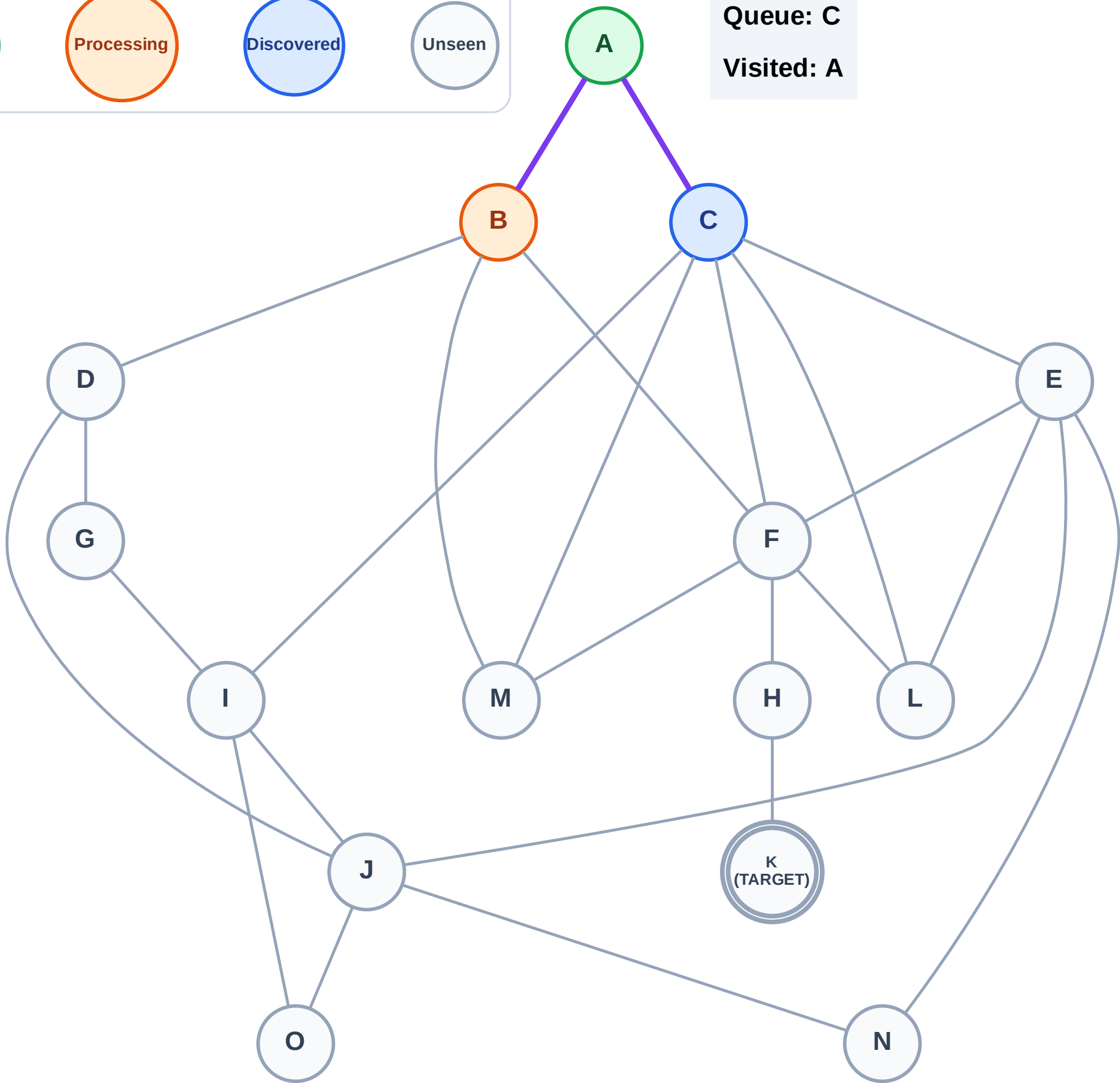
Complete

Processing

Discovered

Unseen

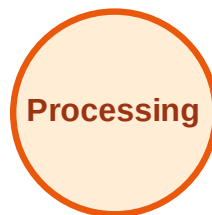
Queue: C  
Visited: A



# BFS Traversal

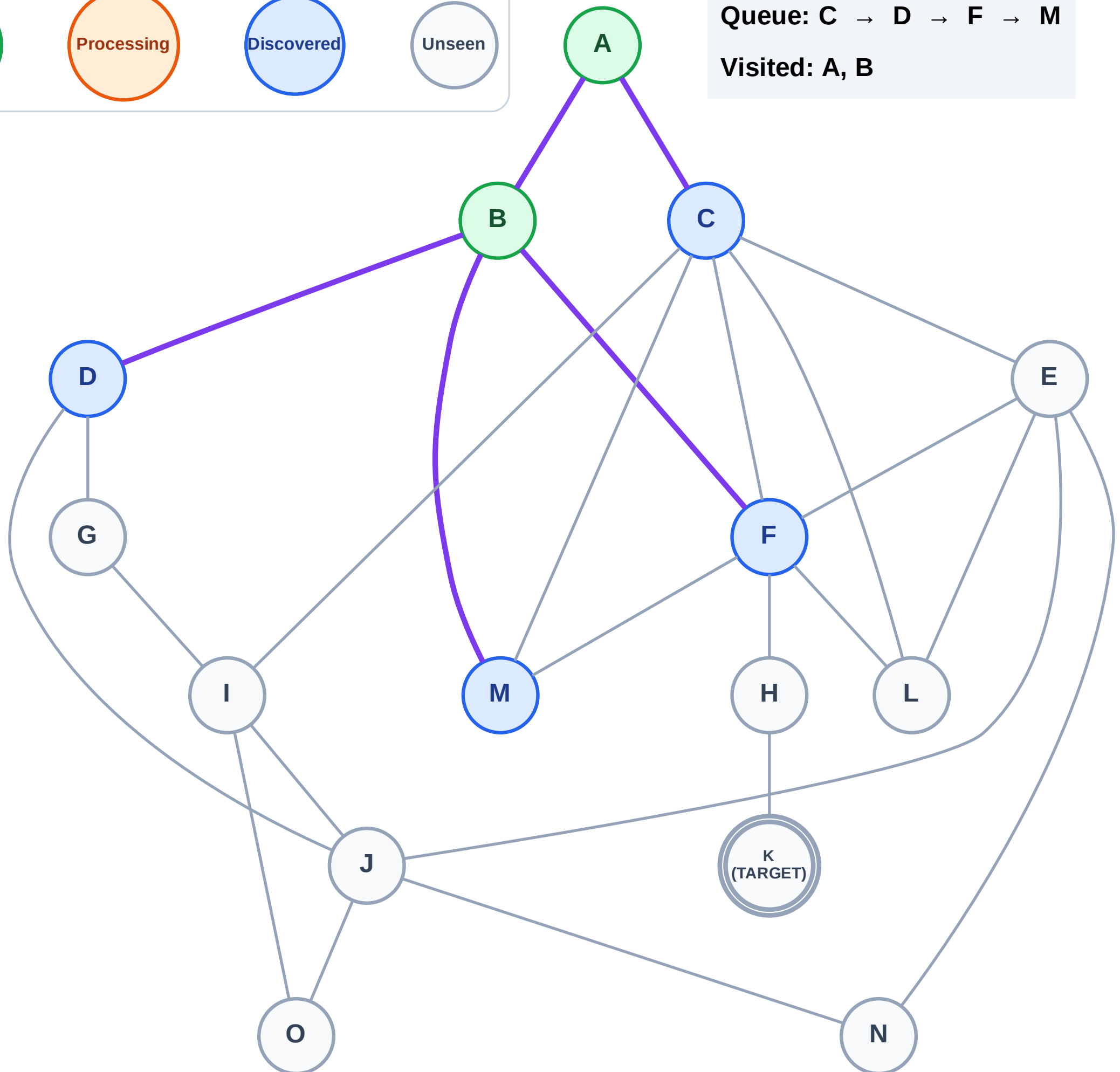
Step 5: Discover D, F, M from B

## Legend



Queue: C → D → F → M

Visited: A, B



# BFS Traversal

Step 6: Process node C

Legend

Complete

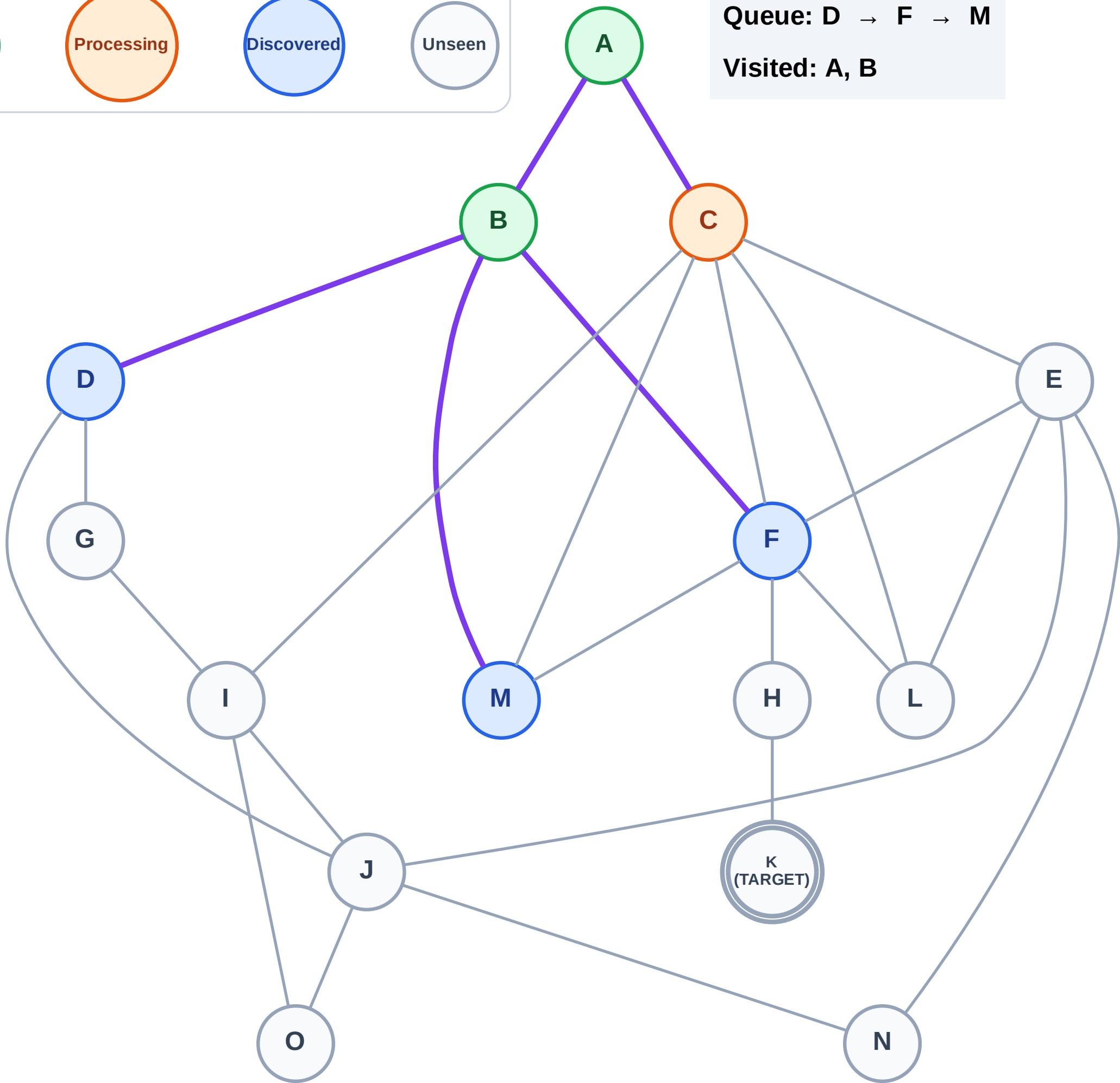
Processing

Discovered

Unseen

Queue: D → F → M

Visited: A, B



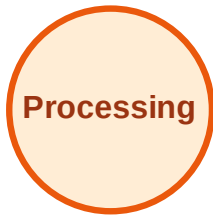
# BFS Traversal

Step 7: Discover E, I, L from C

## Legend



Complete



Processing



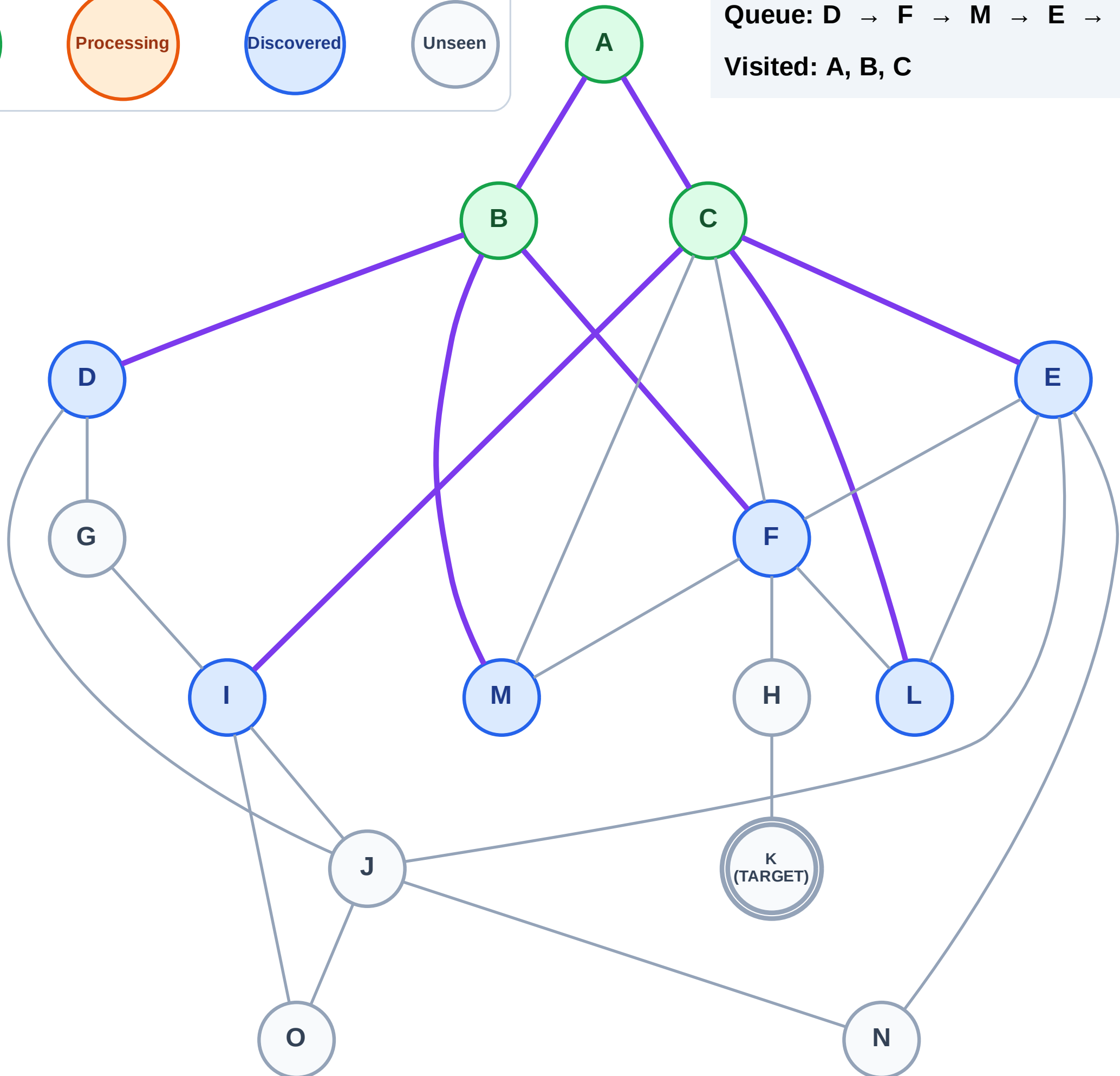
Discovered



Unseen

Queue: D → F → M → E → I → L

Visited: A, B, C



# BFS Traversal

Step 8: Process node D

Legend

Complete

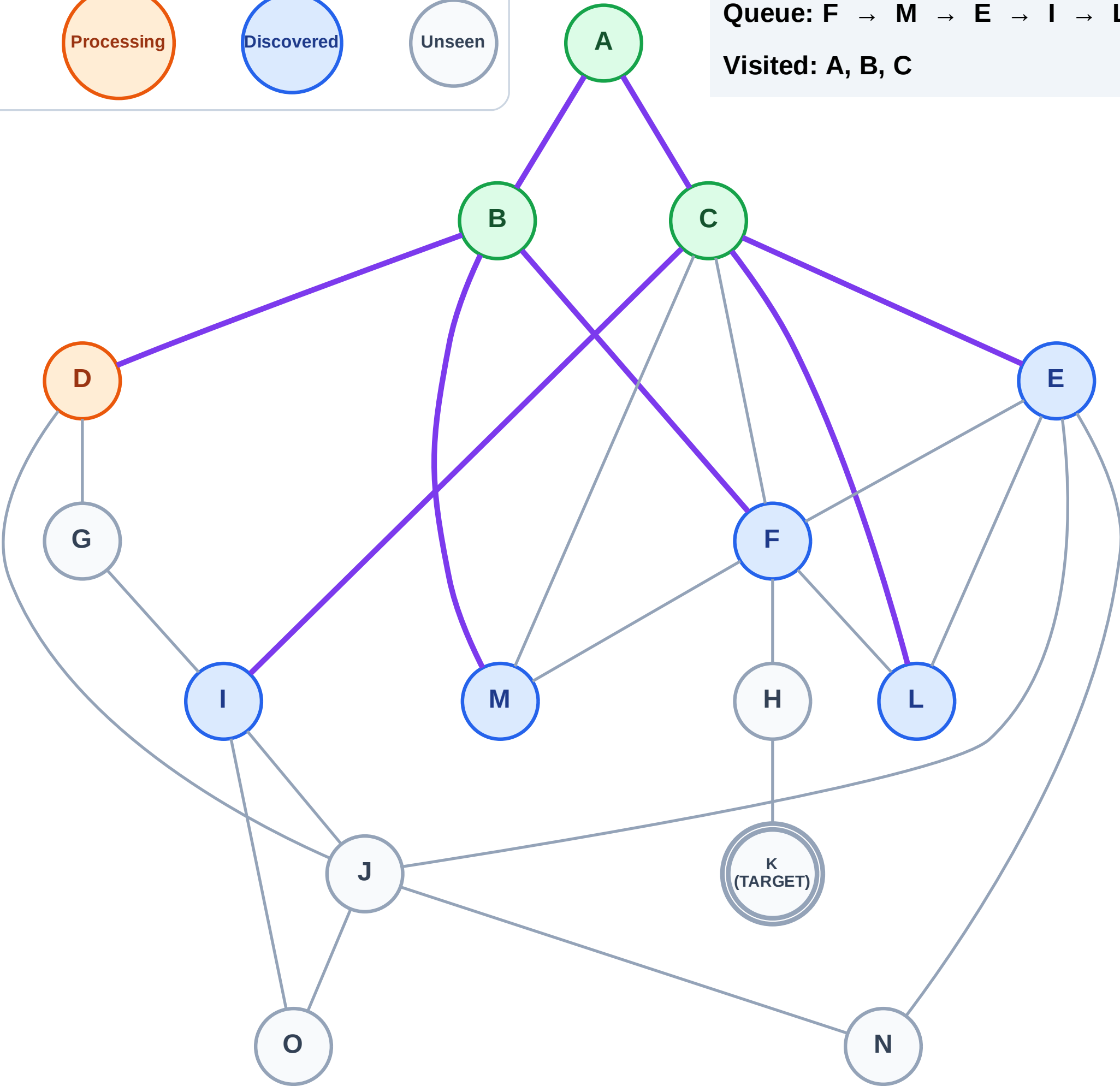
Processing

Discovered

Unseen

Queue: F → M → E → I → L

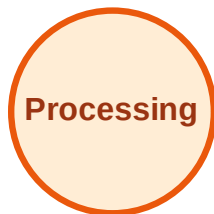
Visited: A, B, C



# BFS Traversal

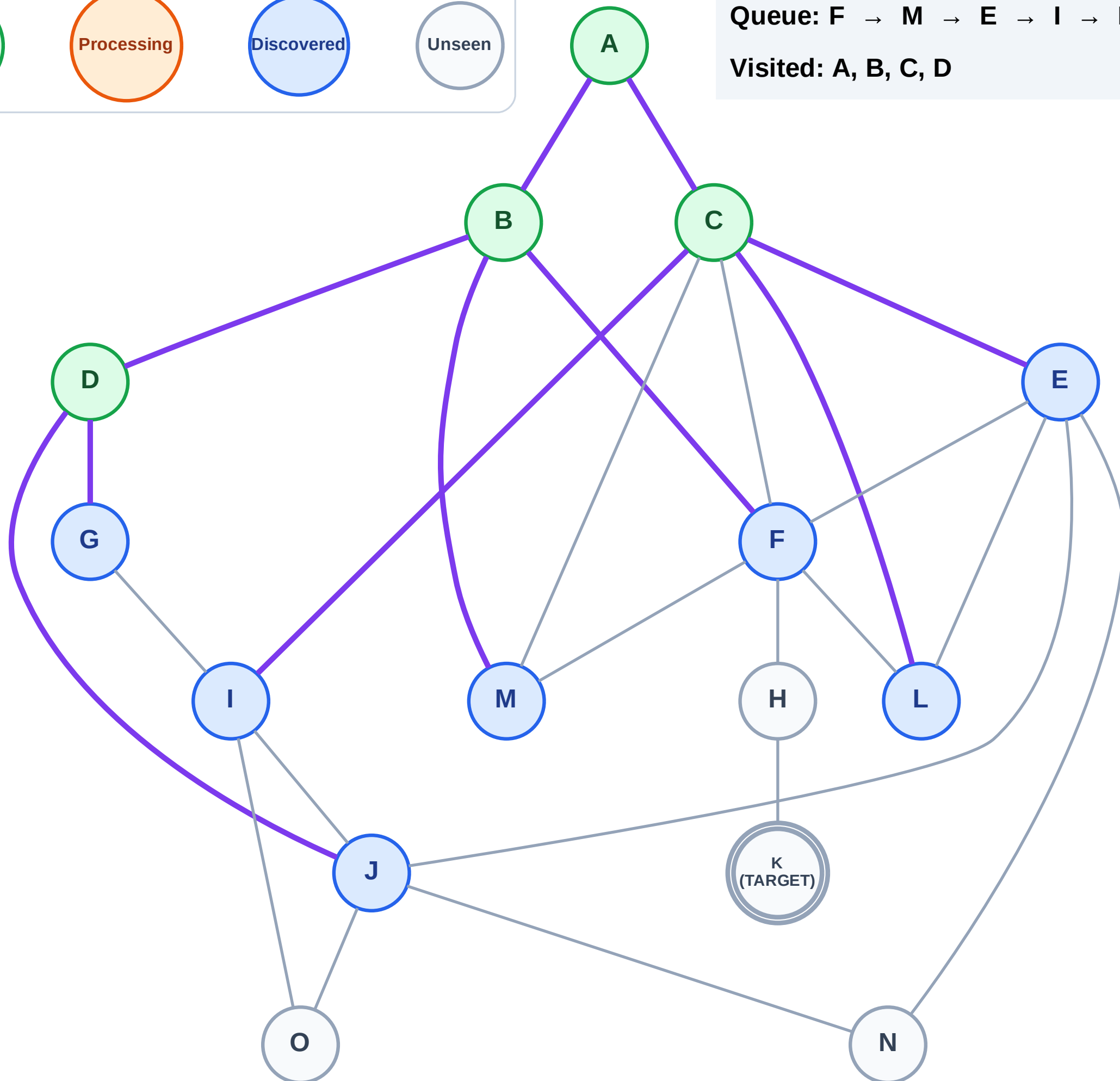
Step 9: Discover G, J from D

## Legend



Queue: F → M → E → I → L → G → J

Visited: A, B, C, D



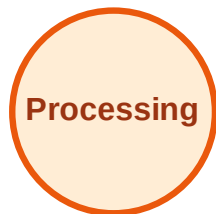
# BFS Traversal

Step 10: Process node F

## Legend



Complete



Processing



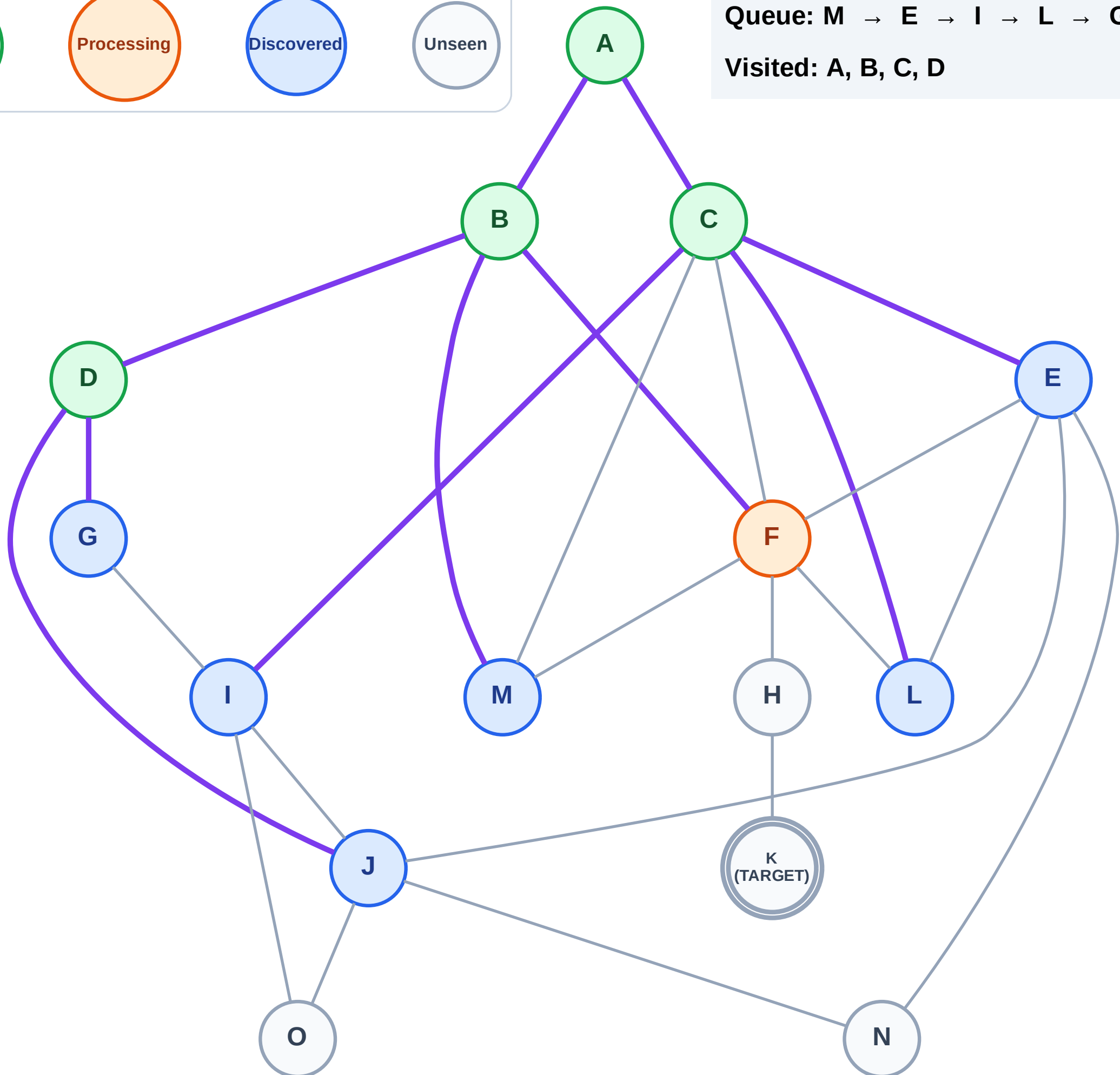
Discovered



Unseen

Queue: M → E → I → L → G → J

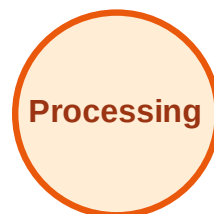
Visited: A, B, C, D



# BFS Traversal

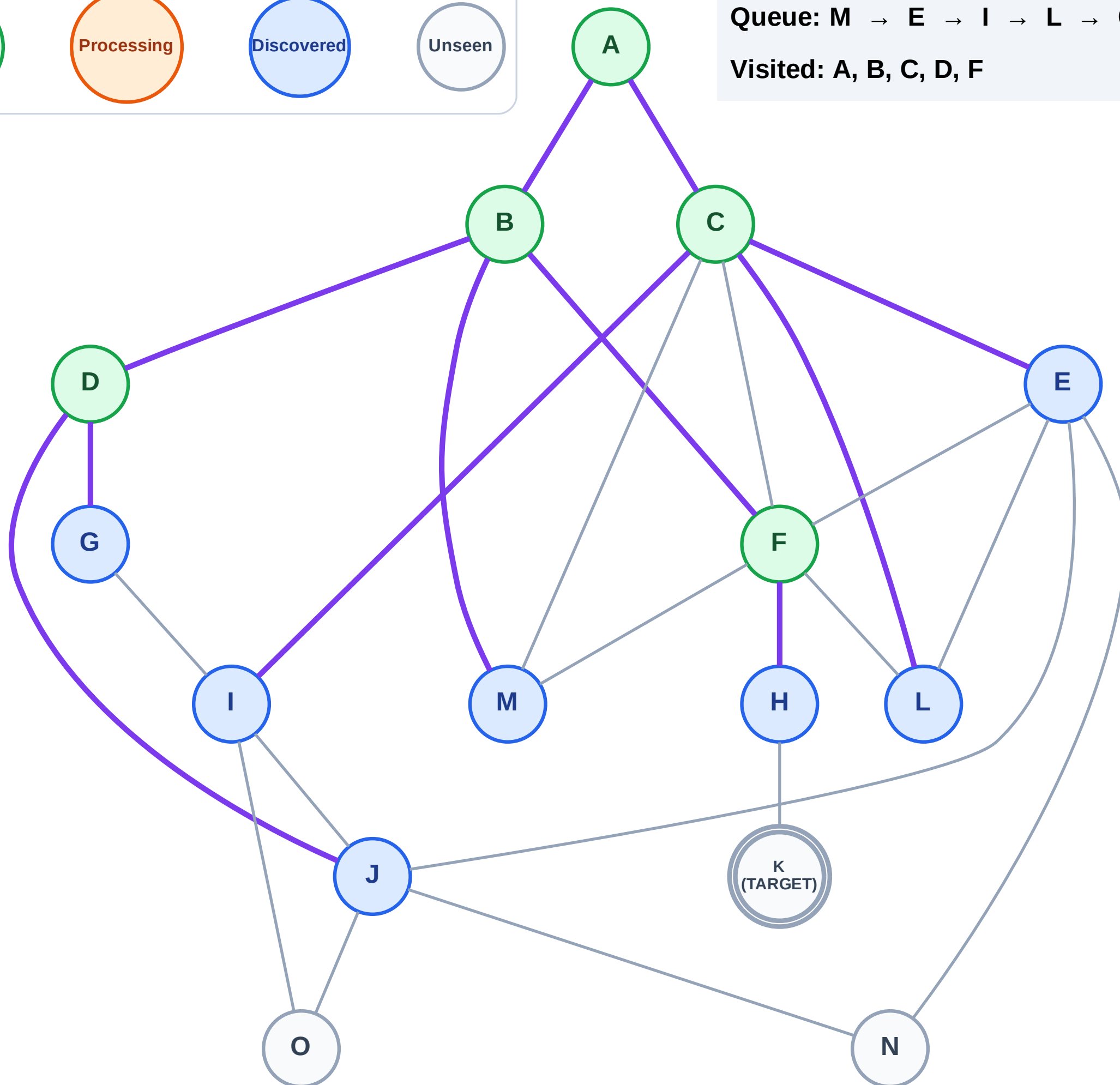
Step 11: Discover H from F

## Legend



Queue: M → E → I → L → G → J → H

Visited: A, B, C, D, F



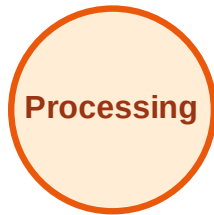
# BFS Traversal

Step 12: Process node M

## Legend



Complete



Processing



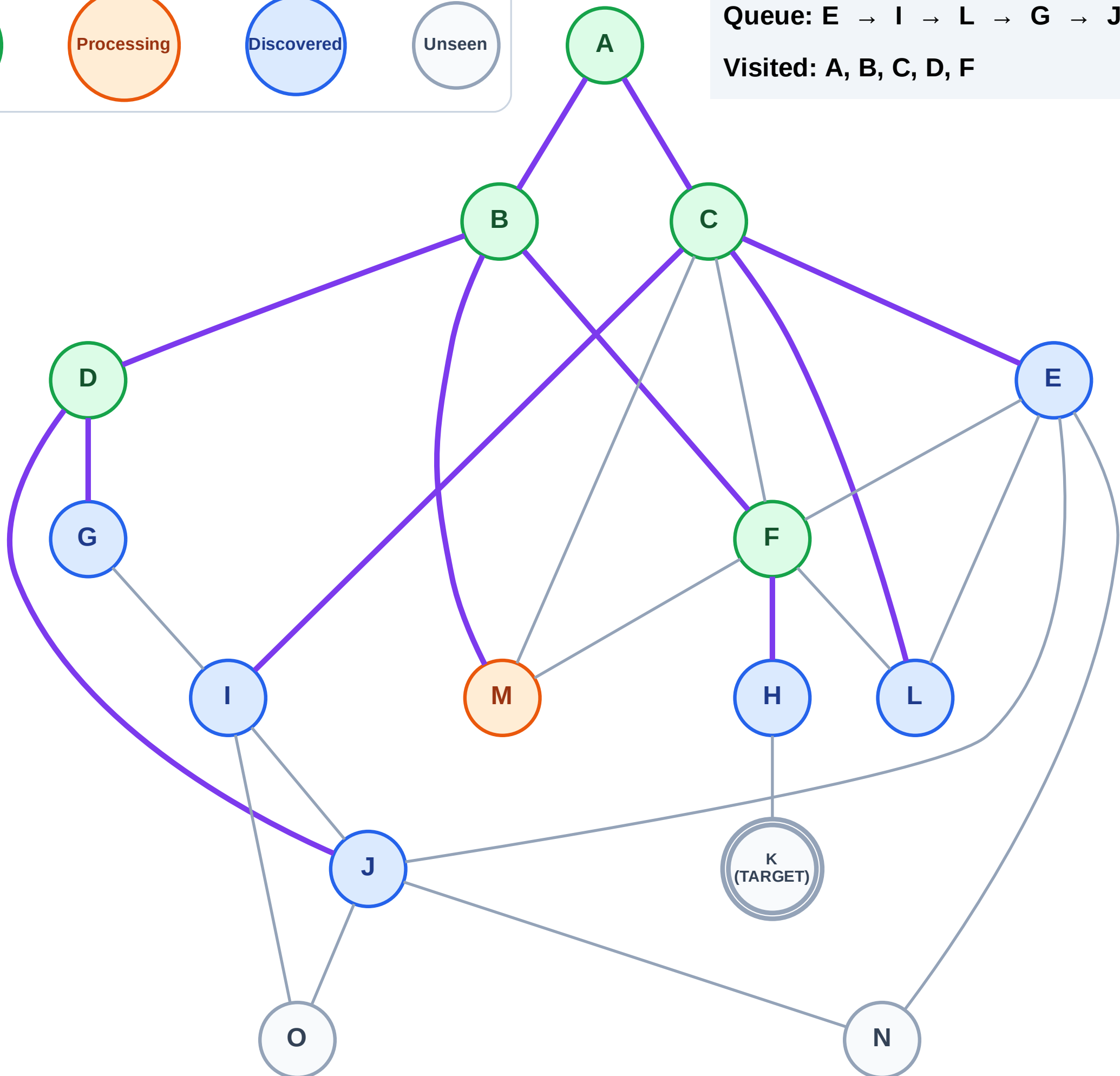
Discovered



Unseen

Queue: E → I → L → G → J → H

Visited: A, B, C, D, F



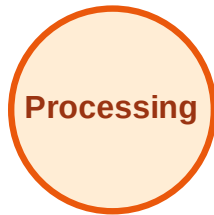
# BFS Traversal

Step 13: No new neighbors from M

## Legend



Complete



Processing



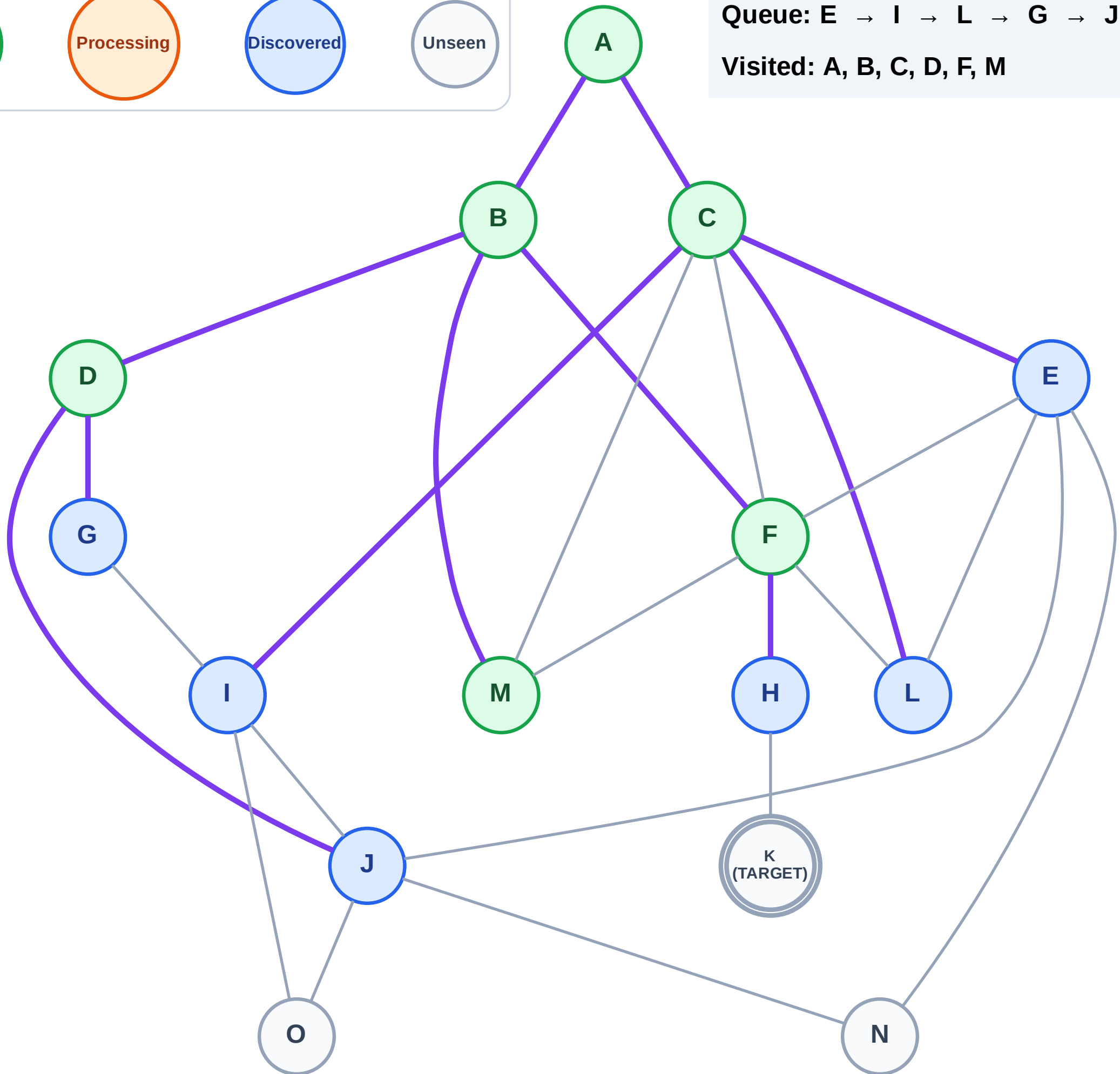
Discovered



Unseen

Queue: E → I → L → G → J → H

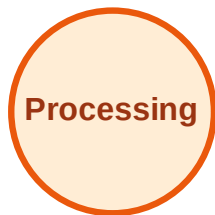
Visited: A, B, C, D, F, M



# BFS Traversal

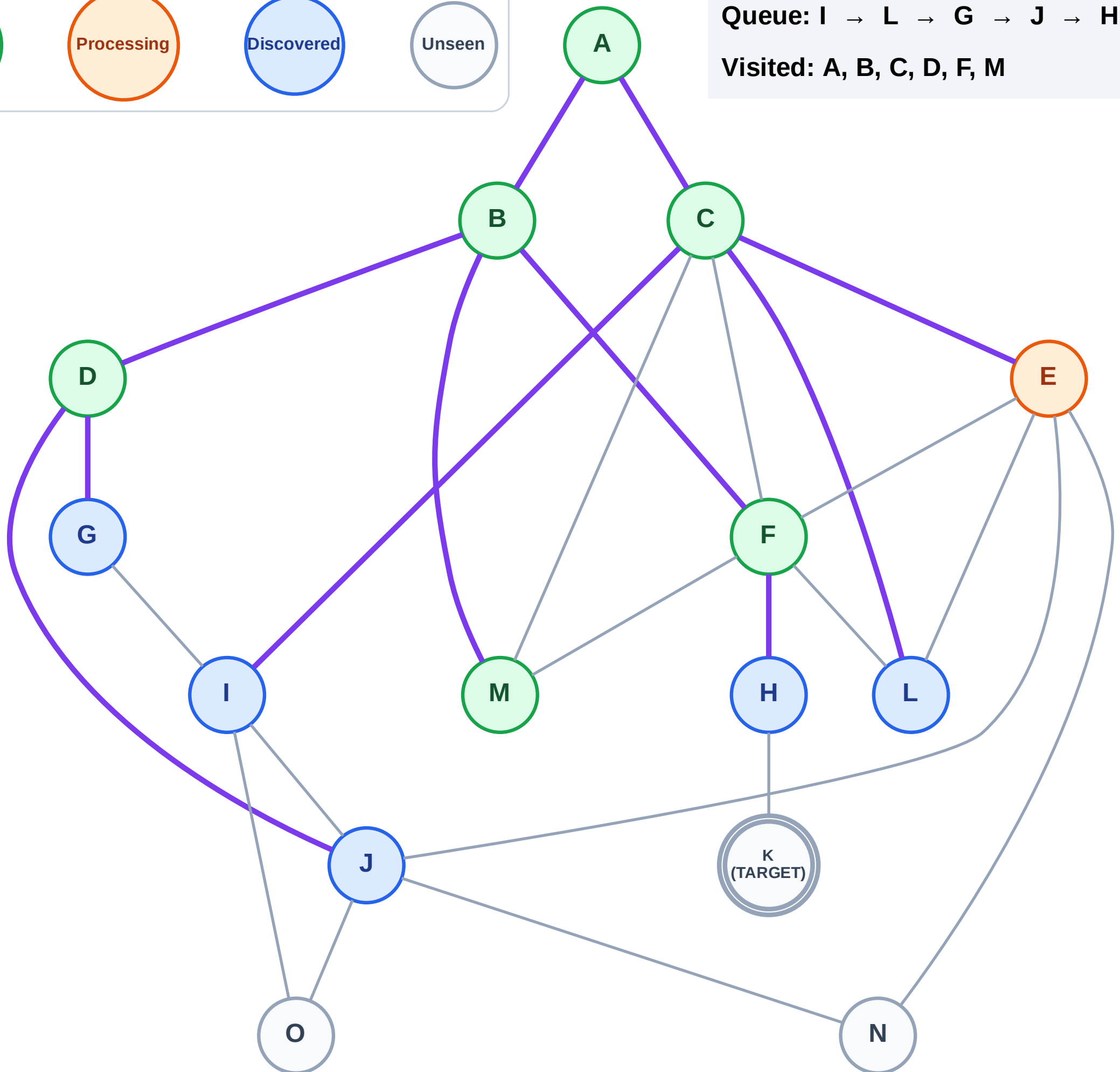
Step 14: Process node E

## Legend



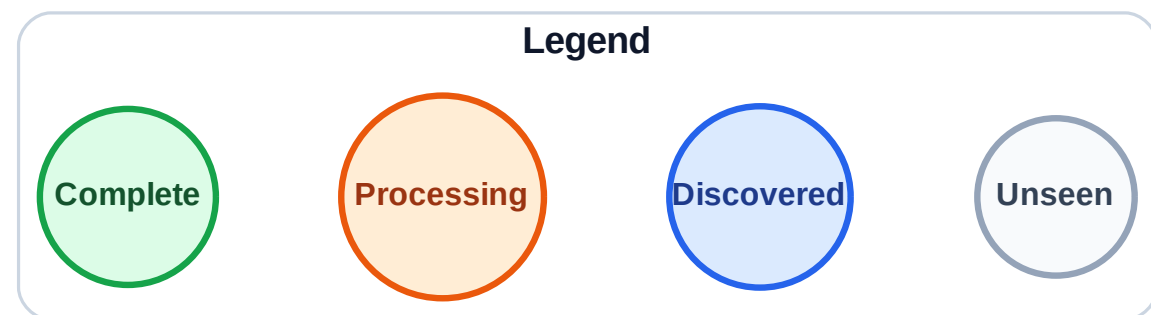
Queue: I → L → G → J → H

Visited: A, B, C, D, F, M



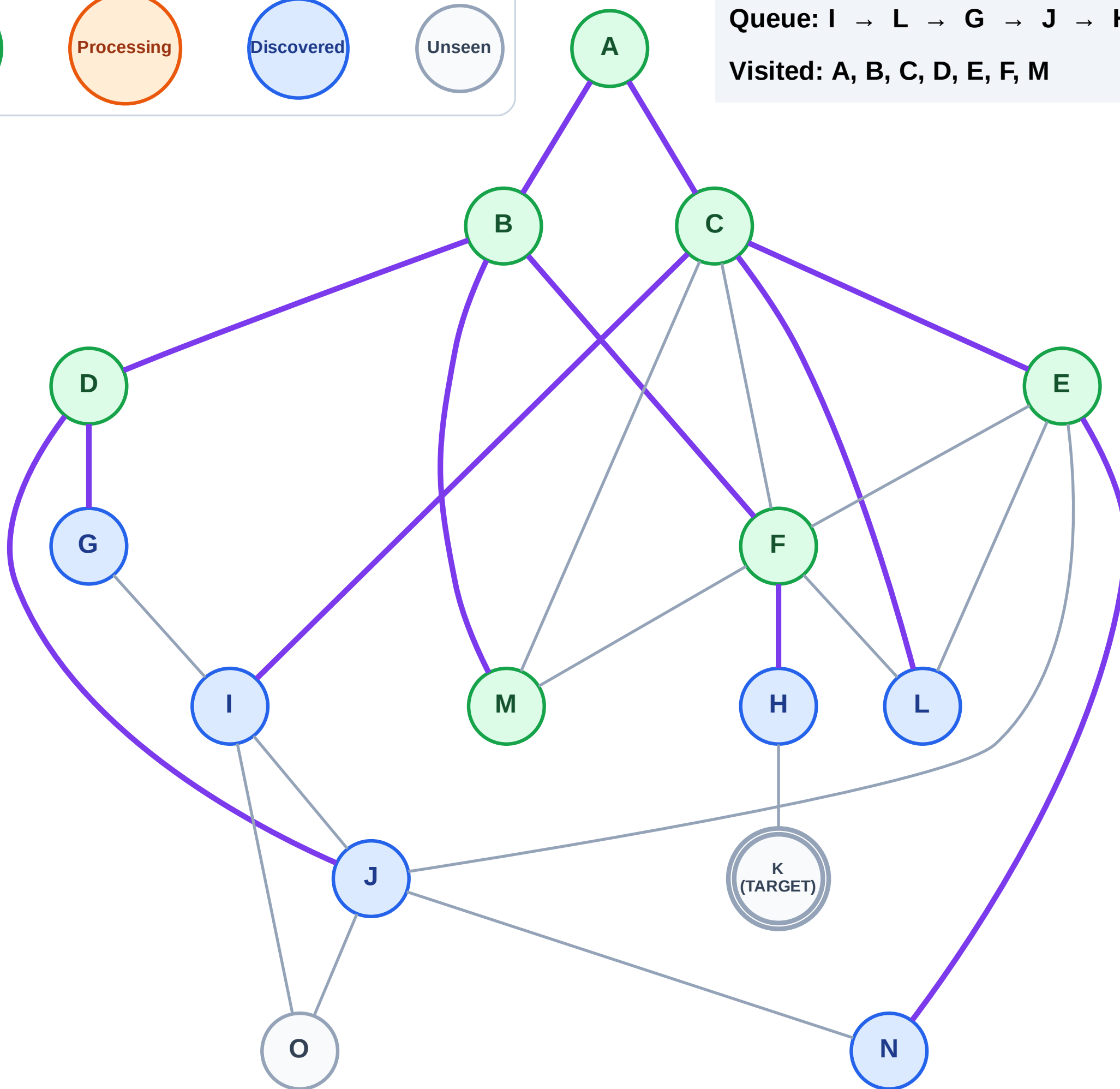
# BFS Traversal

Step 15: Discover N from E



Queue: I → L → G → J → H → N

Visited: A, B, C, D, E, F, M



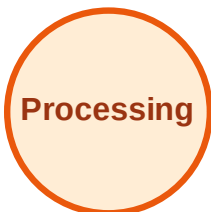
# BFS Traversal

Step 16: Process node I

## Legend



Complete



Processing



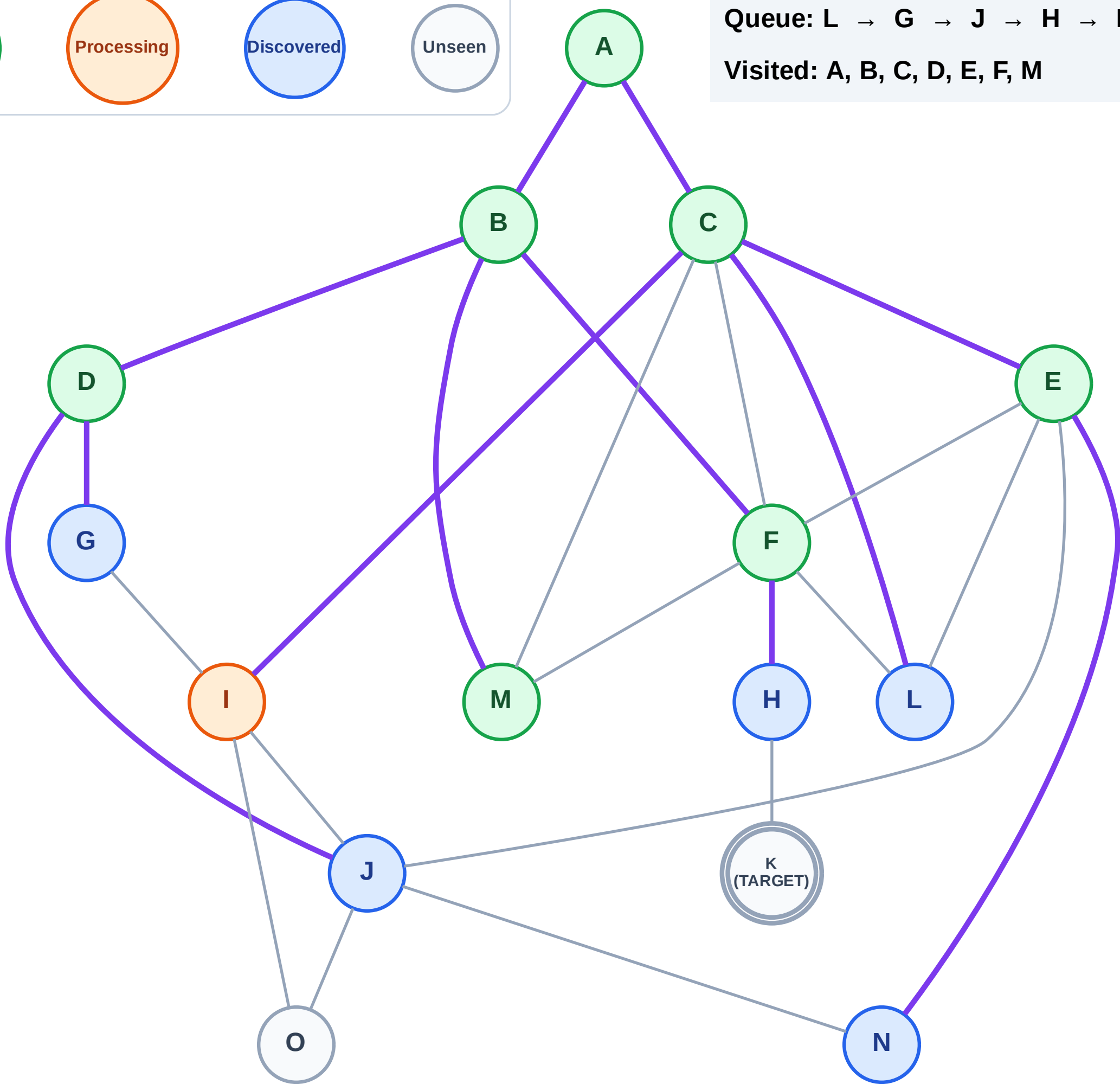
Discovered



Unseen

Queue: L → G → J → H → N

Visited: A, B, C, D, E, F, M



# BFS Traversal

Step 17: Discover O from I

## Legend



Complete



Processing



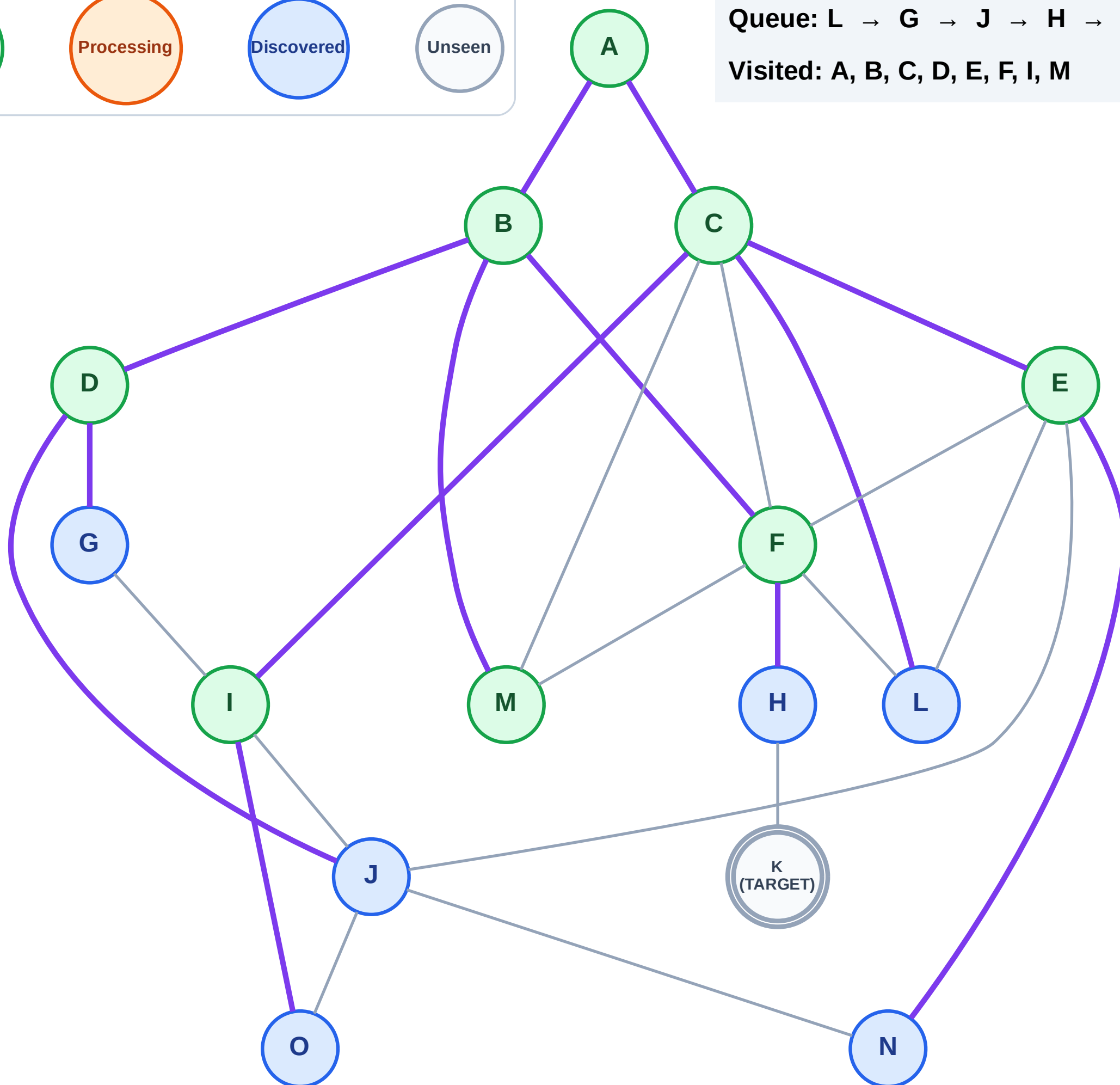
Discovered



Unseen

Queue: L → G → J → H → N → O

Visited: A, B, C, D, E, F, I, M



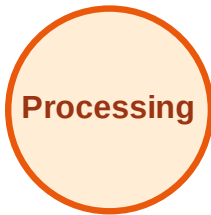
# BFS Traversal

Step 18: Process node L

## Legend



Complete



Processing



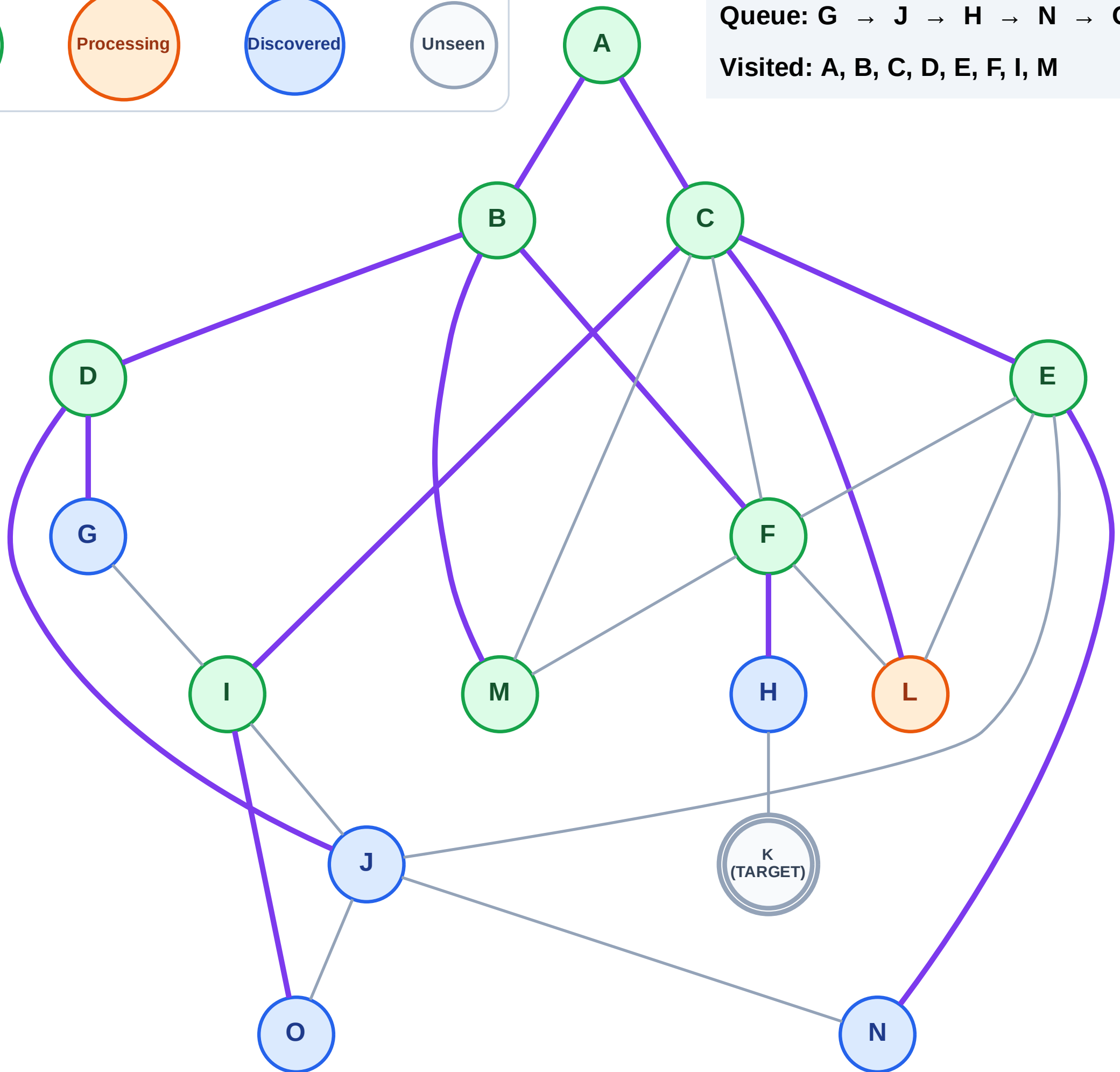
Discovered



Unseen

Queue: G → J → H → N → O

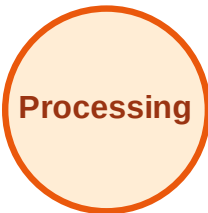
Visited: A, B, C, D, E, F, I, M



# BFS Traversal

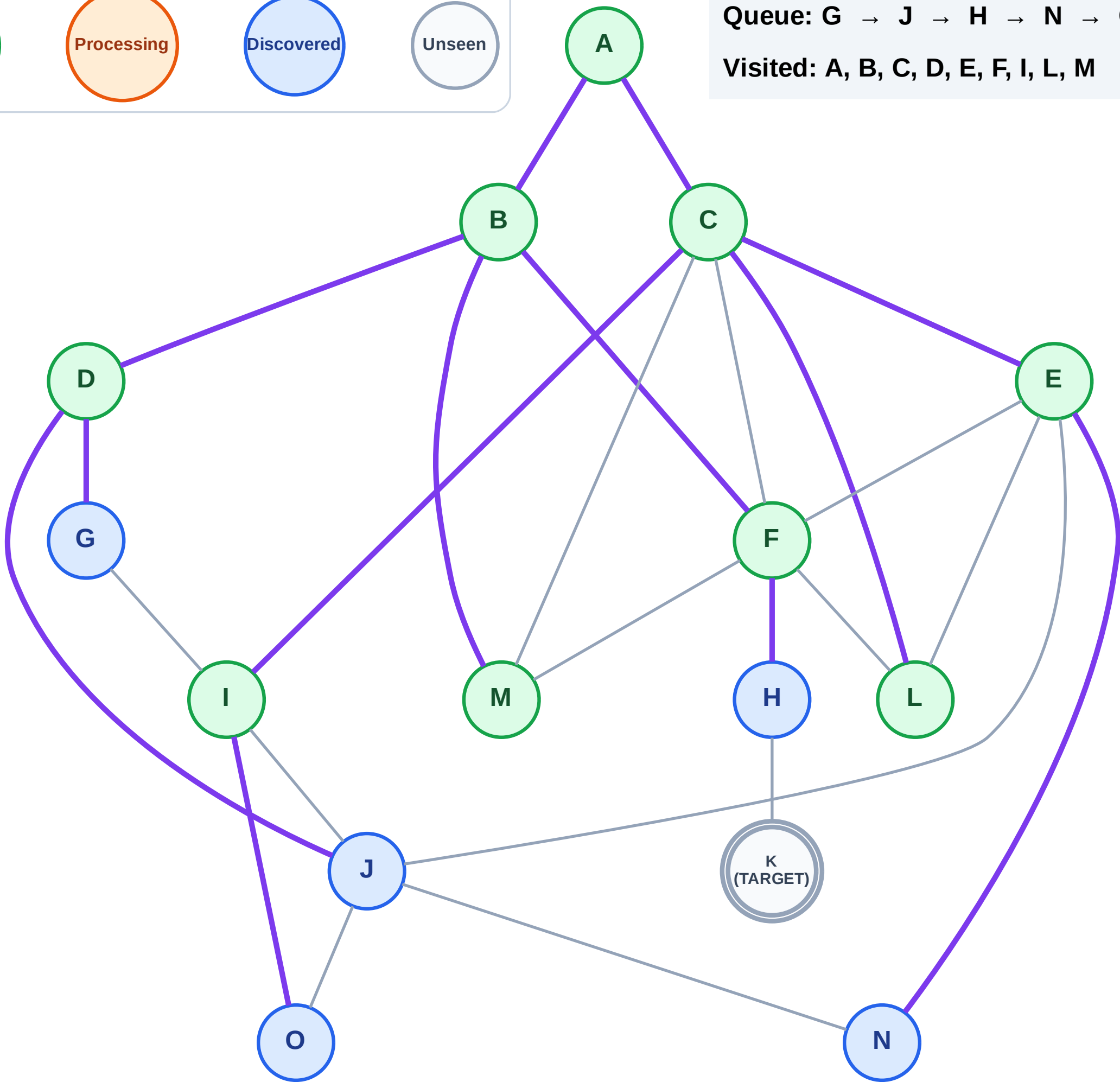
Step 19: No new neighbors from L

## Legend



Queue: G → J → H → N → O

Visited: A, B, C, D, E, F, I, L, M



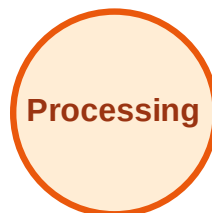
# BFS Traversal

Step 20: Process node G

## Legend



Complete



Processing



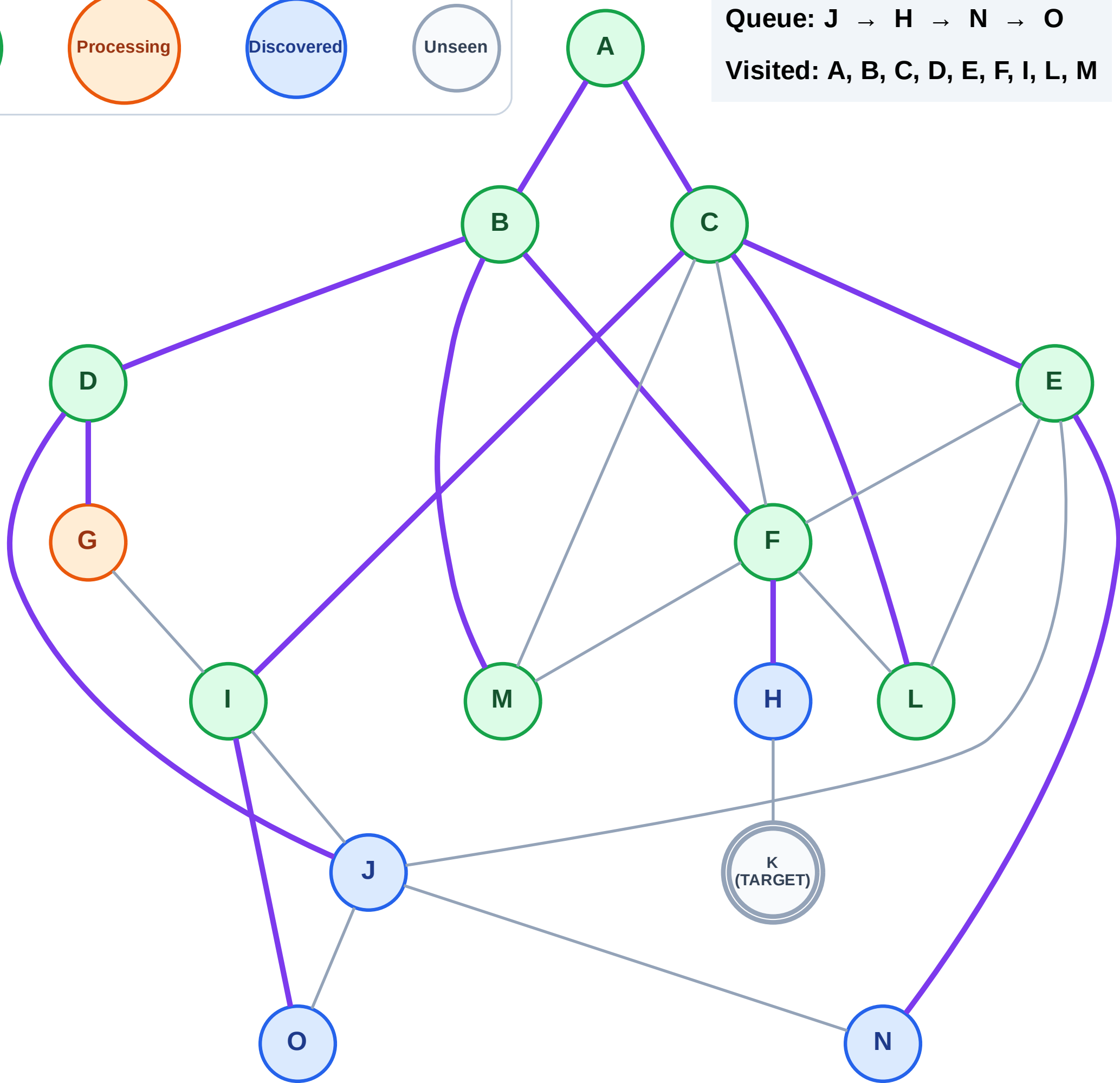
Discovered



Unseen

Queue: J → H → N → O

Visited: A, B, C, D, E, F, I, L, M



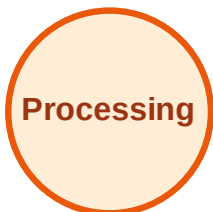
# BFS Traversal

Step 21: No new neighbors from G

## Legend



Complete



Processing



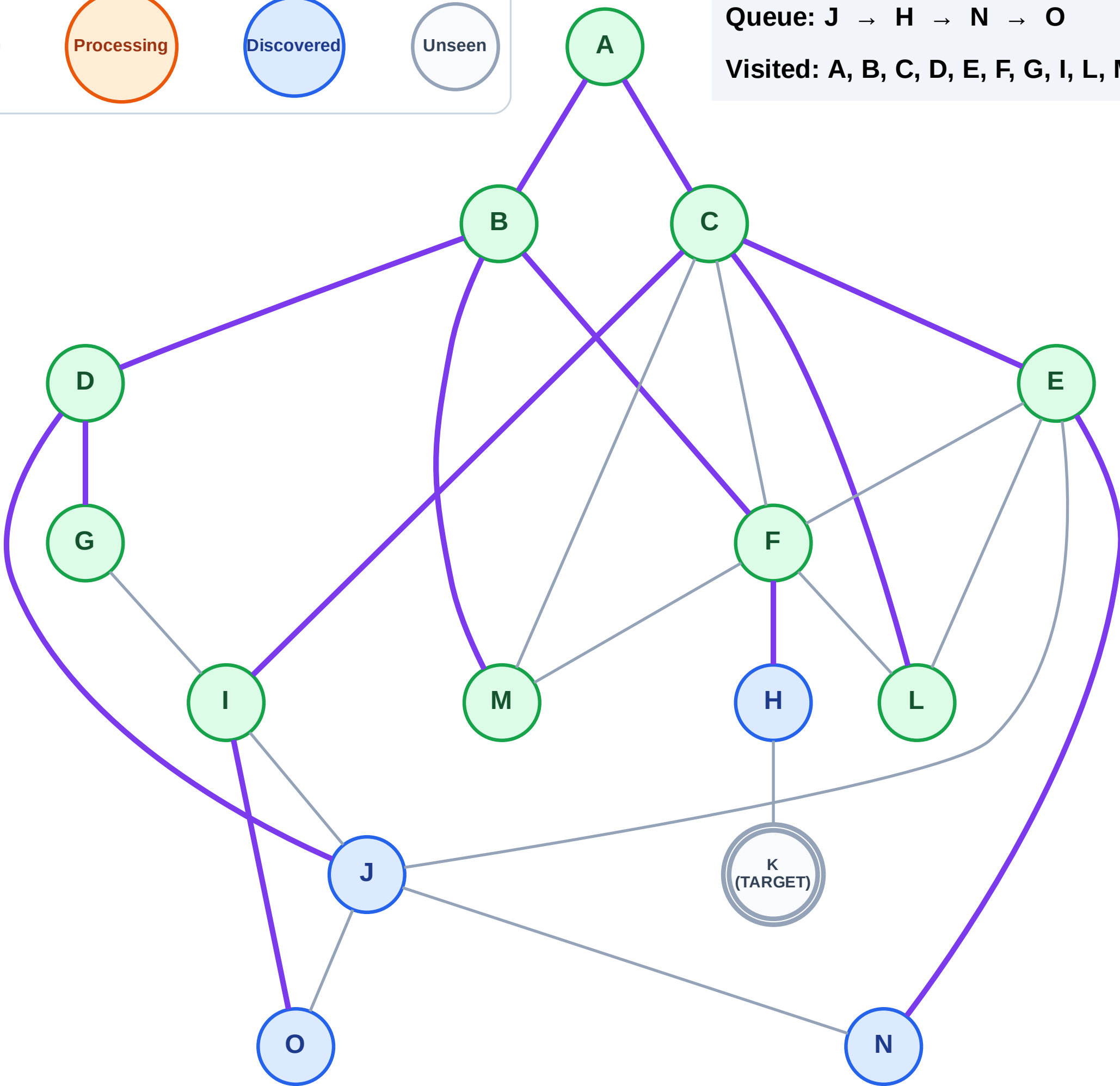
Discovered



Unseen

Queue: J → H → N → O

Visited: A, B, C, D, E, F, G, I, L, M



# BFS Traversal

Step 22: Process node J

Legend

Complete

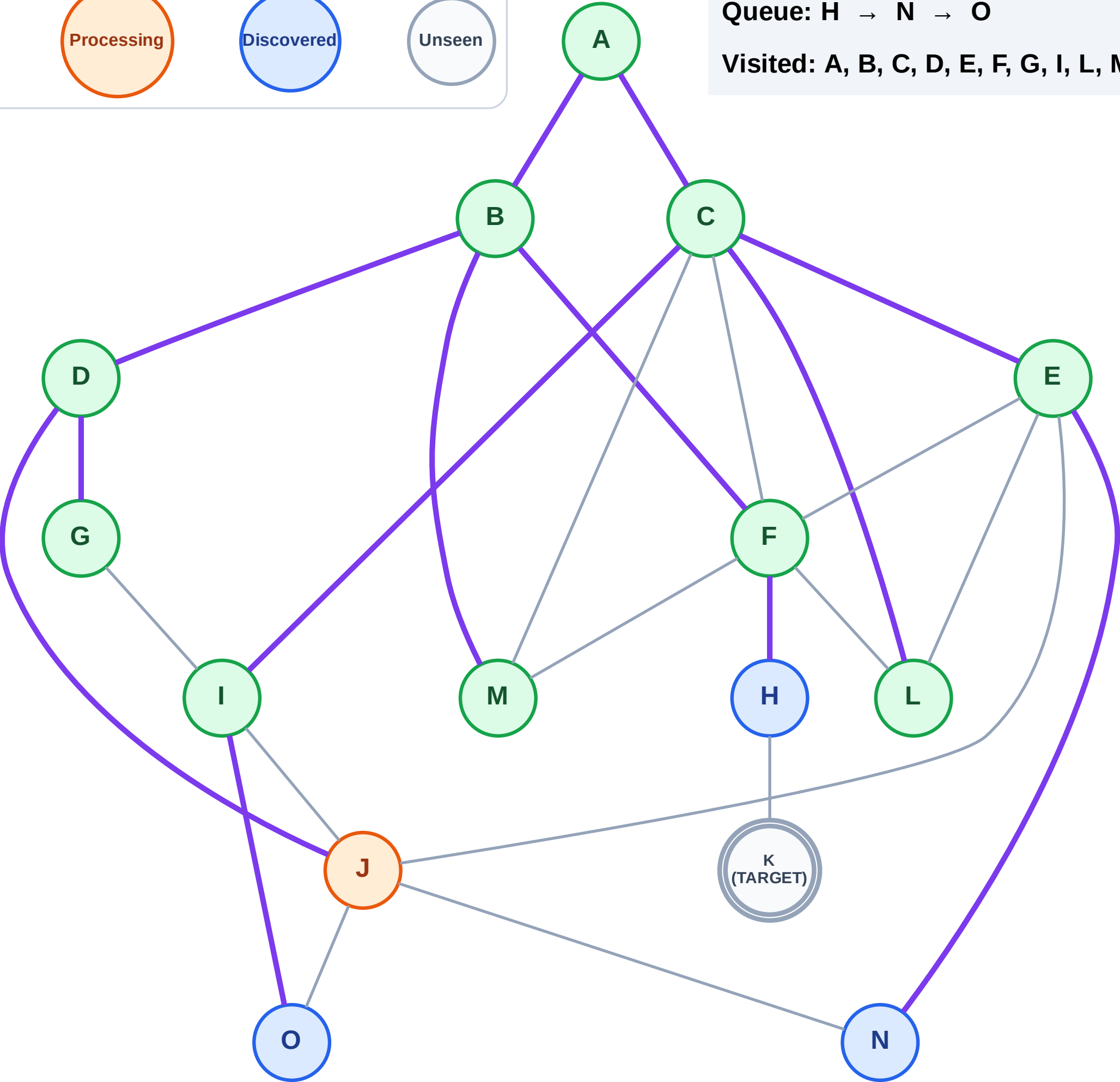
Processing

Discovered

Unseen

Queue: H → N → O

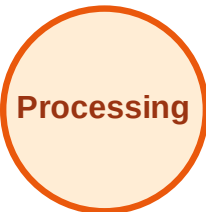
Visited: A, B, C, D, E, F, G, I, L, M



# BFS Traversal

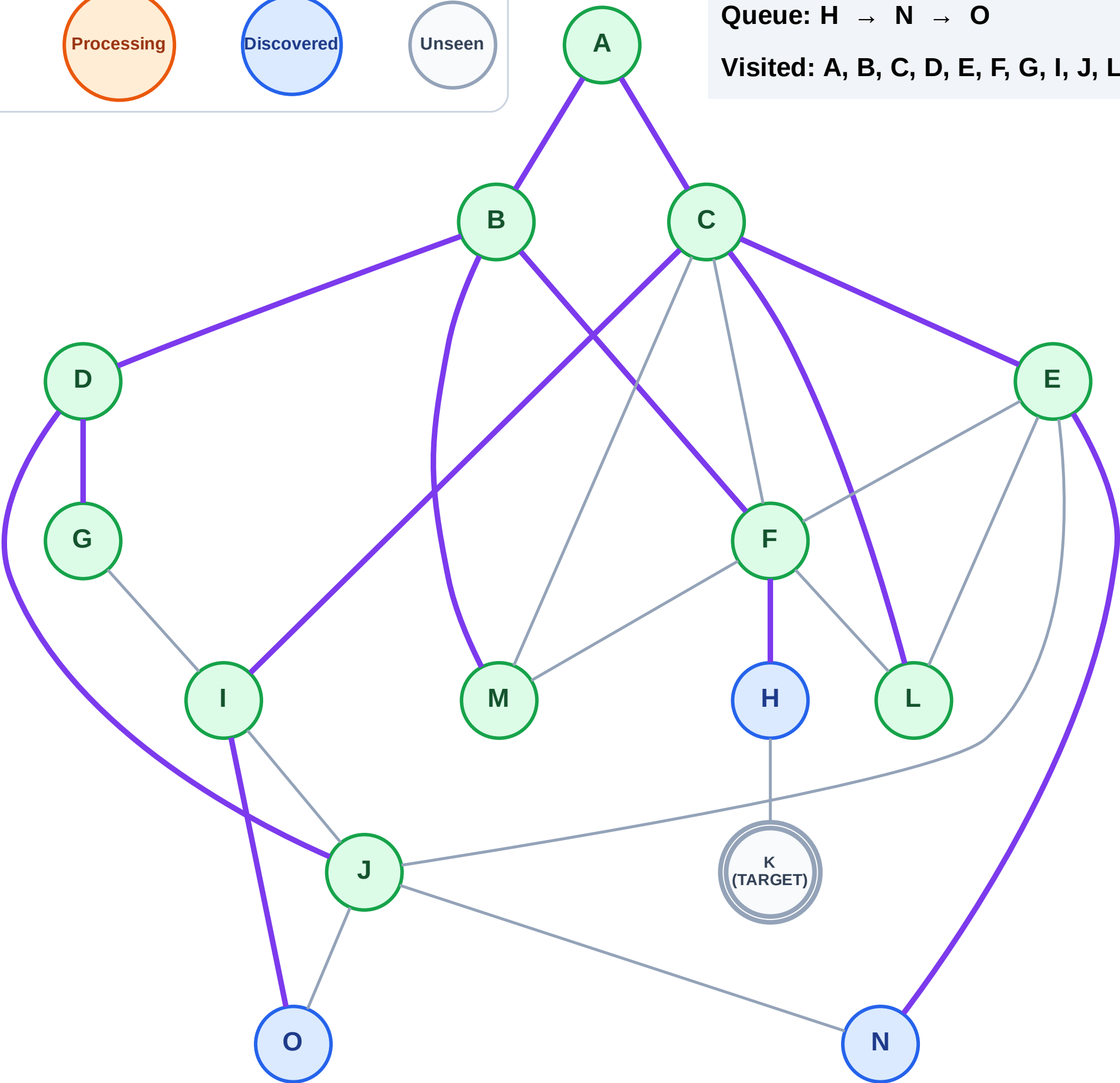
Step 23: No new neighbors from J

## Legend



Queue: H → N → O

Visited: A, B, C, D, E, F, G, I, J, L, M



# BFS Traversal

Step 24: Process node H

Legend

Complete

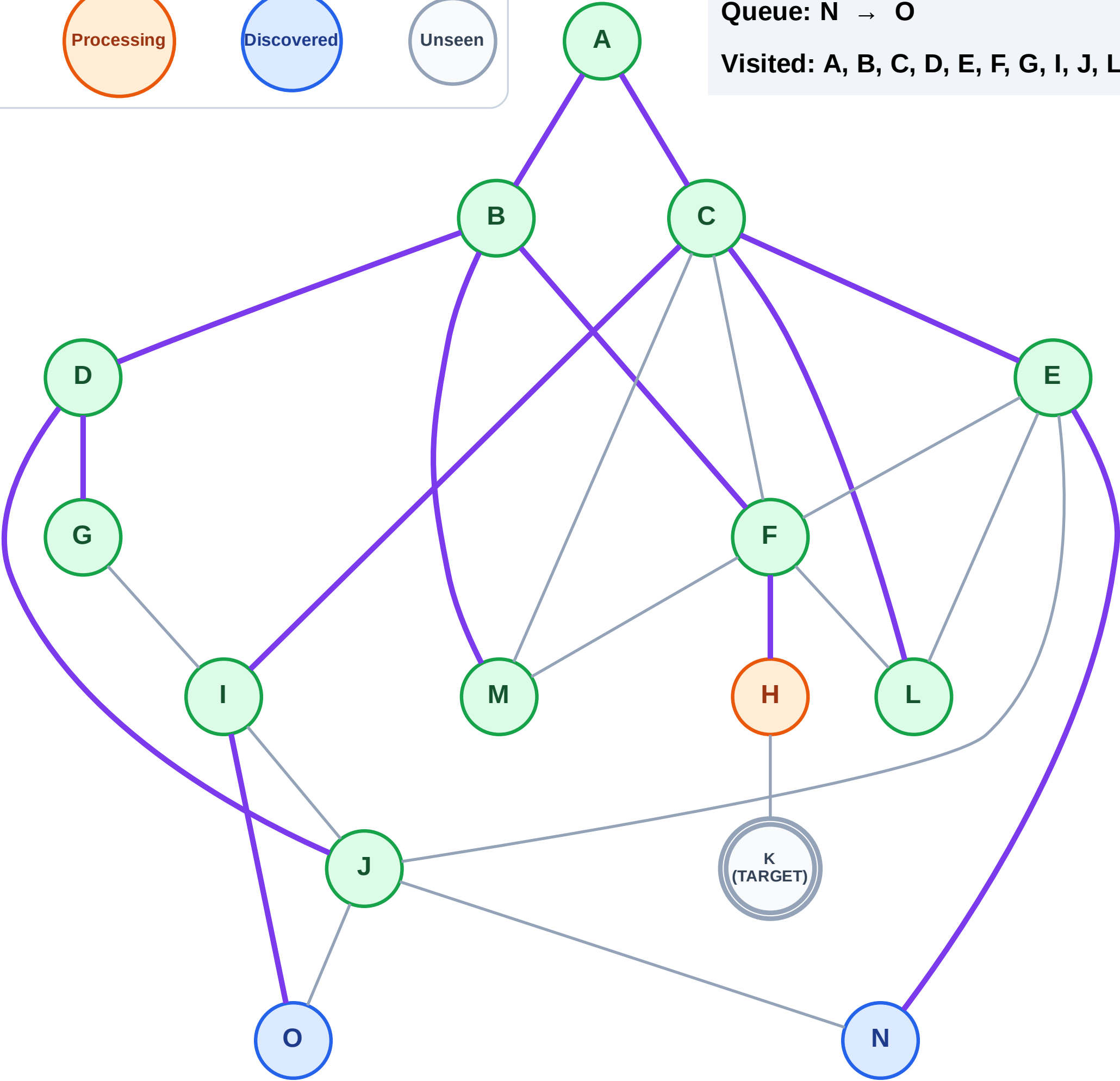
Processing

Discovered

Unseen

Queue: N → O

Visited: A, B, C, D, E, F, G, I, J, L, M



# BFS Traversal

Step 25: Discover K from H

Legend

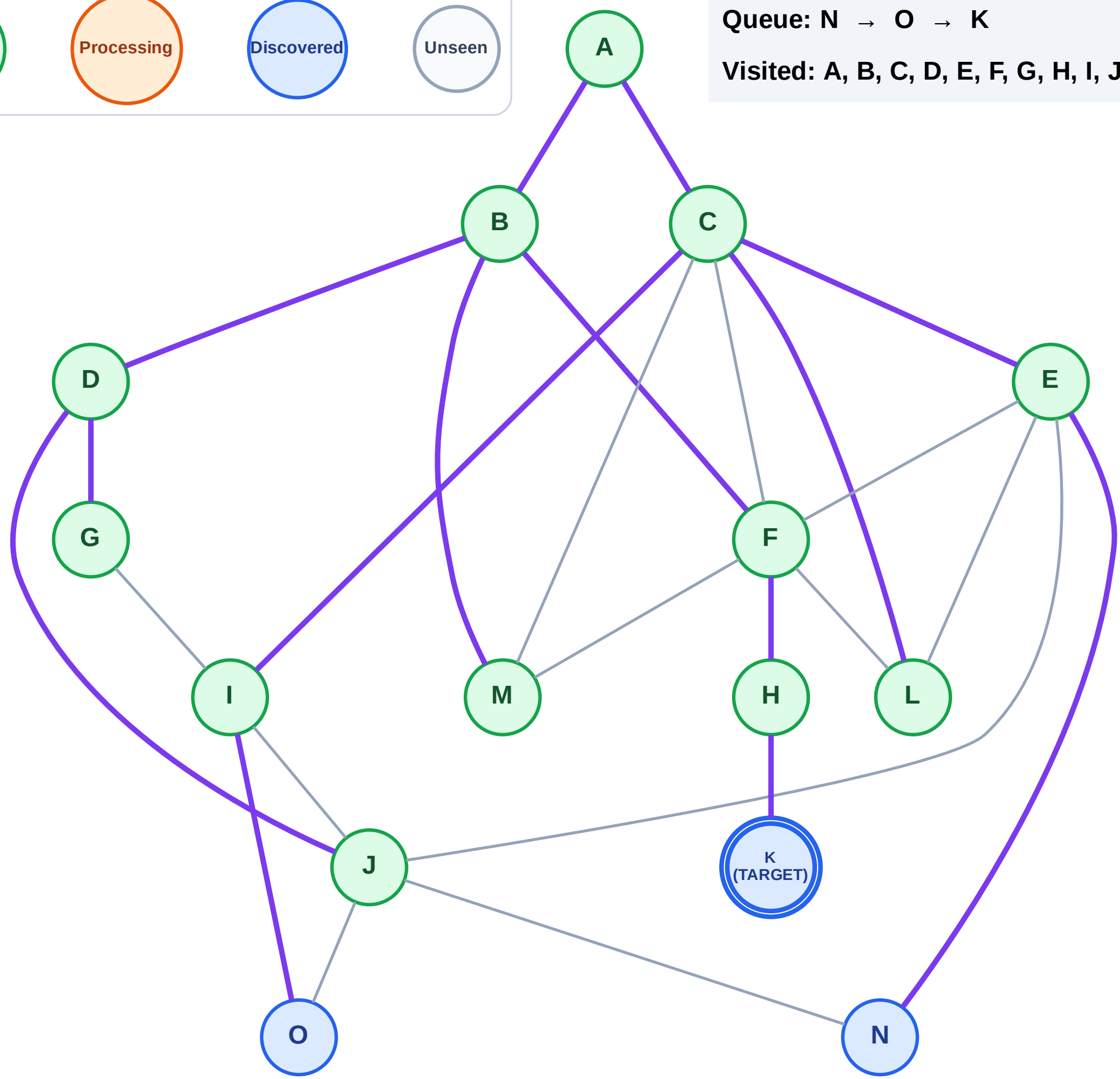
Complete

Processing

Discovered

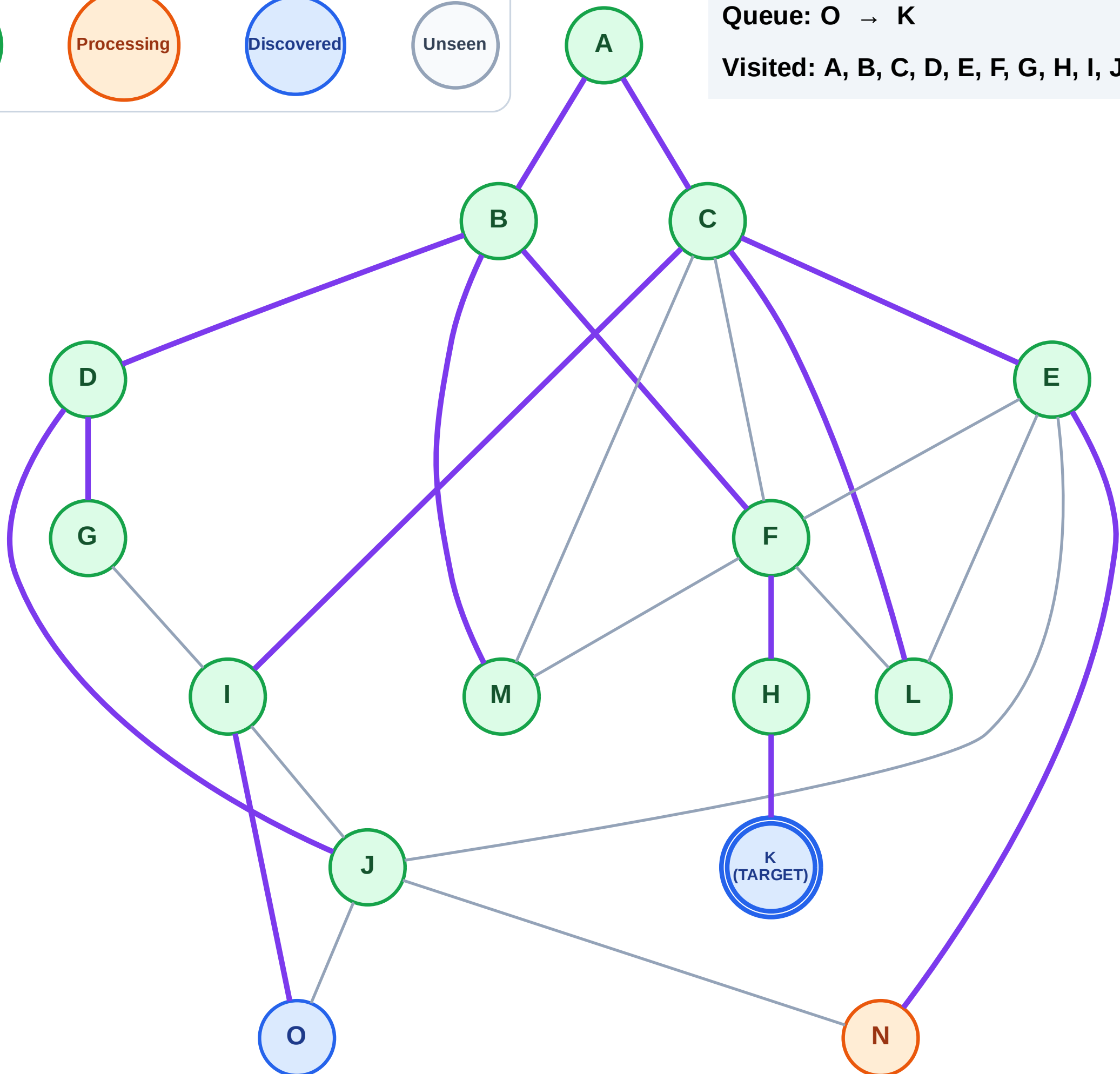
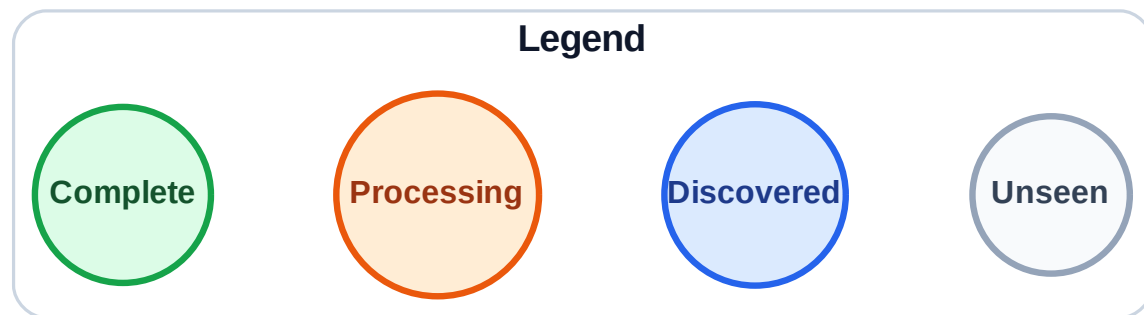
Unseen

Queue: N → O → K  
Visited: A, B, C, D, E, F, G, H, I, J, L, M



## Step 26: Process node N

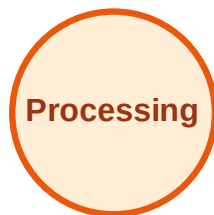
### Step 26: Process node N



# BFS Traversal

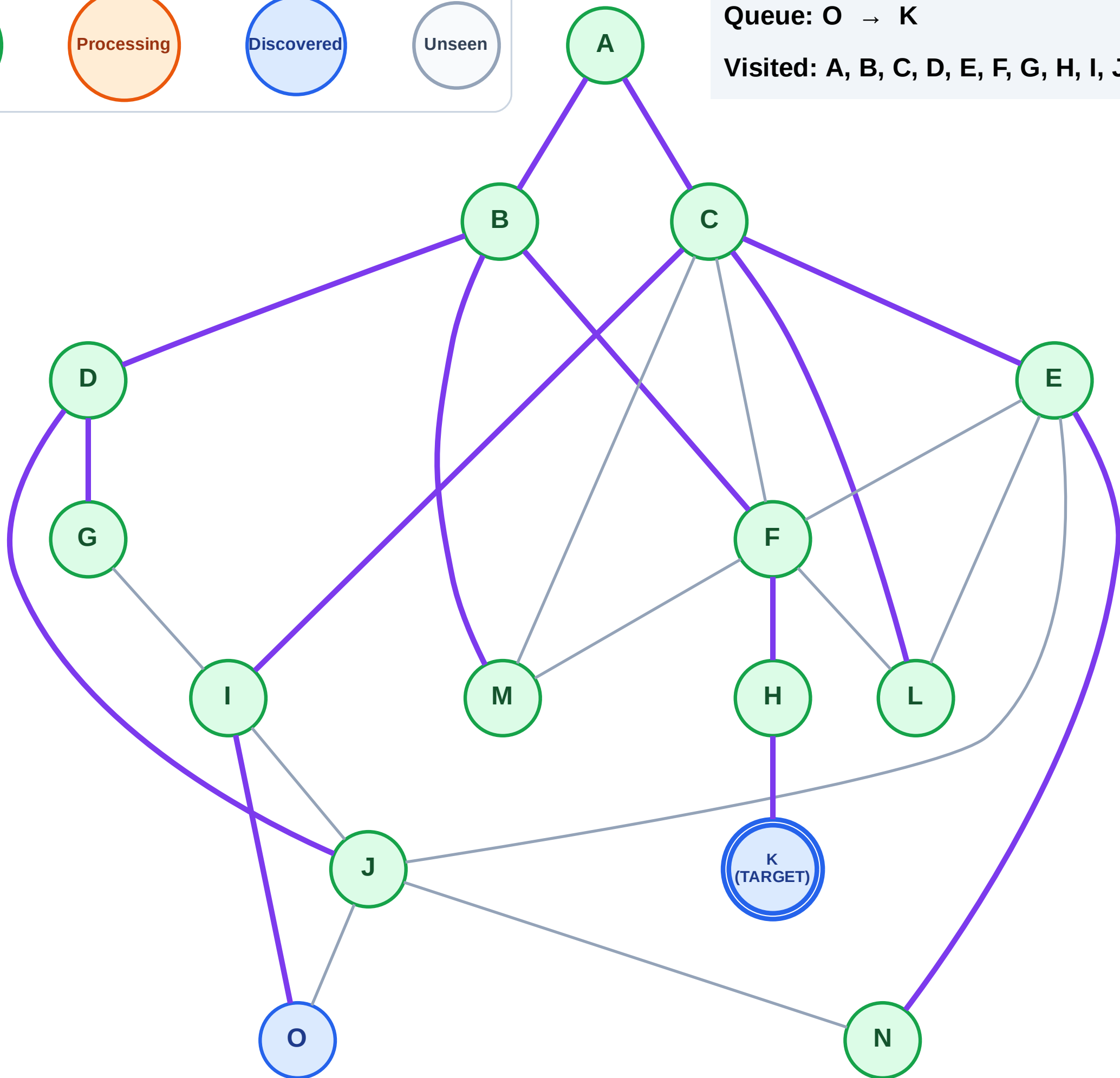
Step 27: No new neighbors from N

## Legend

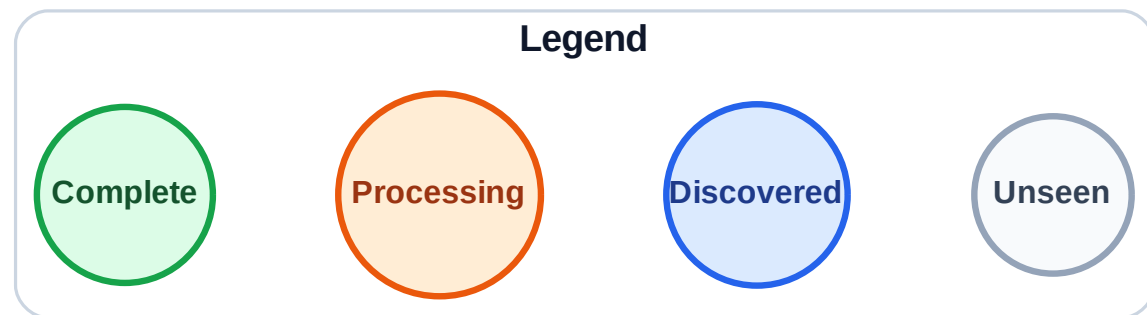


Queue: O → K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N

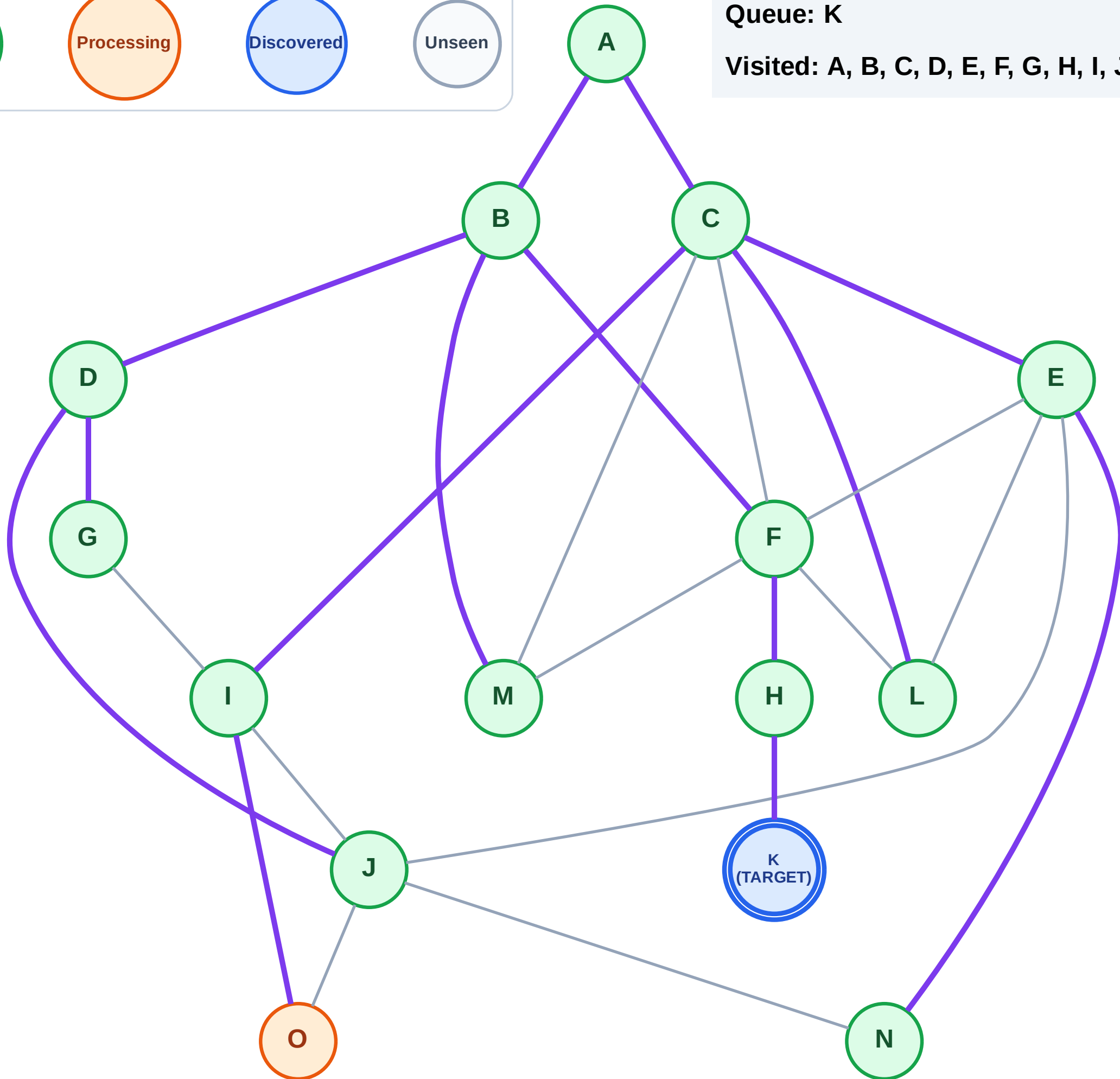


## Step 28: Process node O



**Queue: K**

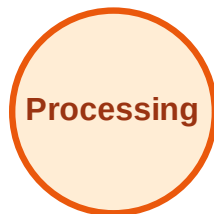
**Visited: A, B, C, D, E, F, G, H, I, J, L, M, N**



# BFS Traversal

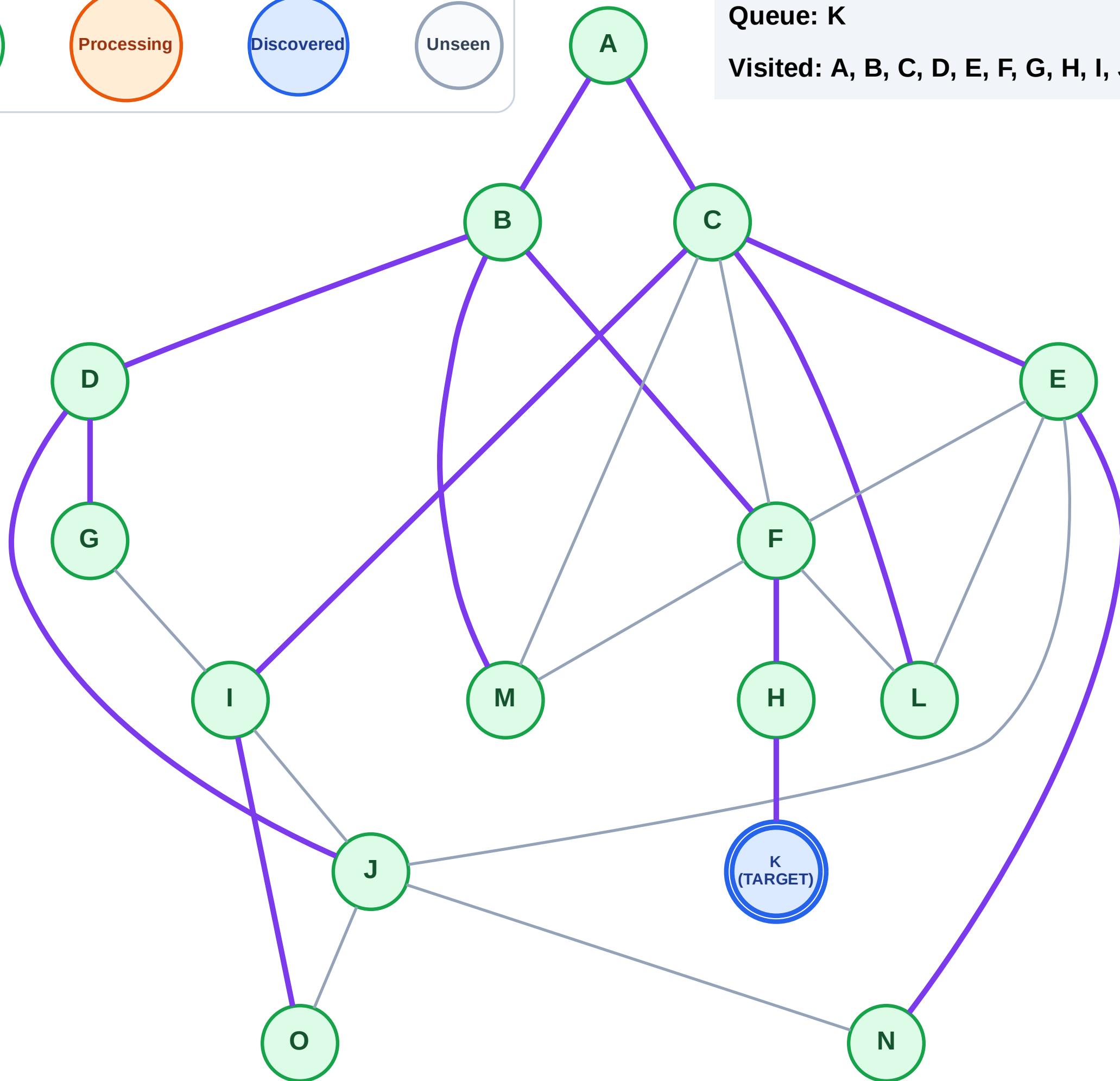
Step 29: No new neighbors from O

## Legend



Queue: K

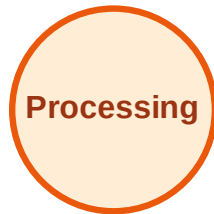
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



# BFS Traversal

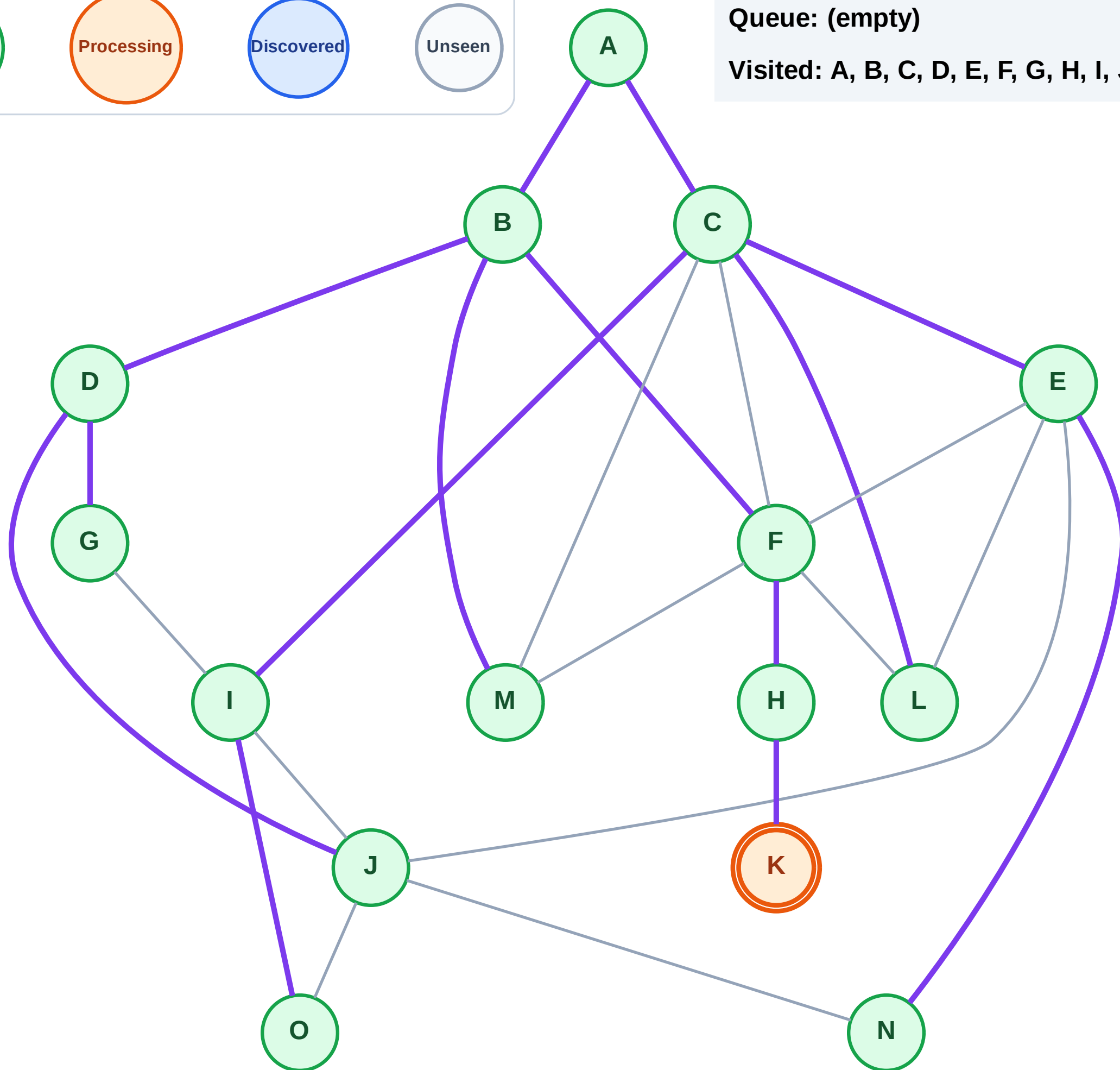
Step 30: Process node K

## Legend



Queue: (empty)

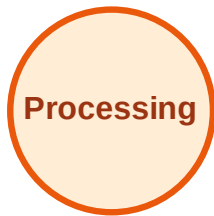
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



# BFS Traversal

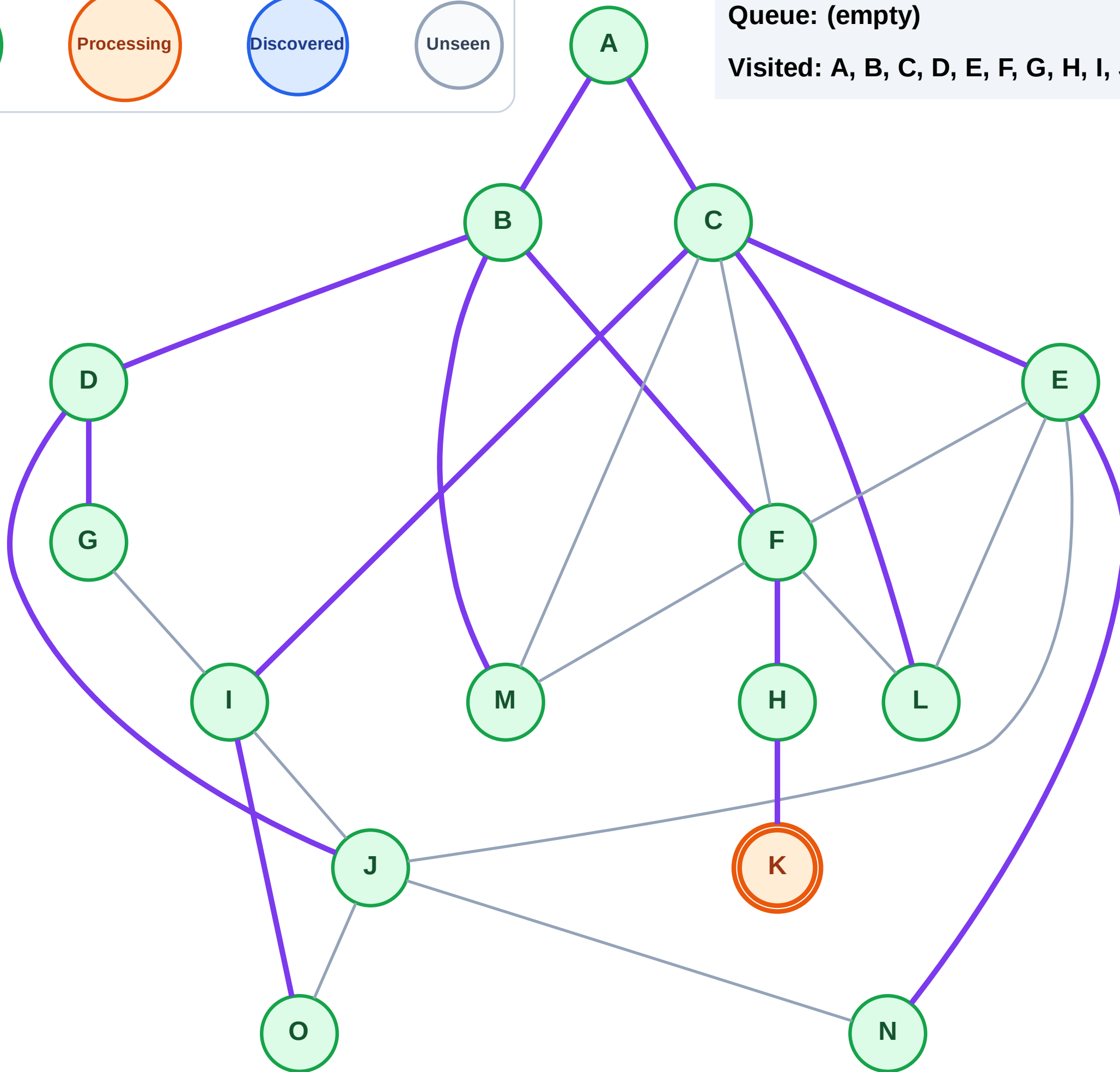
Step 31: TARGET NODE K REACHED! Path (4 hops): A → B → F → H → K

## Legend



Queue: (empty)

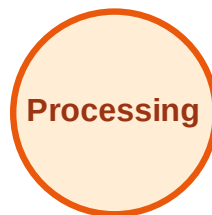
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



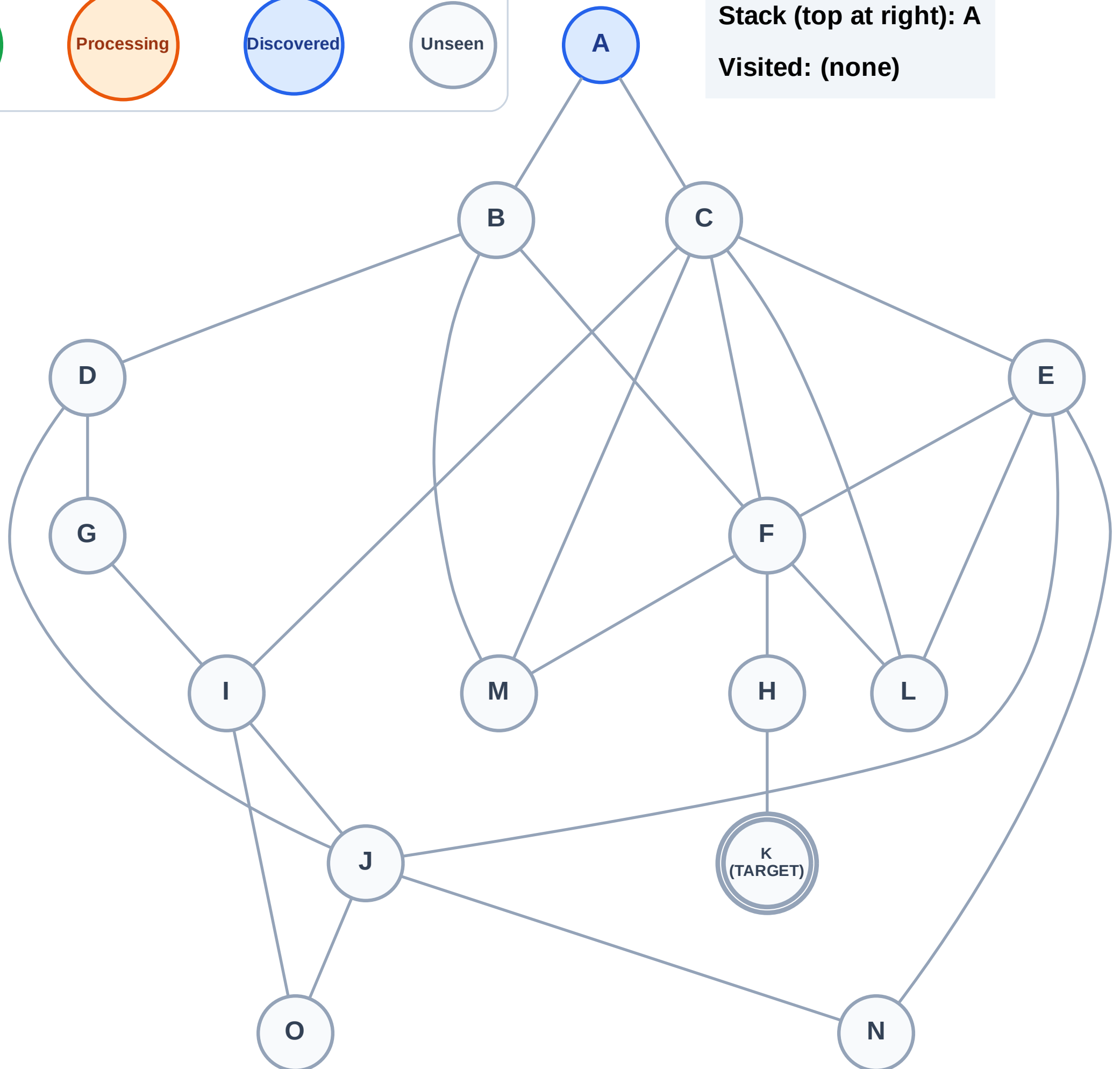
# DFS Traversal

Step 1: Start at A (Seeking target node K)

## Legend



**Stack (top at right): A**  
**Visited: (none)**



# DFS Traversal

Step 2: Process node A

Legend

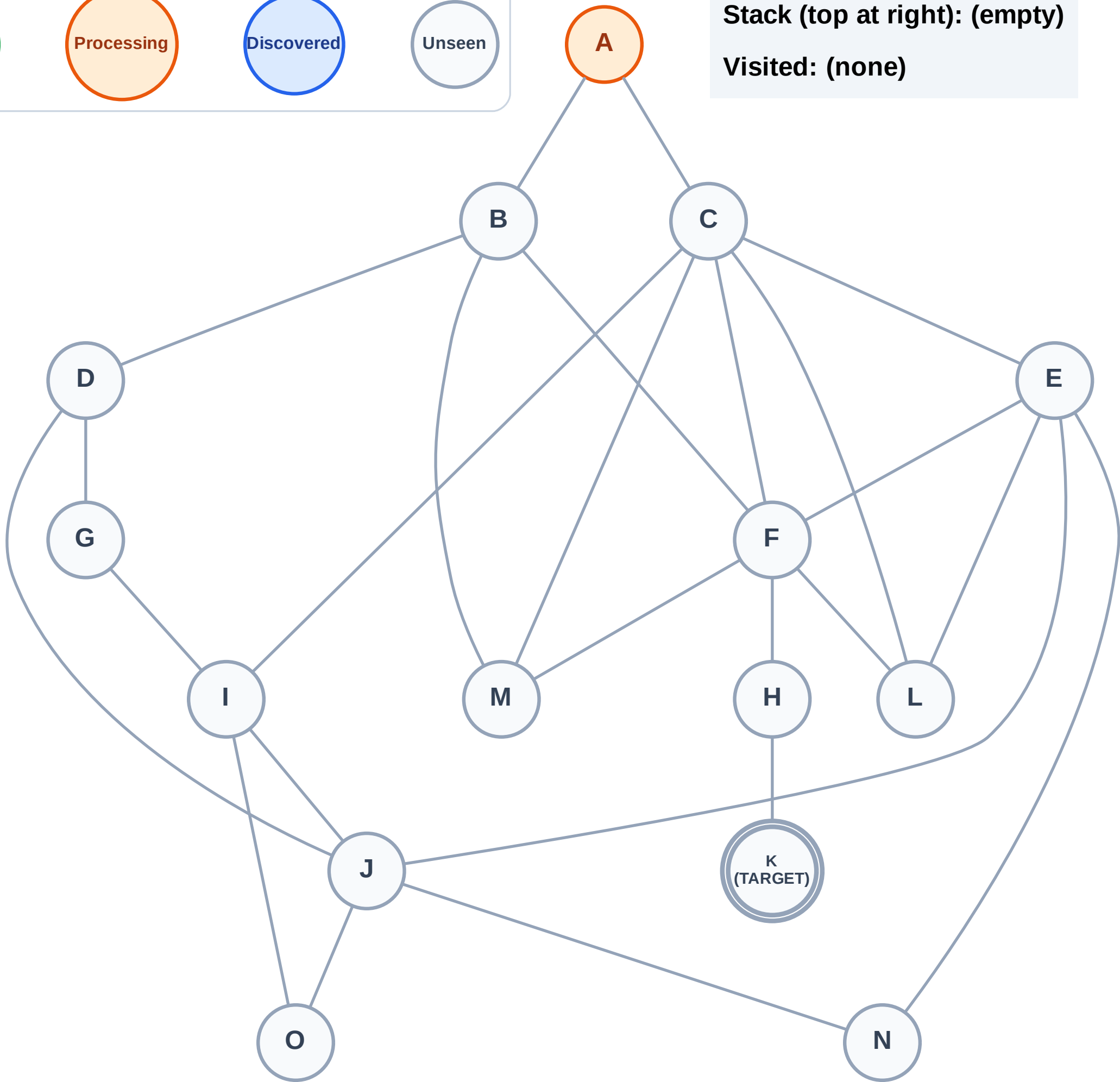
Complete

Processing

Discovered

Unseen

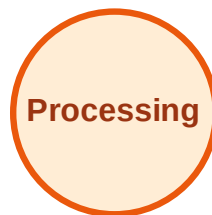
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

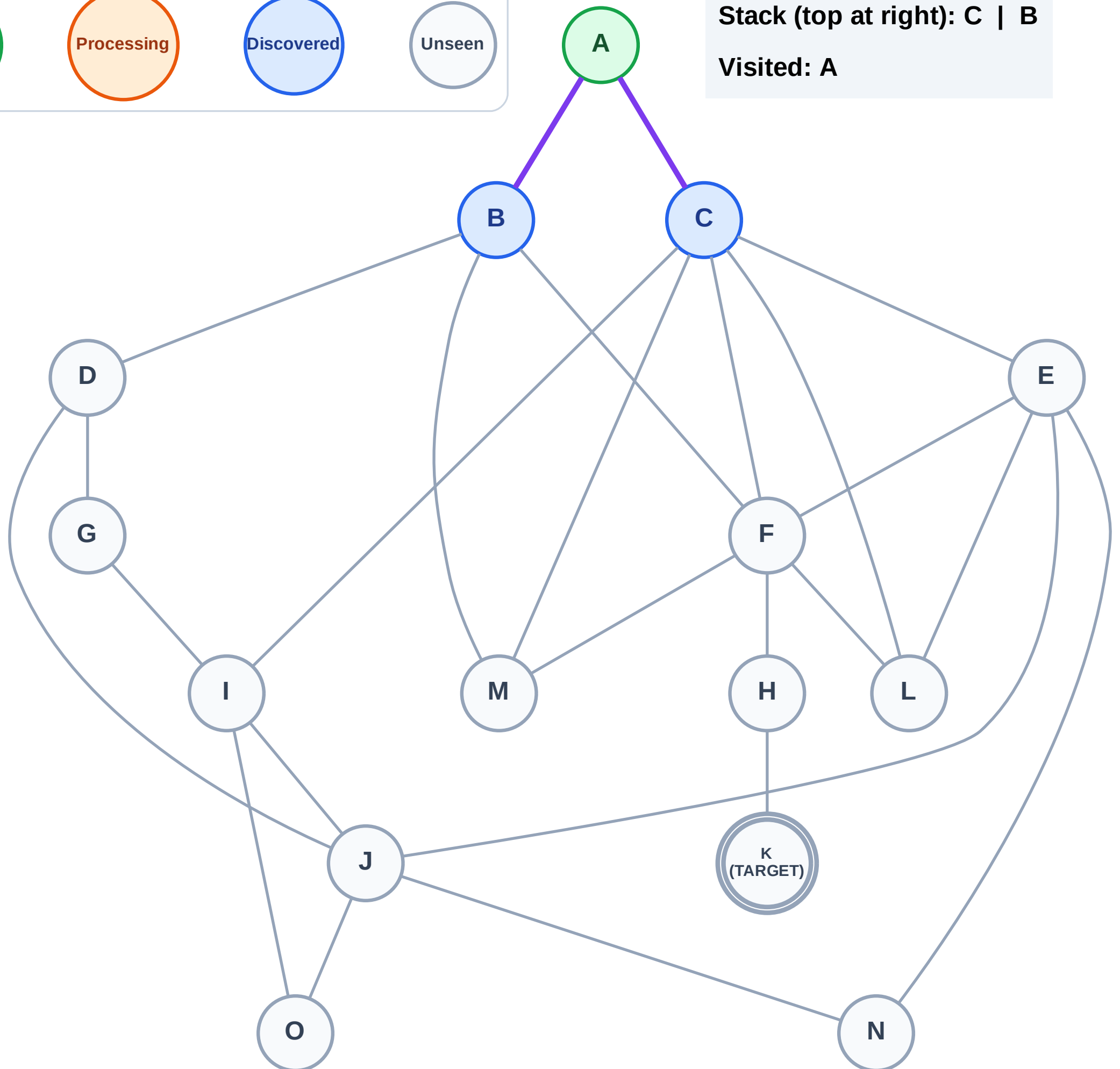
Step 3: Discover C, B from A

## Legend



Stack (top at right): C | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

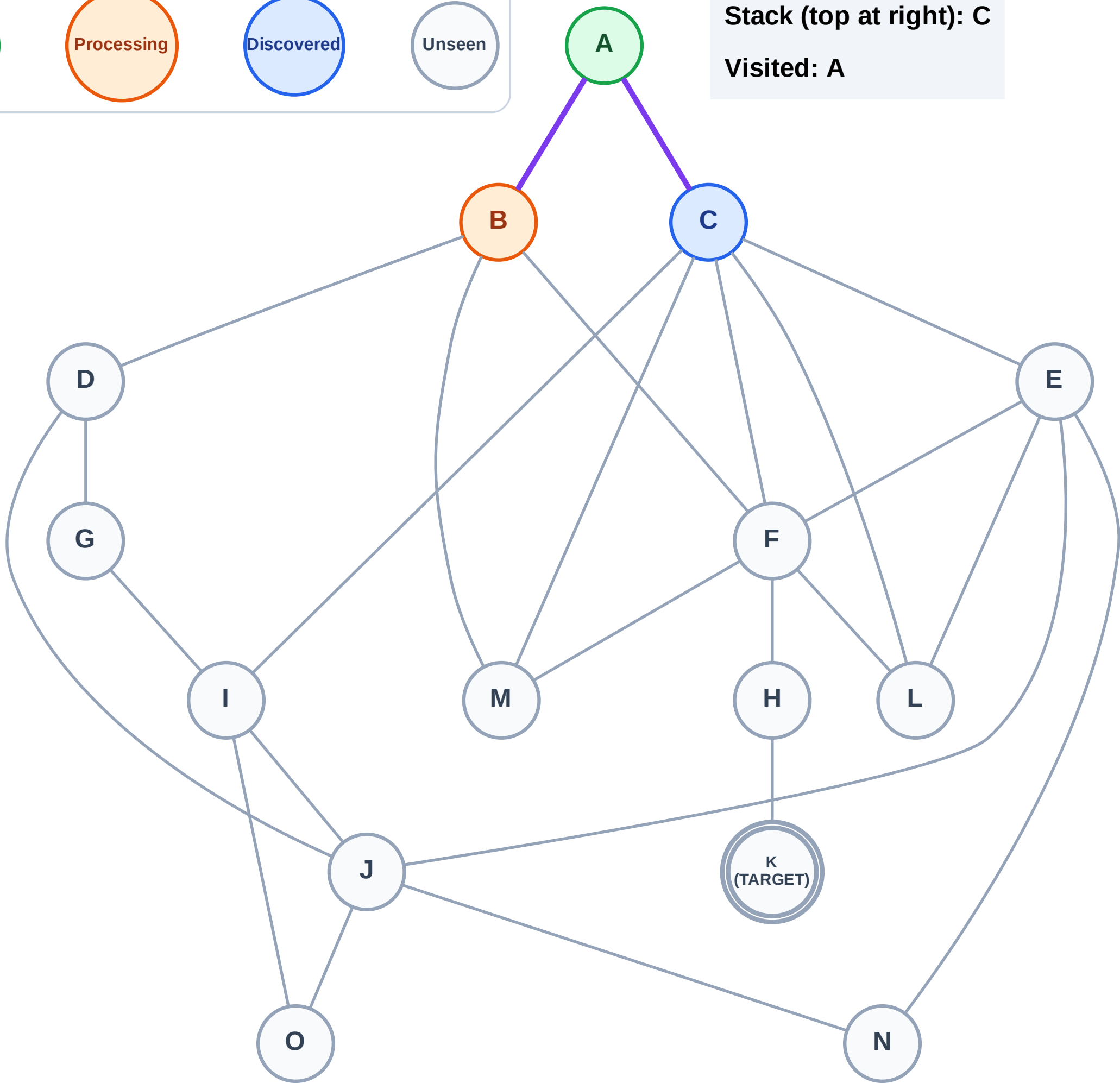
Complete

Processing

Discovered

Unseen

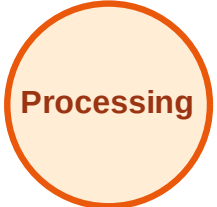
Stack (top at right): C  
Visited: A



# DFS Traversal

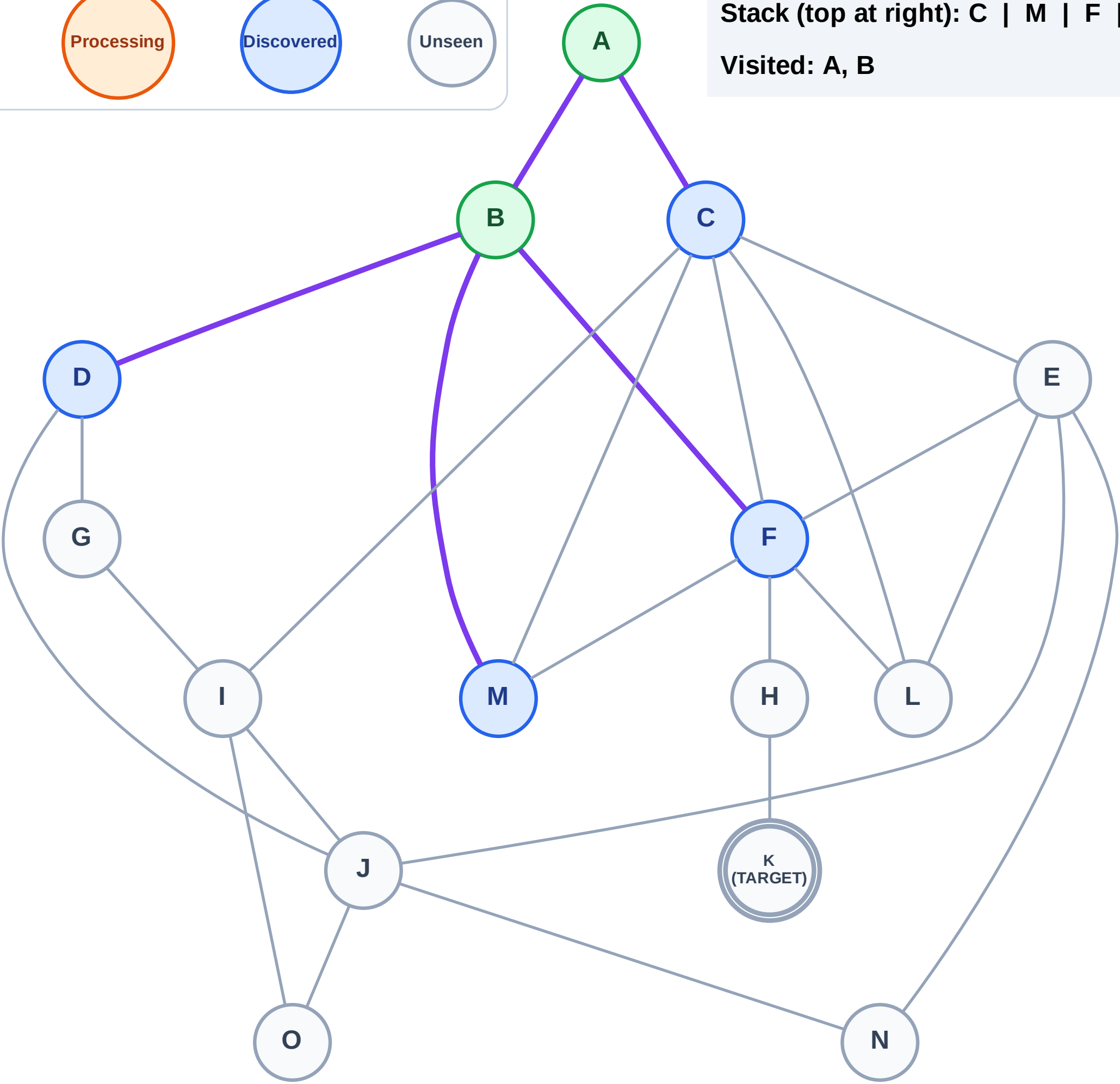
Step 5: Discover M, F, D from B

## Legend



Stack (top at right): C | M | F | D

Visited: A, B



# DFS Traversal

Step 6: Process node D

Legend

Complete

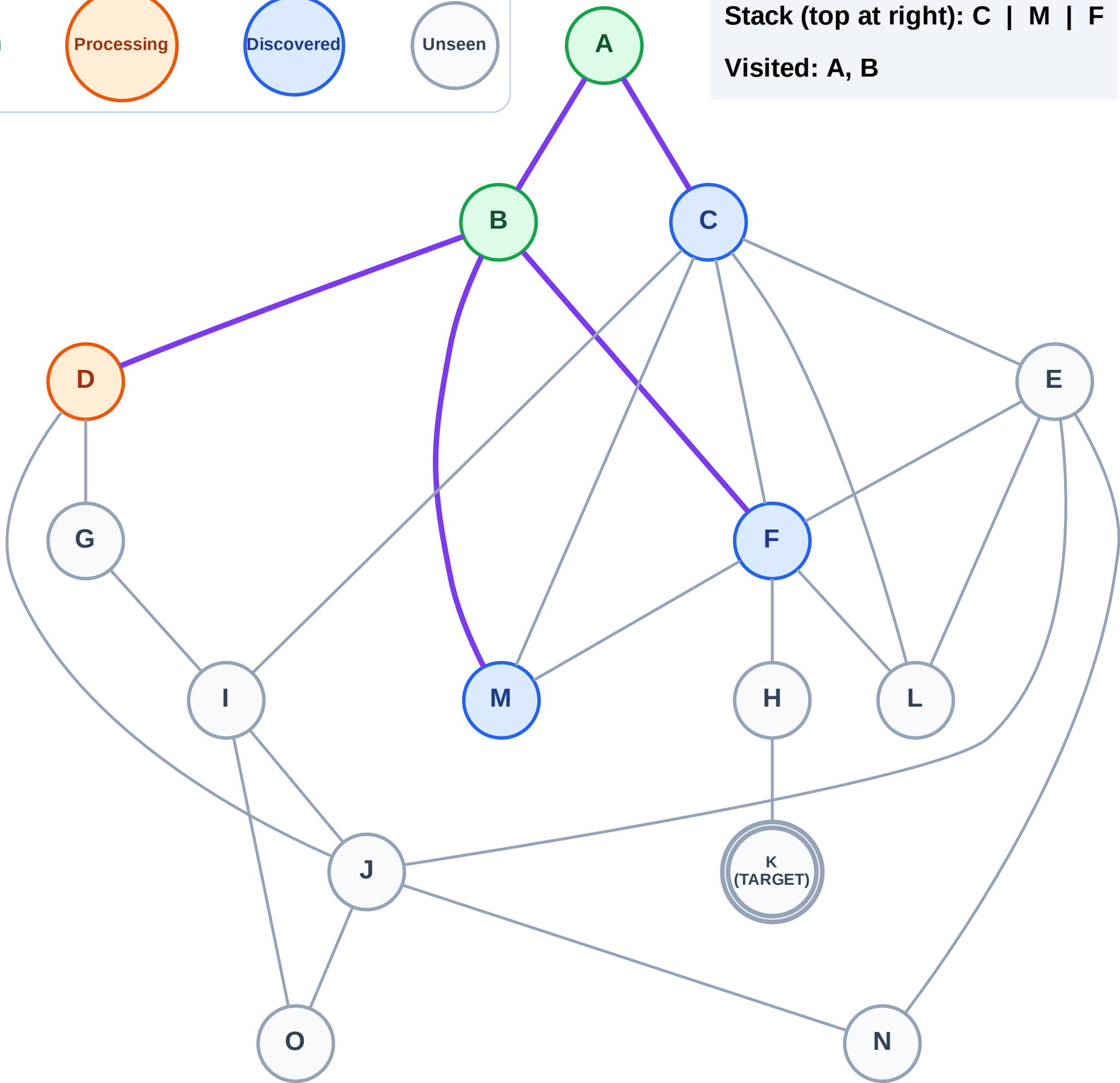
Processing

Discovered

Unseen

Stack (top at right): C | M | F

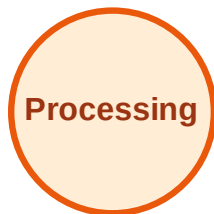
Visited: A, B



# DFS Traversal

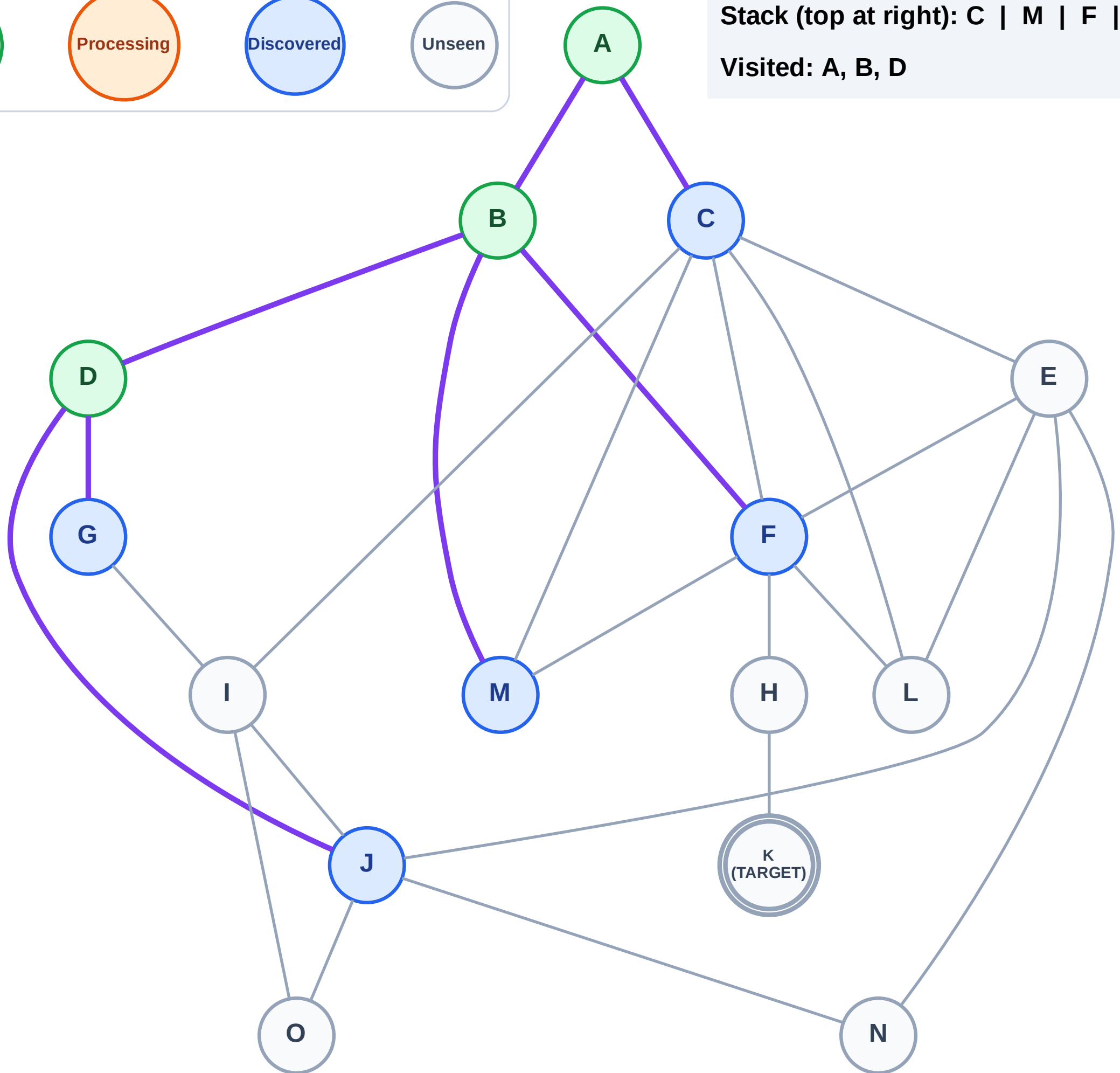
Step 7: Discover J, G from D

## Legend



Stack (top at right): C | M | F | J | G

Visited: A, B, D



# DFS Traversal

Step 8: Process node G

Legend

Complete

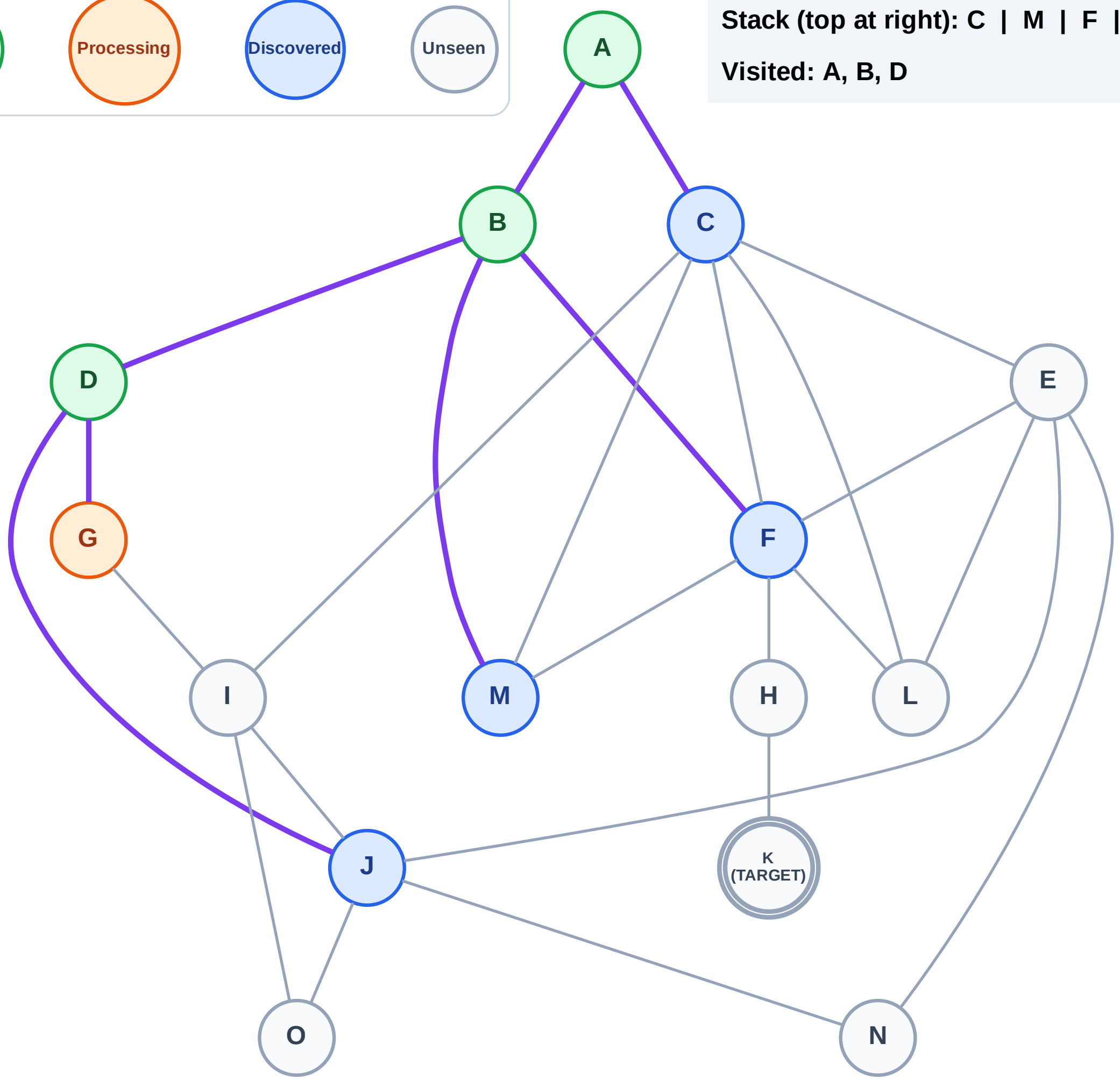
Processing

Discovered

Unseen

Stack (top at right): C | M | F | J

Visited: A, B, D



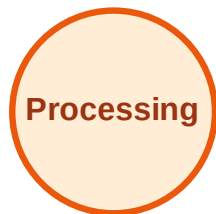
# DFS Traversal

Step 9: Discover I from G

## Legend



Complete



Processing



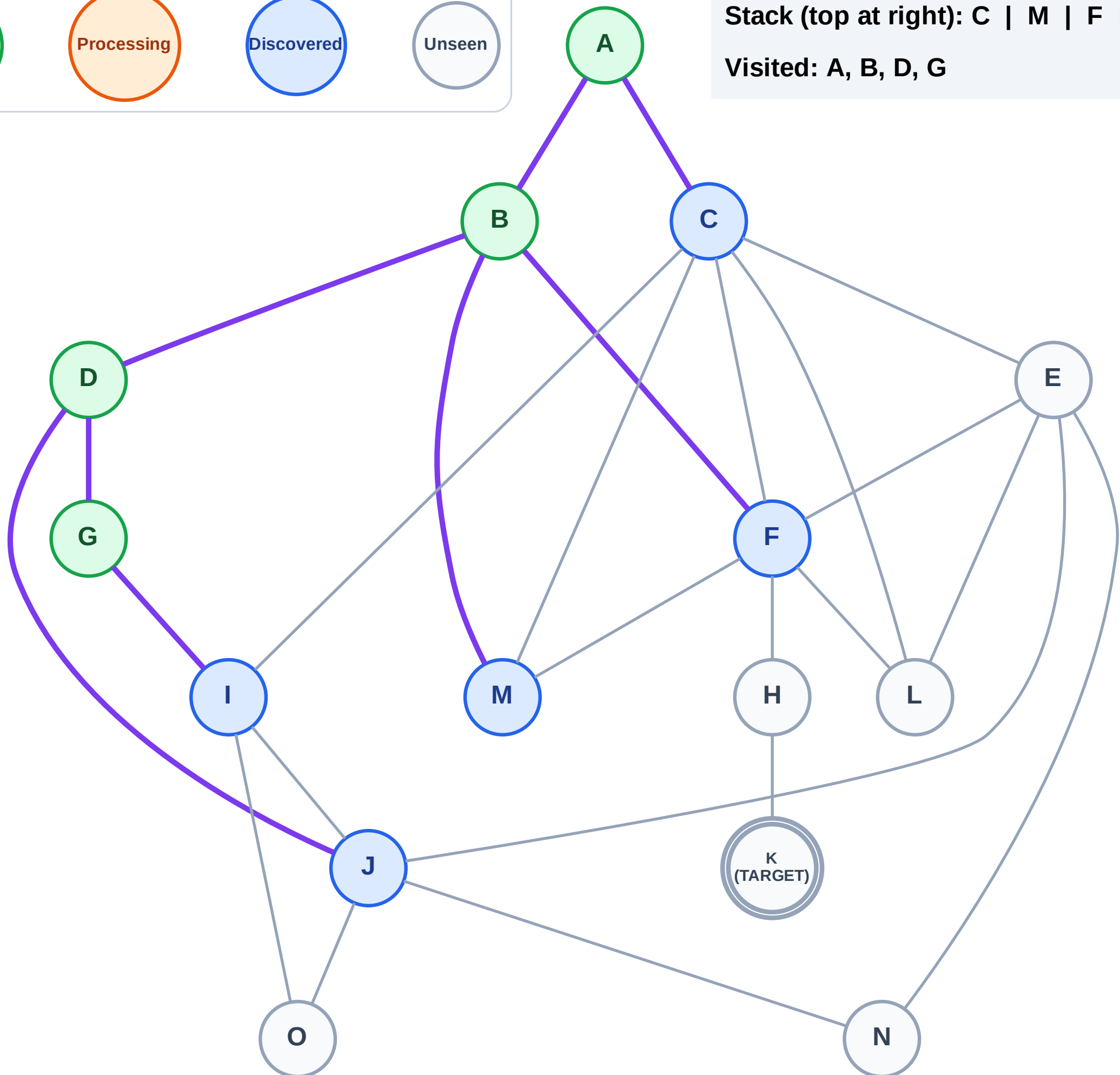
Discovered



Unseen

Stack (top at right): C | M | F | J | I

Visited: A, B, D, G



# DFS Traversal

Step 10: Process node I

Legend

Complete

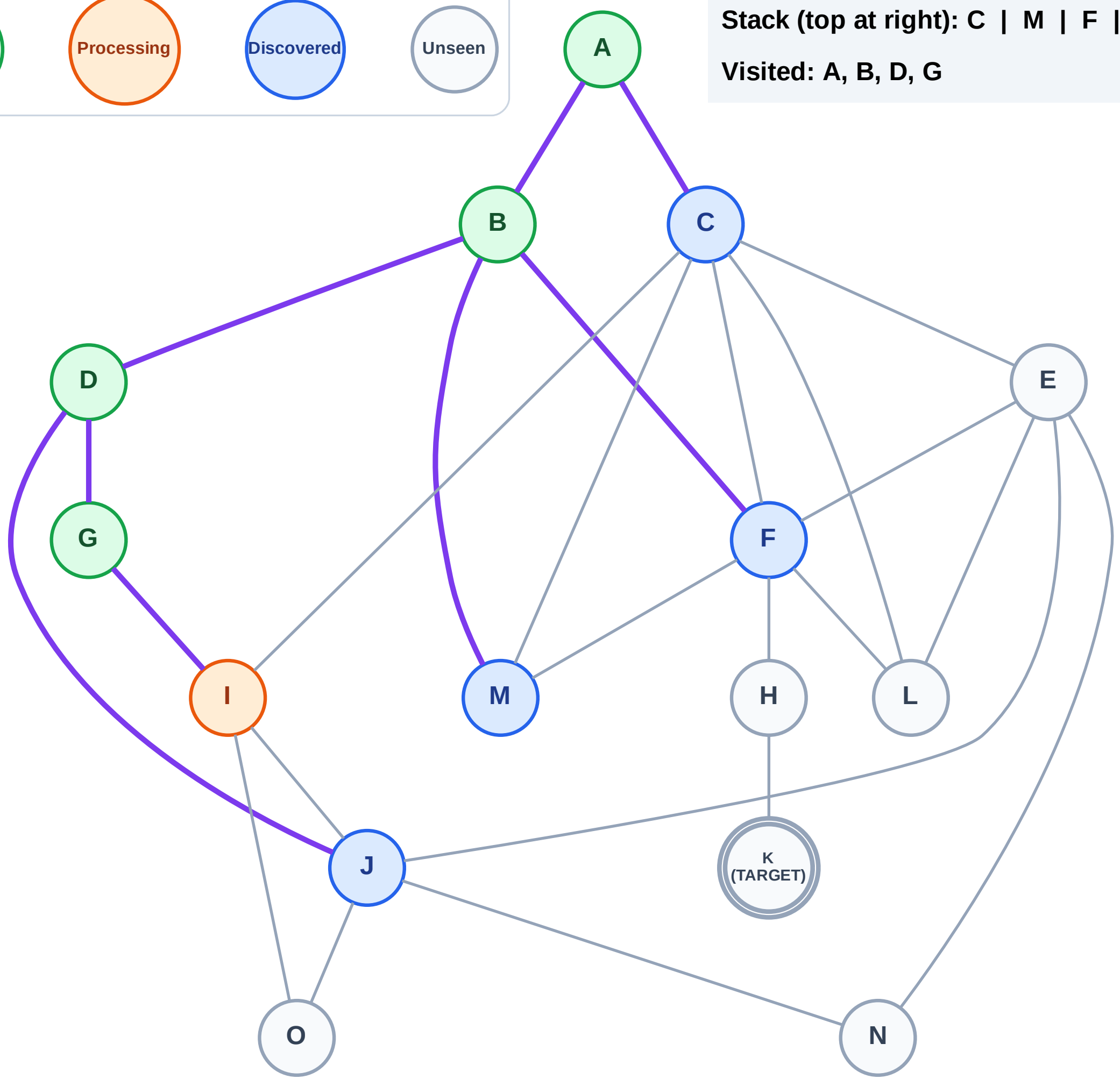
Processing

Discovered

Unseen

Stack (top at right): C | M | F | J

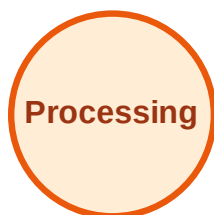
Visited: A, B, D, G



# DFS Traversal

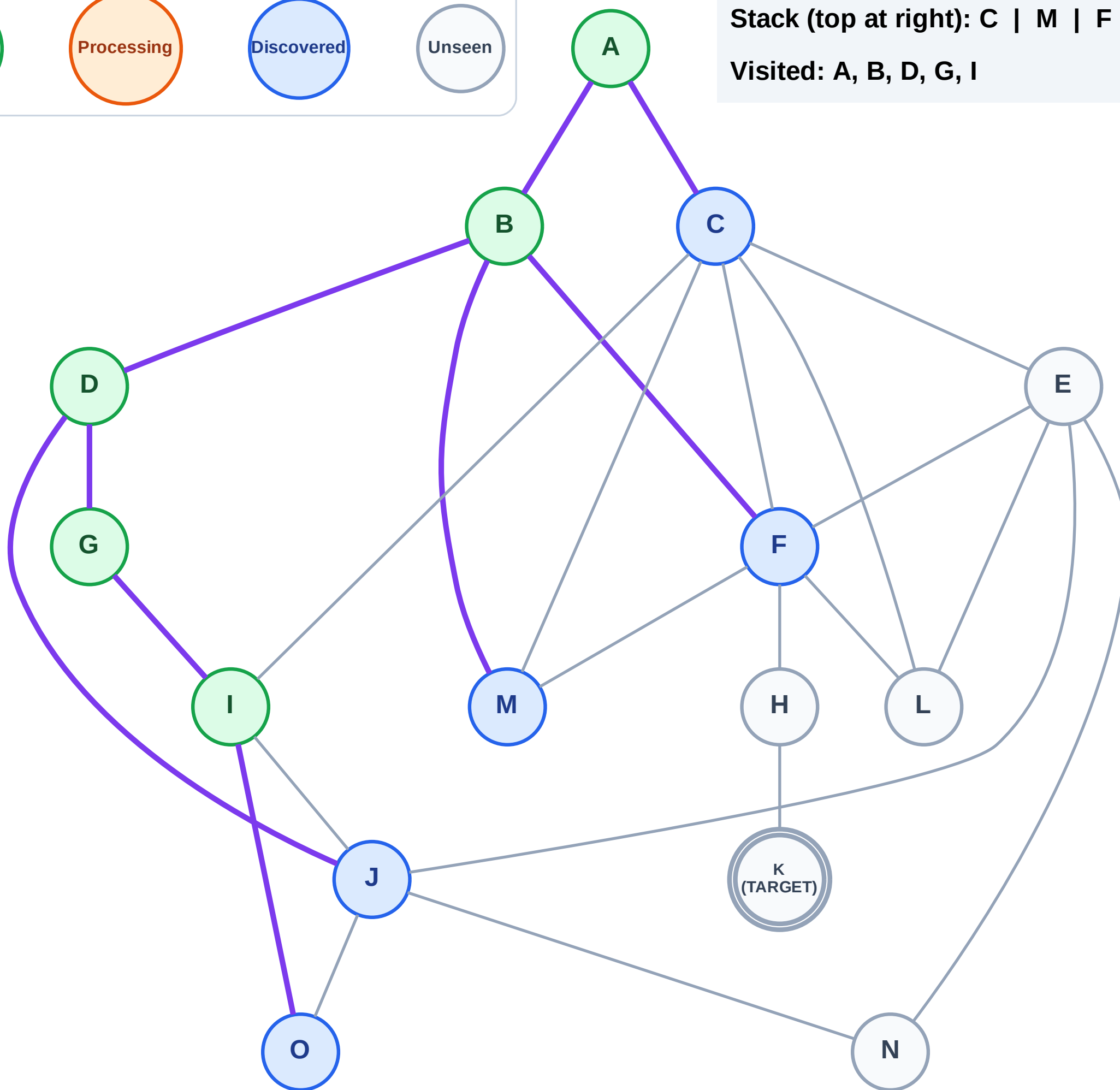
Step 11: Discover O from I

## Legend



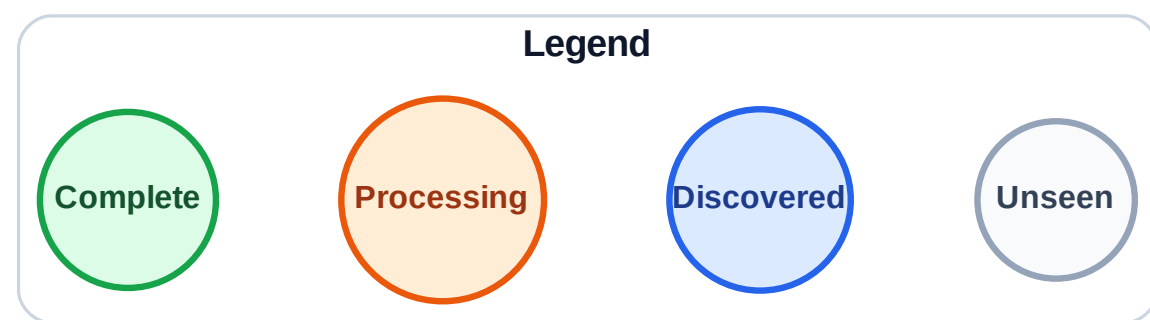
Stack (top at right): C | M | F | J | O

Visited: A, B, D, G, I

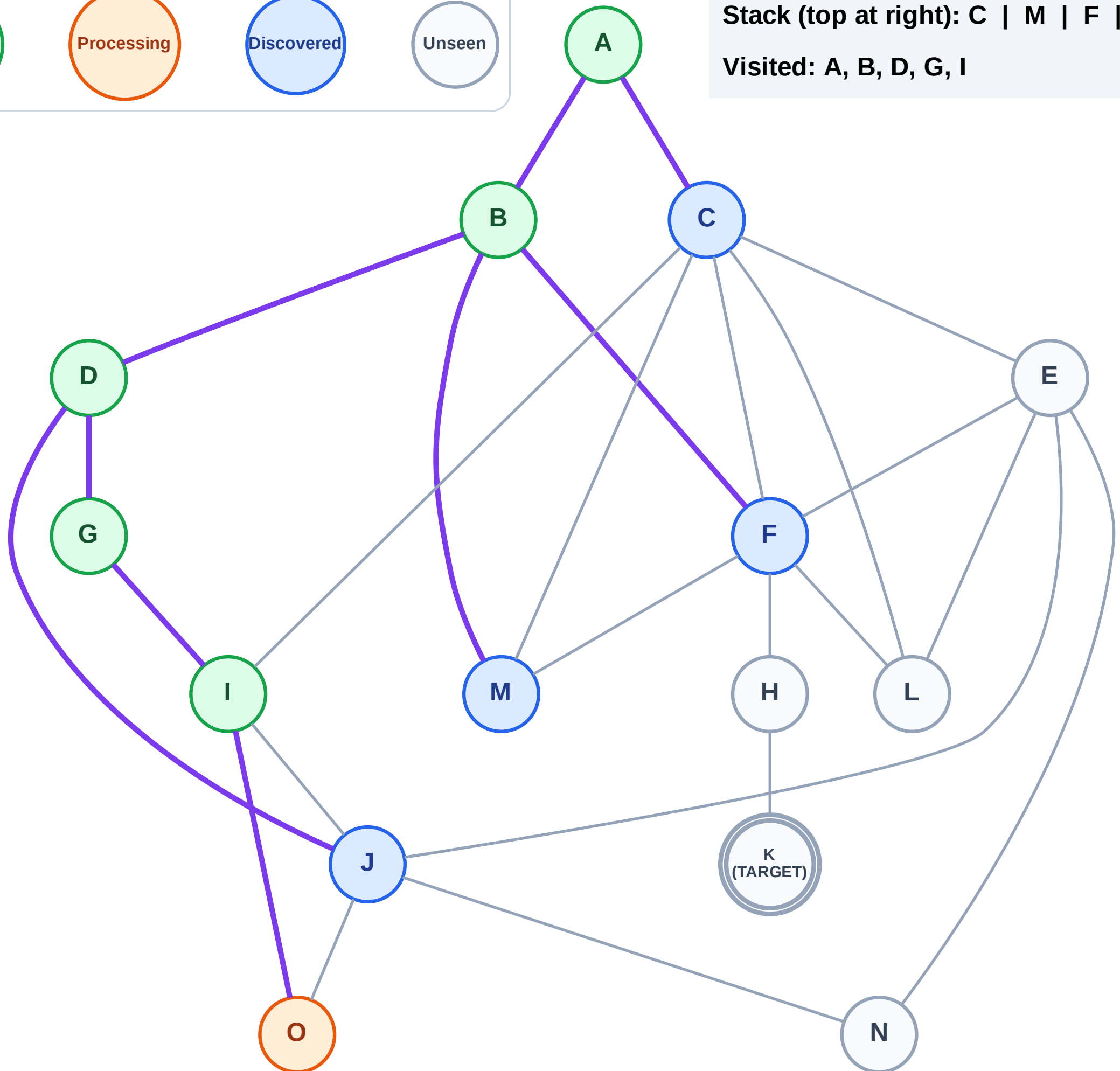


# DFS Traversal

Step 12: Process node O



Stack (top at right): C | M | F | J  
Visited: A, B, D, G, I



# DFS Traversal

Step 13: No new neighbors from O

Legend

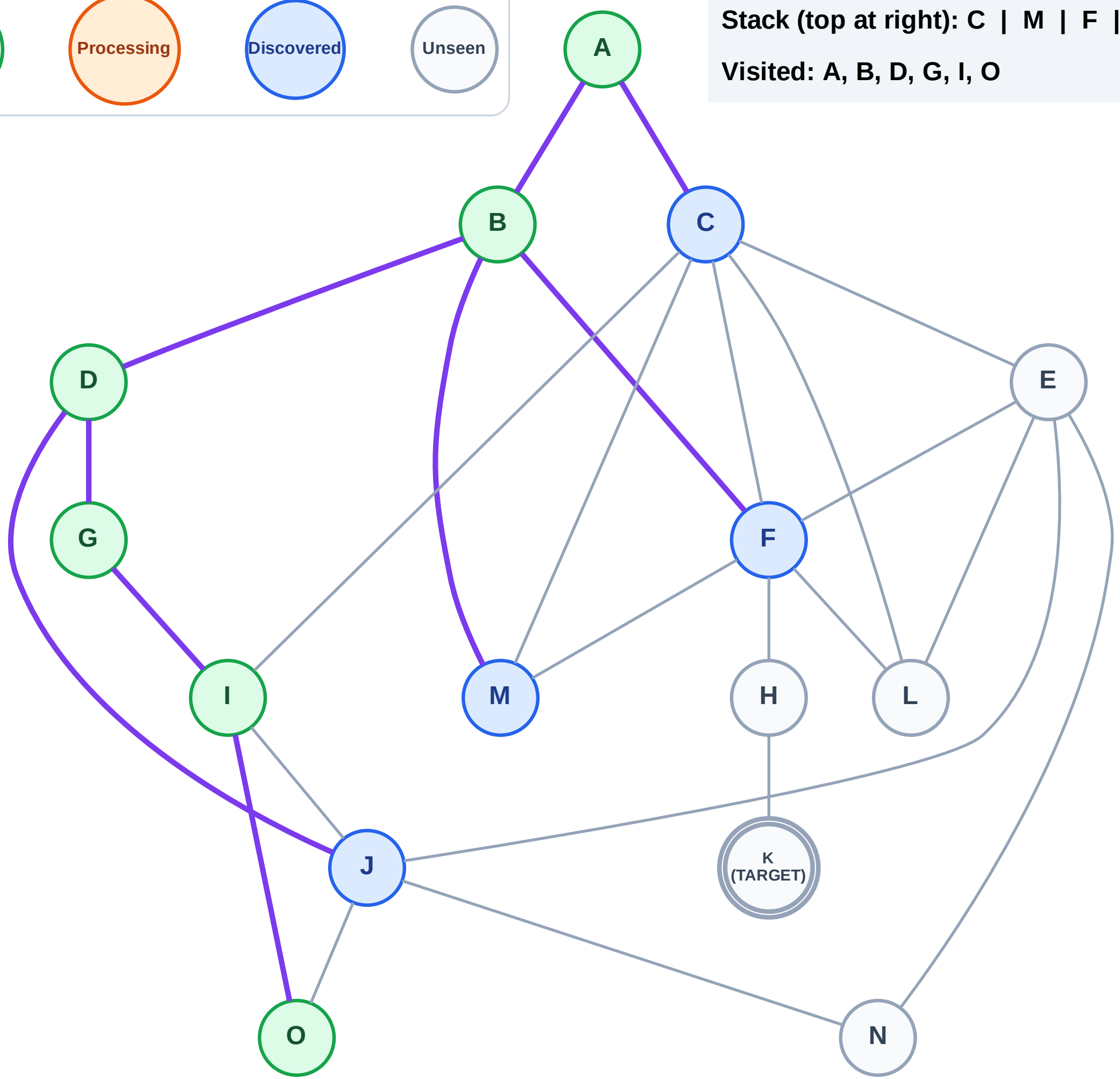
Complete

Processing

Discovered

Unseen

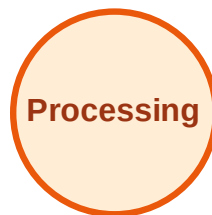
Stack (top at right): C | M | F | J  
Visited: A, B, D, G, I, O



# DFS Traversal

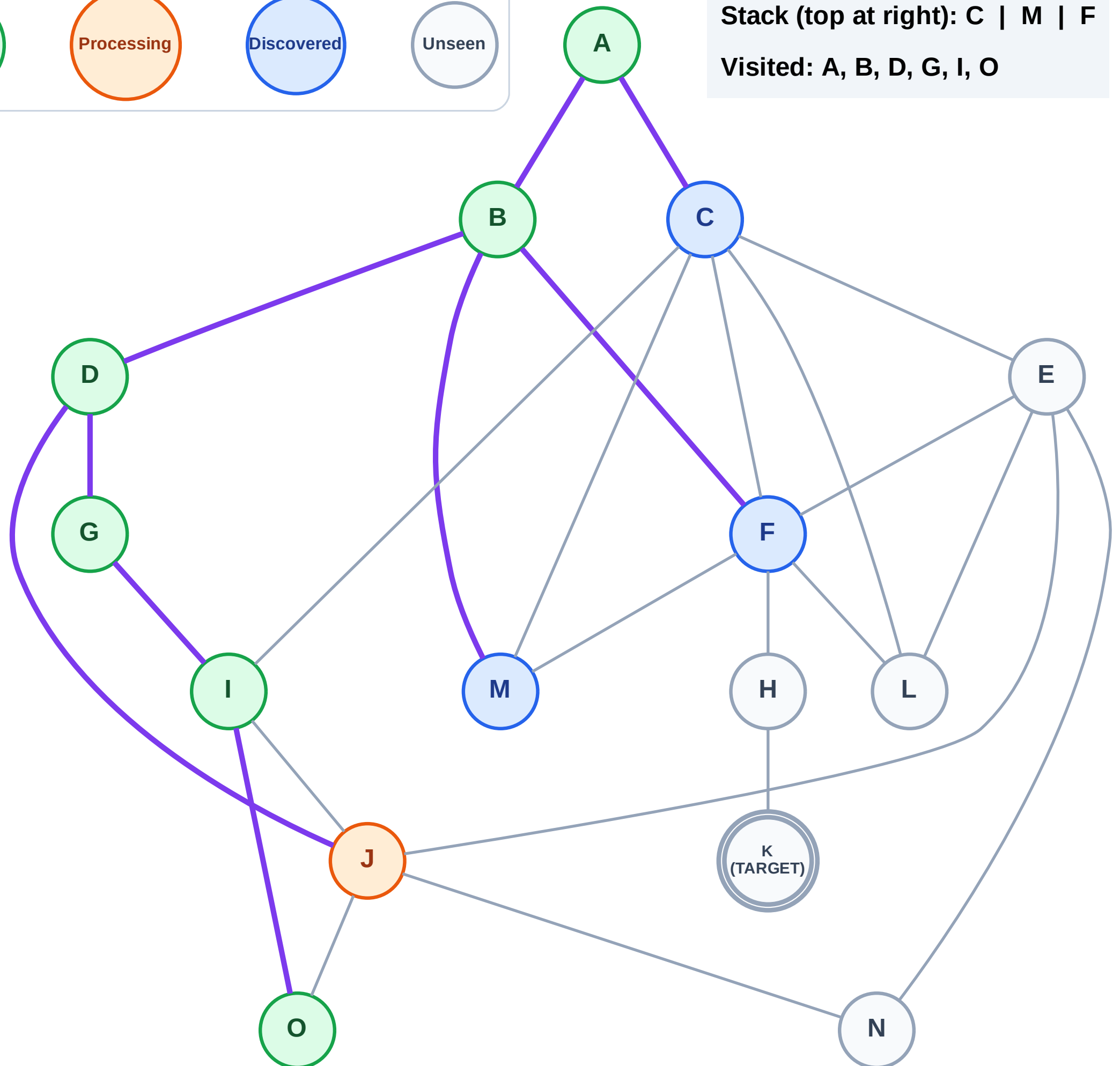
Step 14: Process node J

## Legend



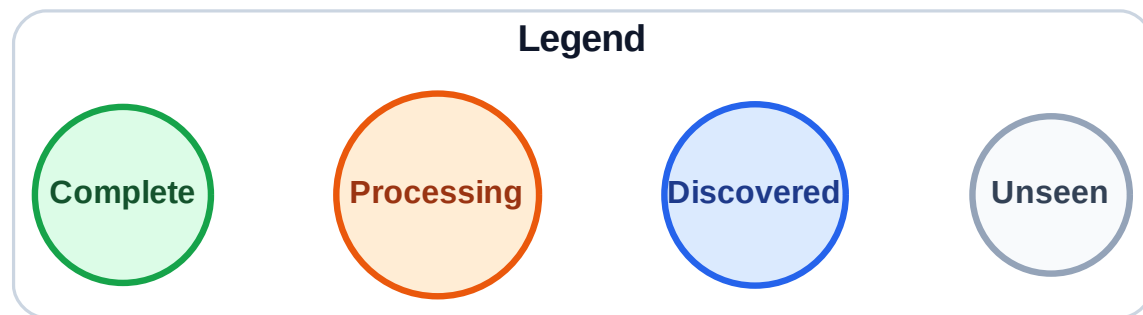
Stack (top at right): C | M | F

Visited: A, B, D, G, I, O



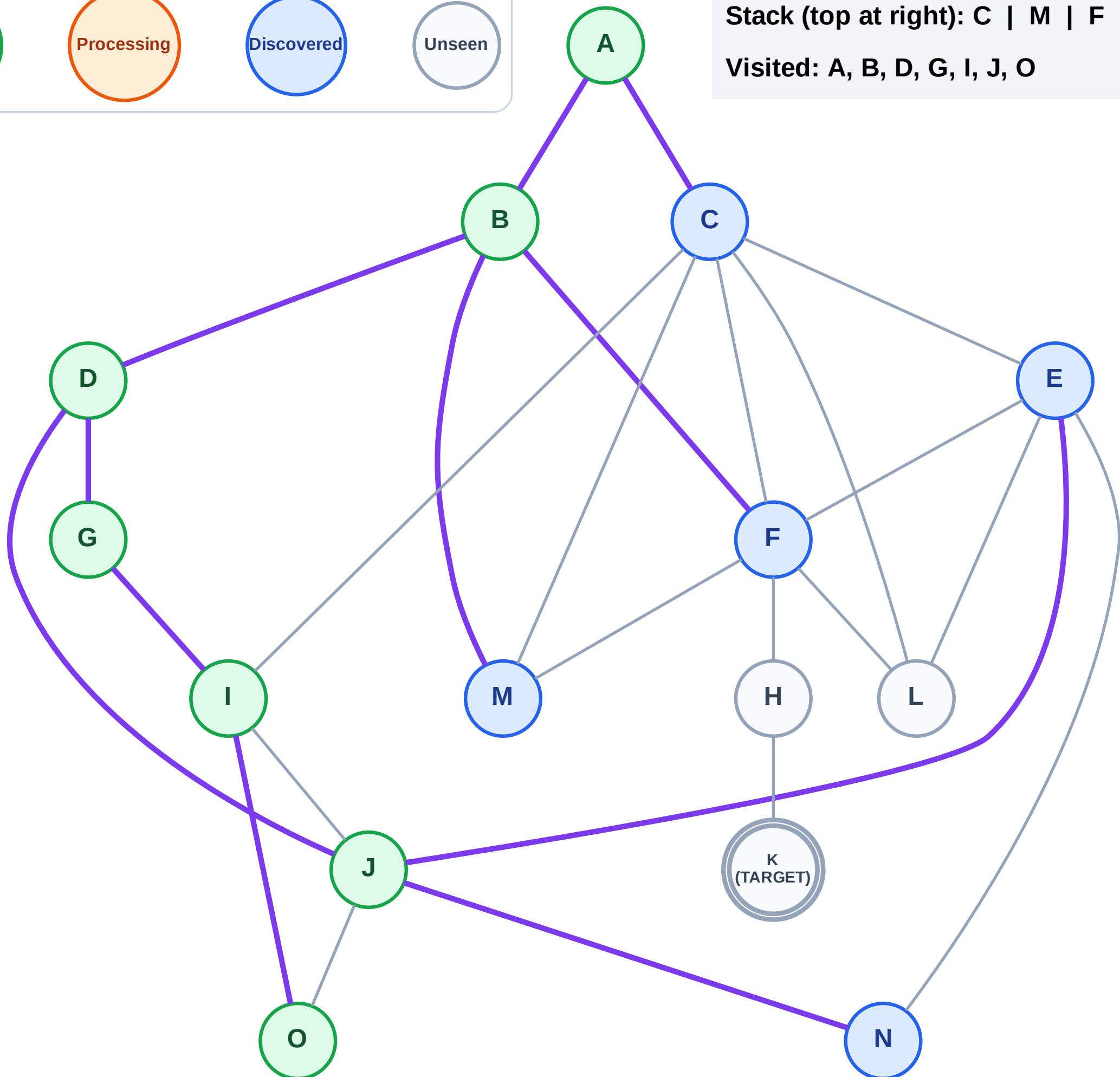
### Step 15: Discover N, E from J

### Step 15: Discover N, E from J



**Stack (top at right):** C | M | F | N | E

**Visited: A, B, D, G, I, J, O**



# DFS Traversal

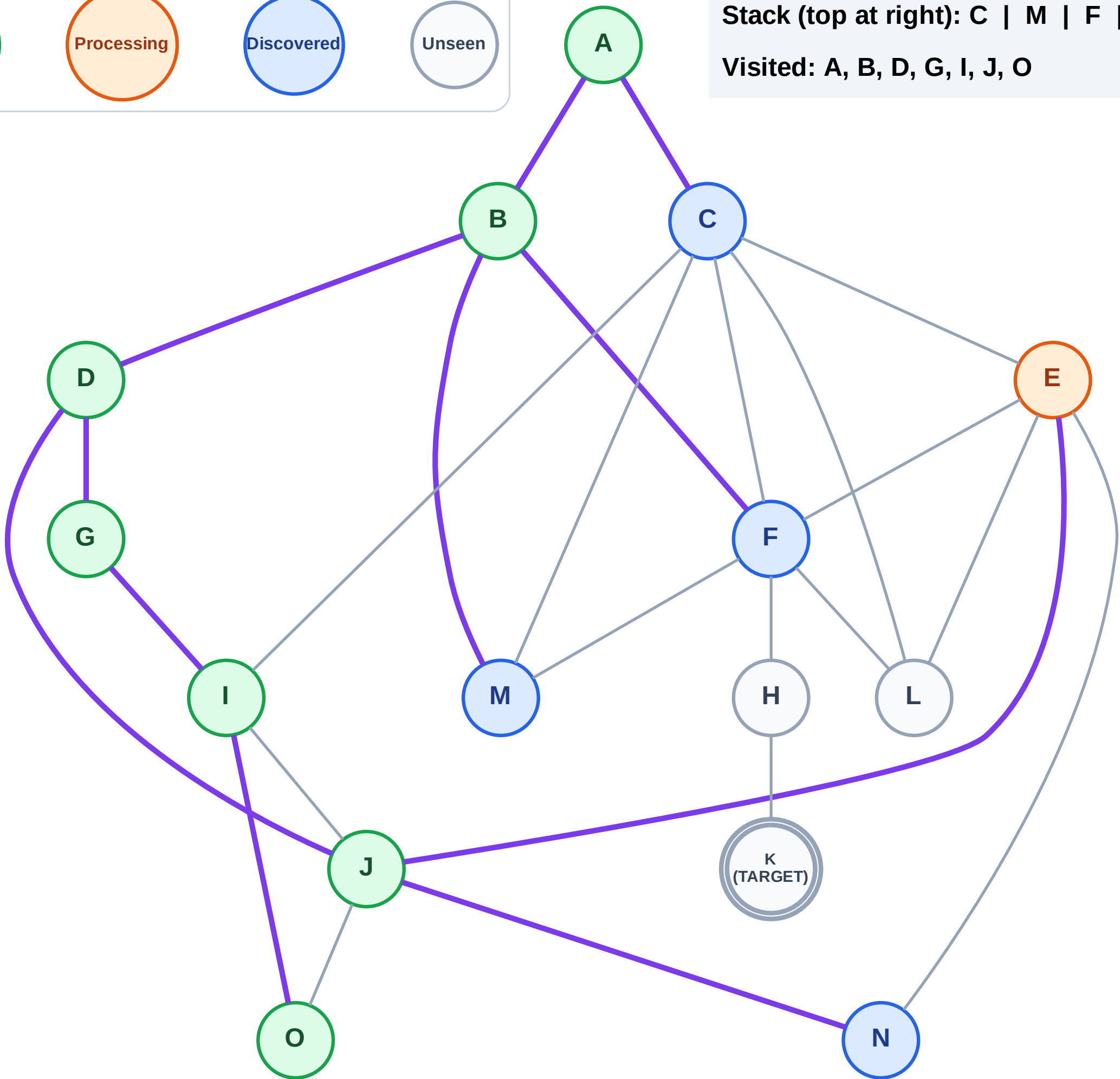
Step 16: Process node E

## Legend



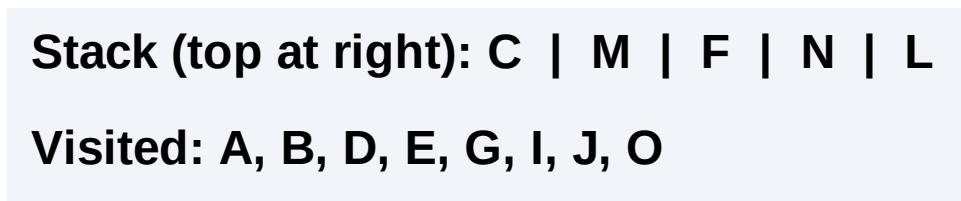
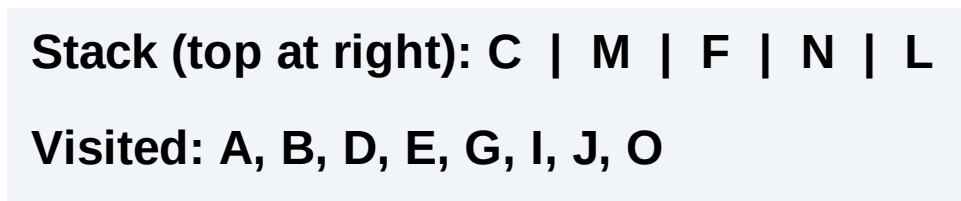
Stack (top at right): C | M | F | N

Visited: A, B, D, G, I, J, O



### Step 17: Discover L from E

### Step 17: Discover L from E



# DFS Traversal

Step 18: Process node L

Legend

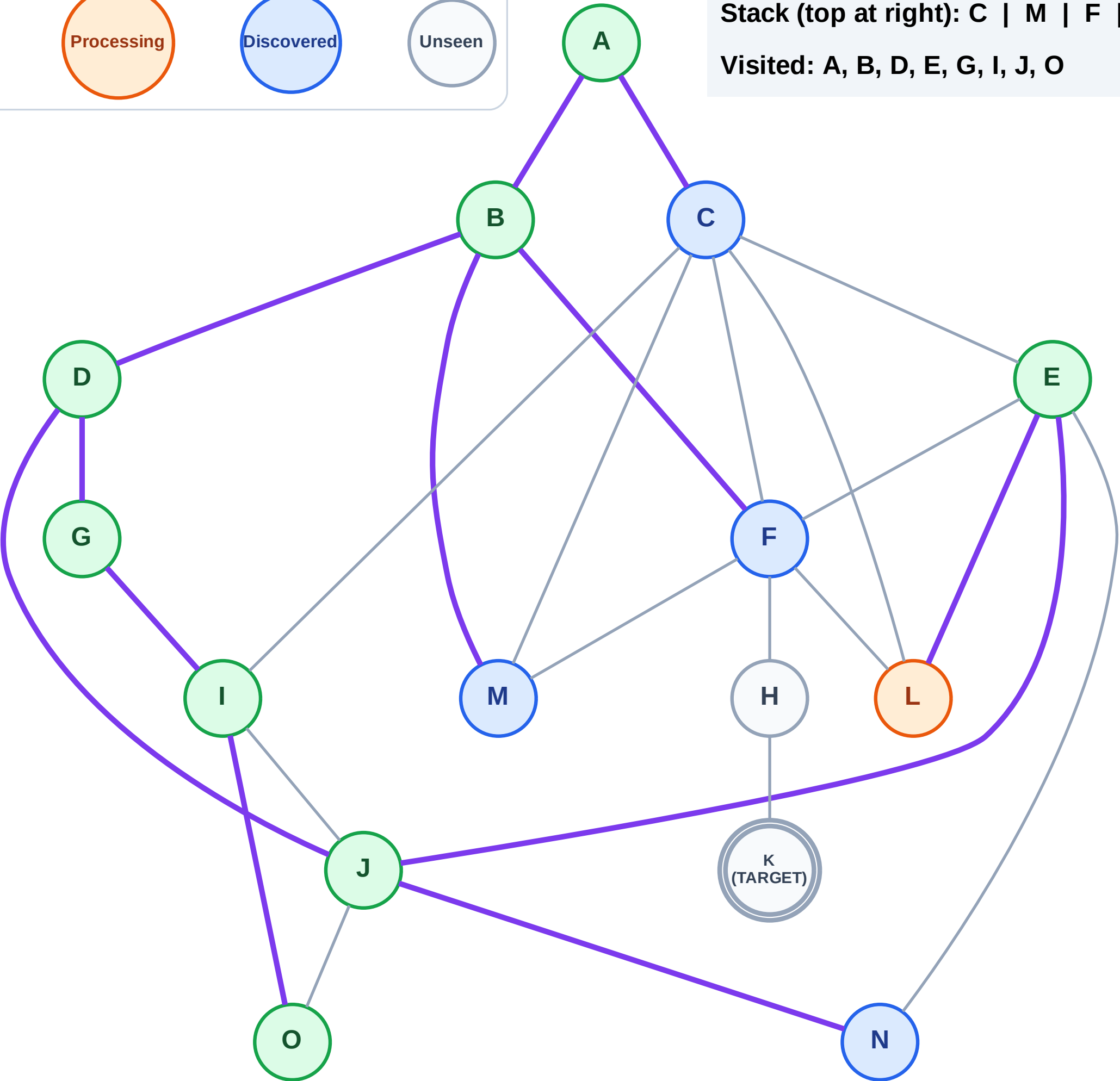
Complete

Processing

Discovered

Unseen

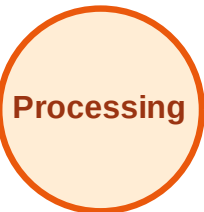
Stack (top at right): C | M | F | N  
Visited: A, B, D, E, G, I, J, O



# DFS Traversal

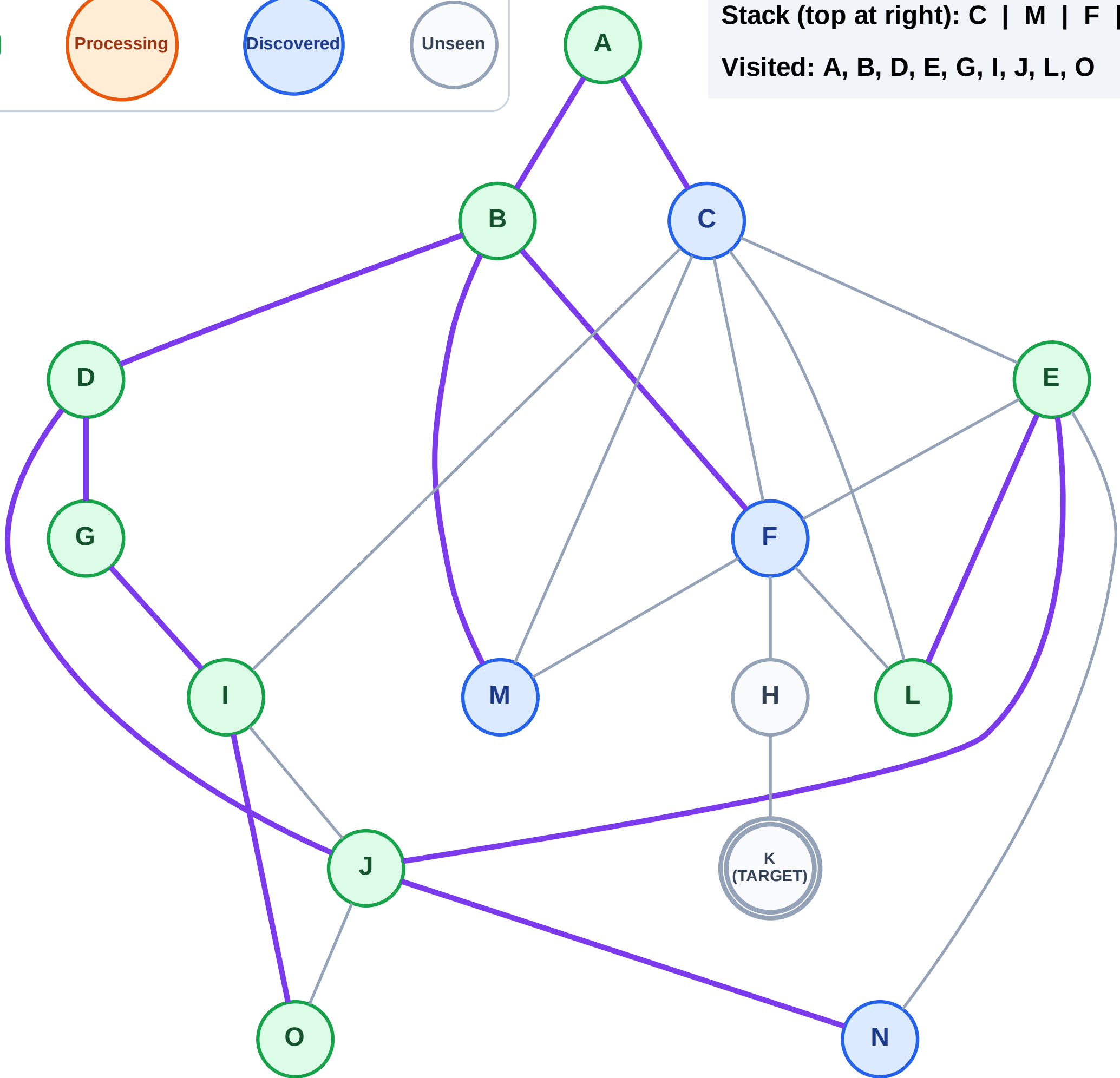
Step 19: No new neighbors from L

## Legend



Stack (top at right): C | M | F | N

Visited: A, B, D, E, G, I, J, L, O



# DFS Traversal

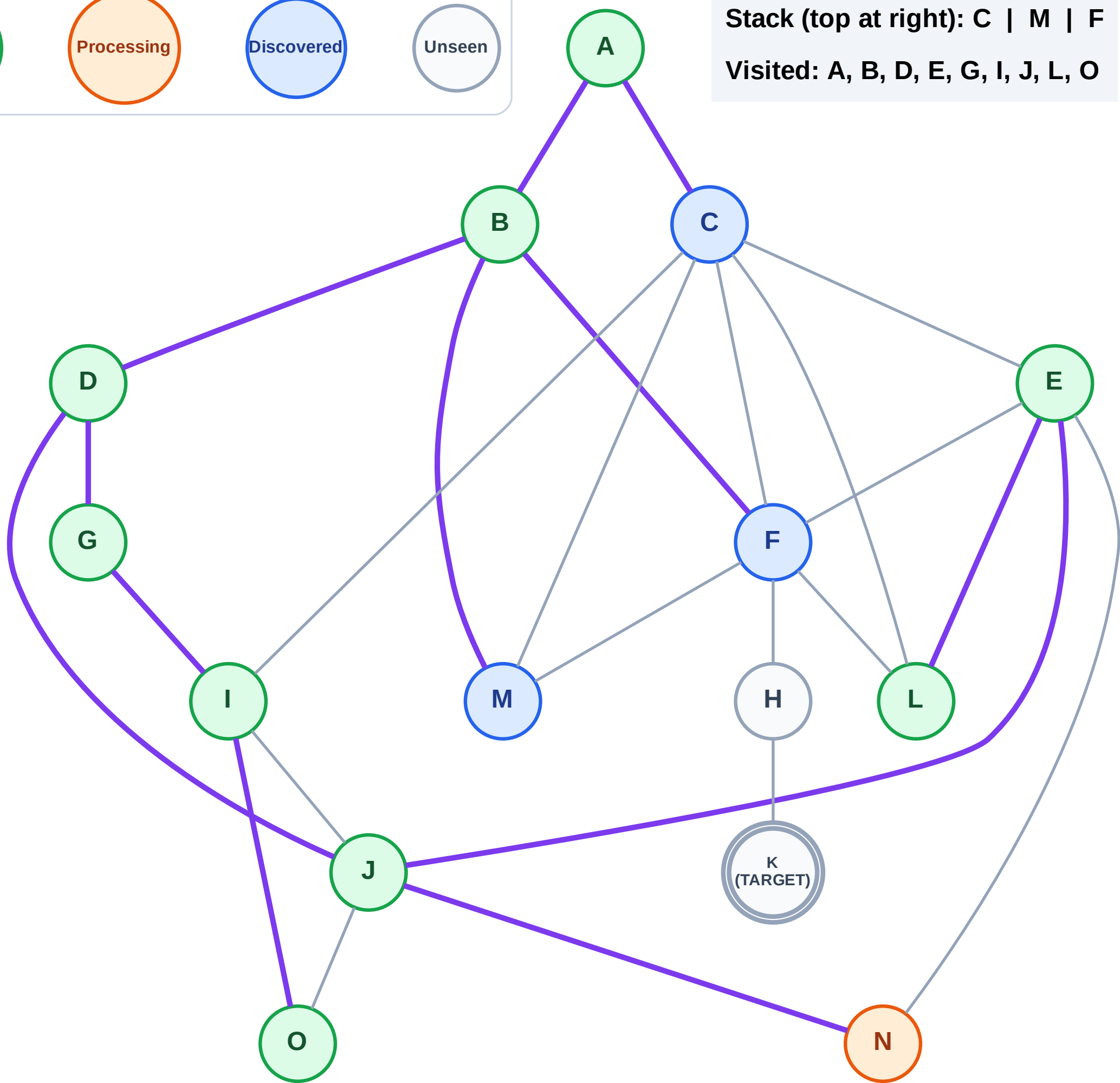
Step 20: Process node N

## Legend



Stack (top at right): C | M | F

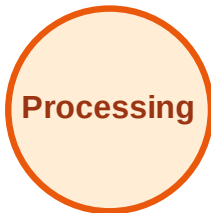
Visited: A, B, D, E, G, I, J, L, O



# DFS Traversal

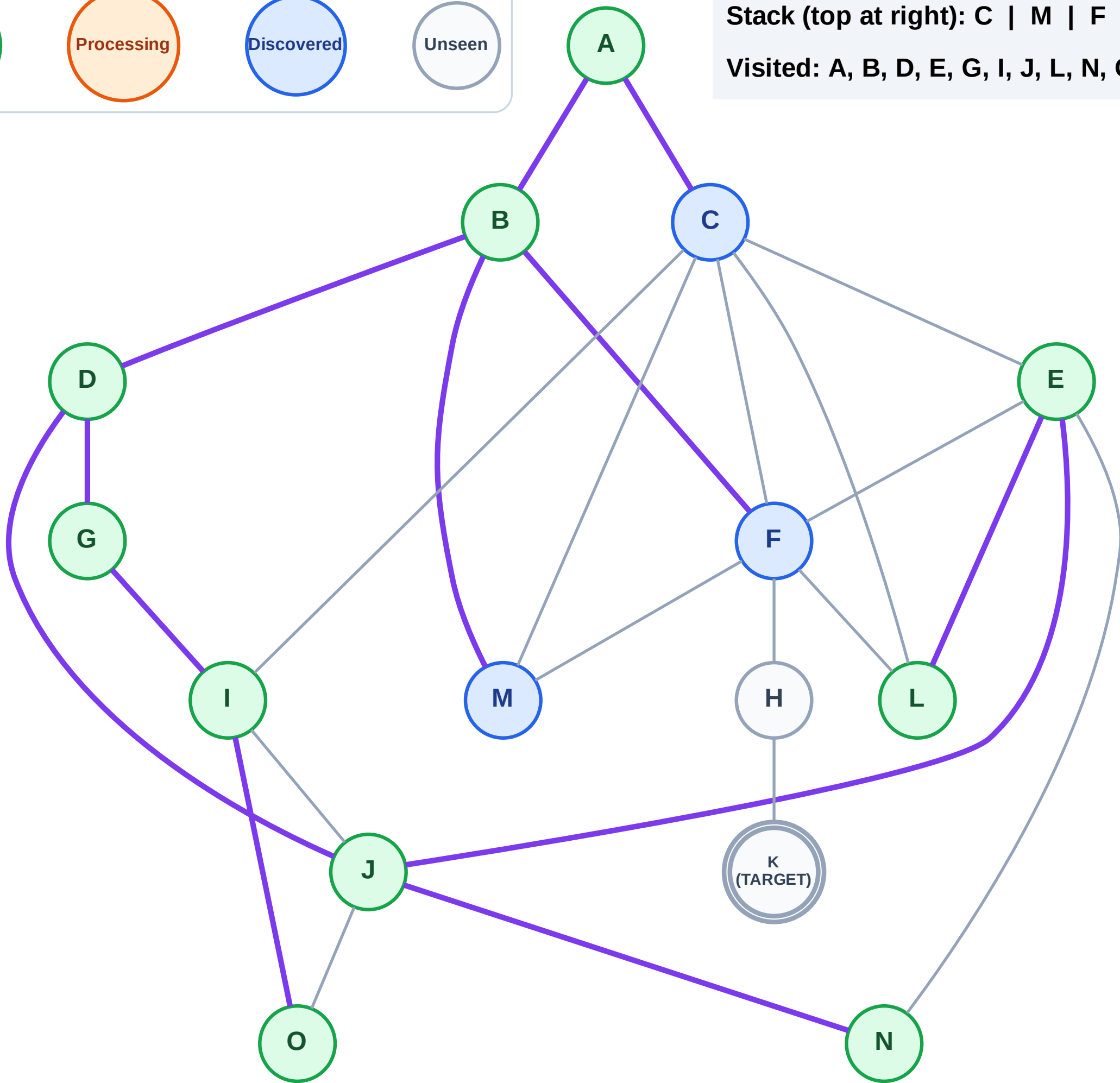
Step 21: No new neighbors from N

## Legend



Stack (top at right): C | M | F

Visited: A, B, D, E, G, I, J, L, N, O



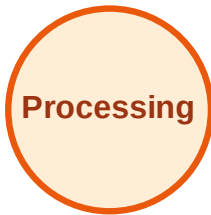
# DFS Traversal

Step 22: Process node F

## Legend



Complete



Processing



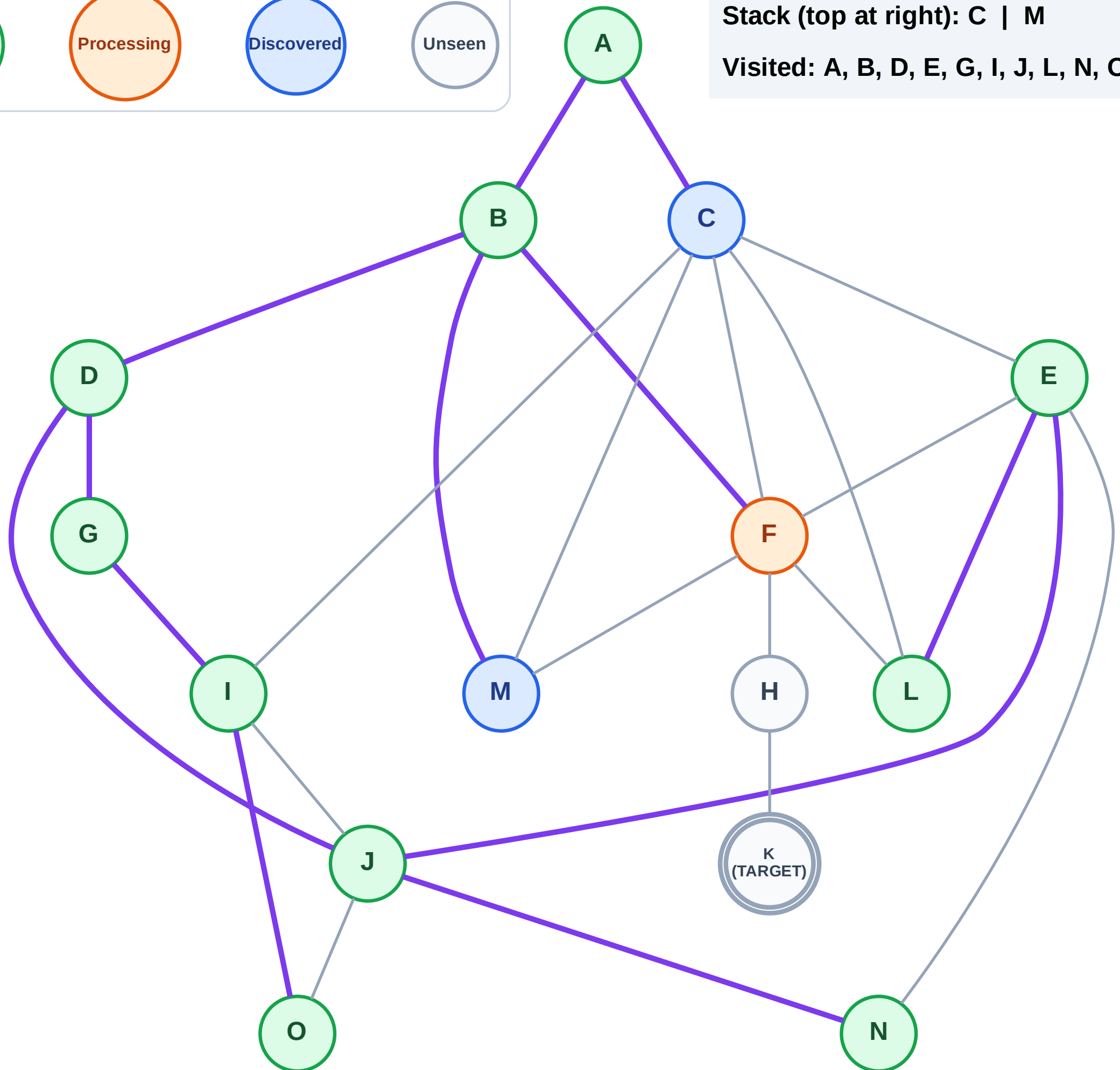
Discovered



Unseen

Stack (top at right): C | M

Visited: A, B, D, E, G, I, J, L, N, O



# DFS Traversal

Step 23: Discover H from F

Legend

Complete

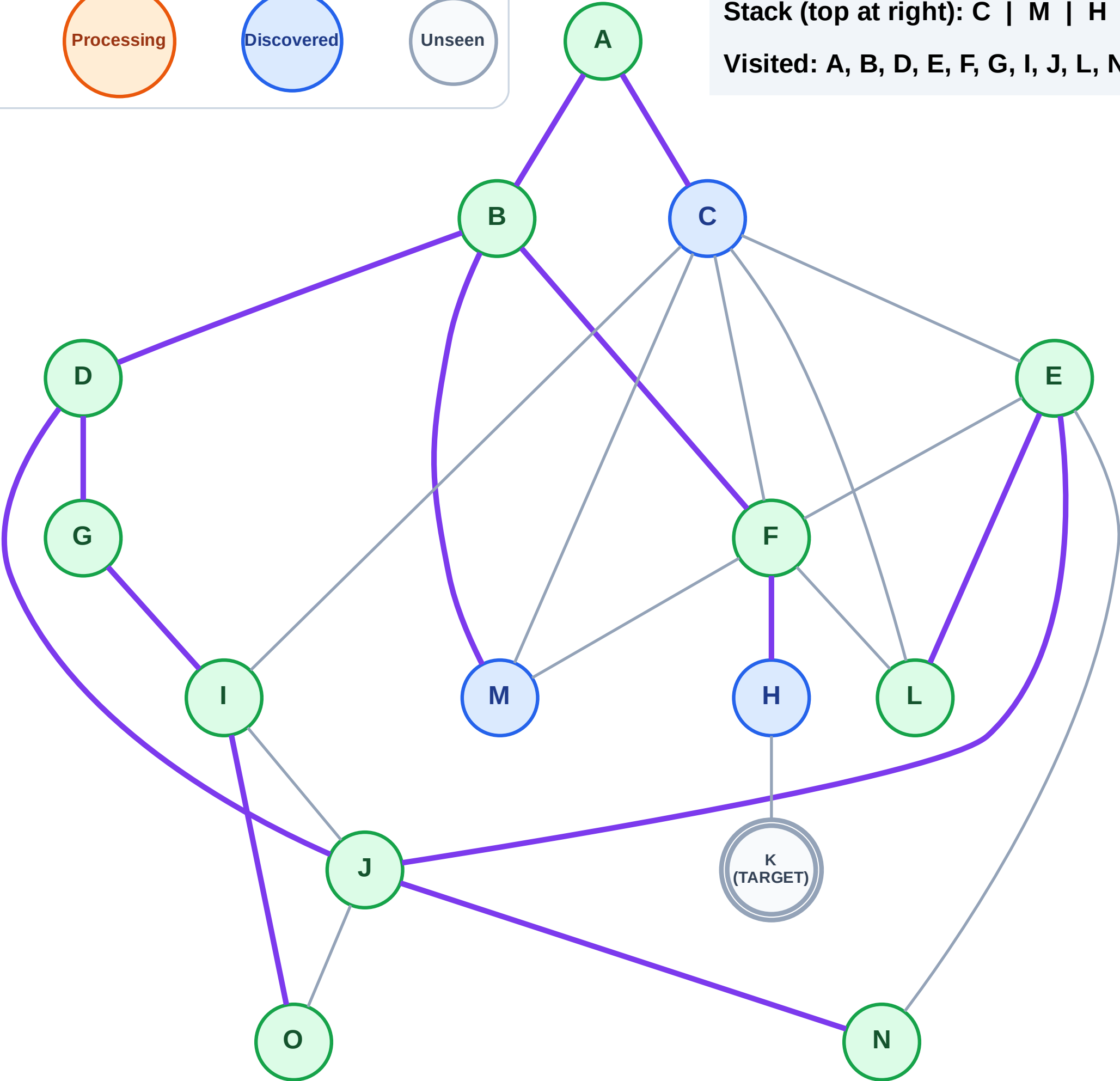
Processing

Discovered

Unseen

Stack (top at right): C | M | H

Visited: A, B, D, E, F, G, I, J, L, N, O



# DFS Traversal

Step 24: Process node H

Legend

Complete

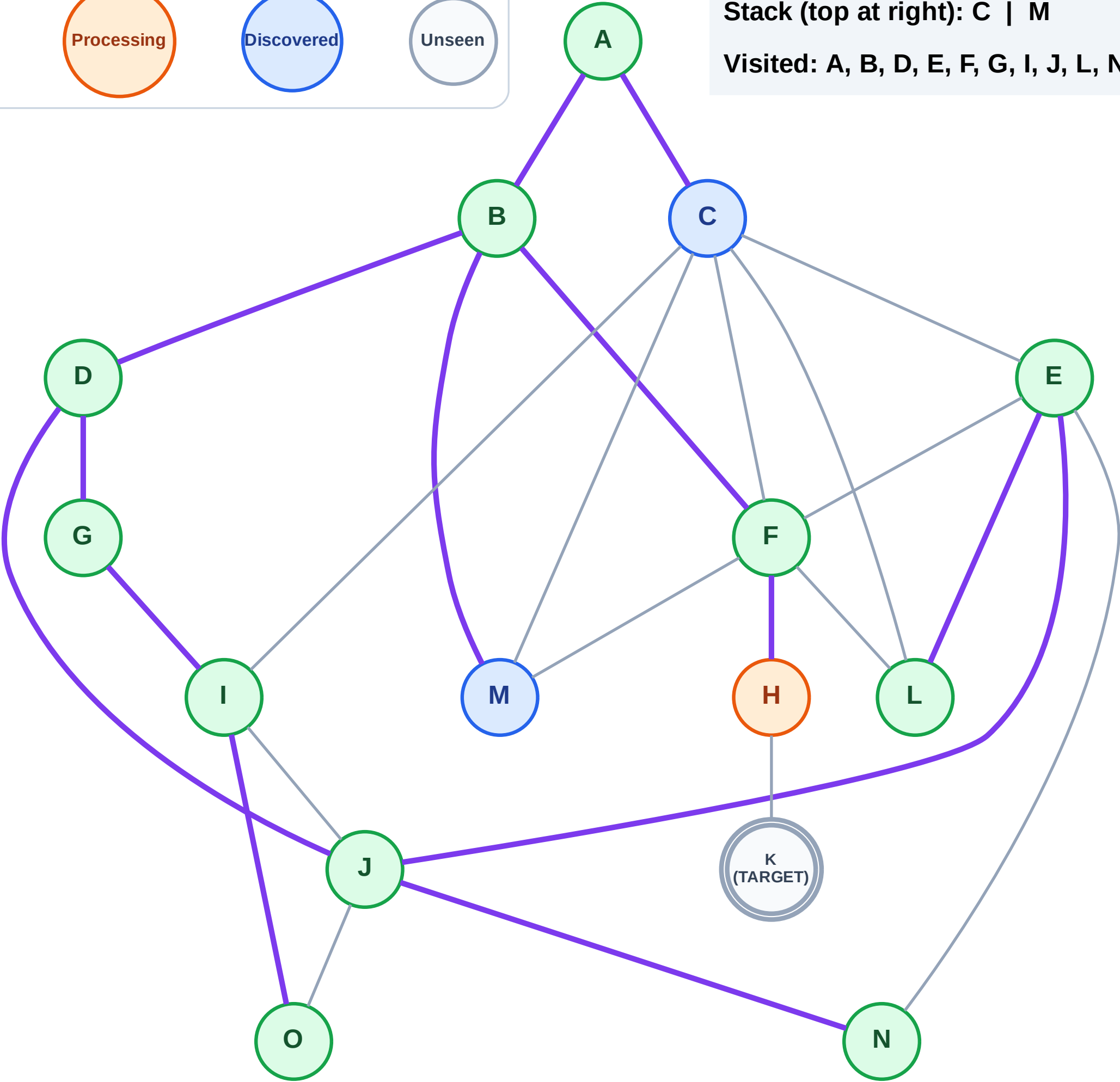
Processing

Discovered

Unseen

Stack (top at right): C | M

Visited: A, B, D, E, F, G, I, J, L, N, O



# DFS Traversal

Step 25: Discover K from H

Legend

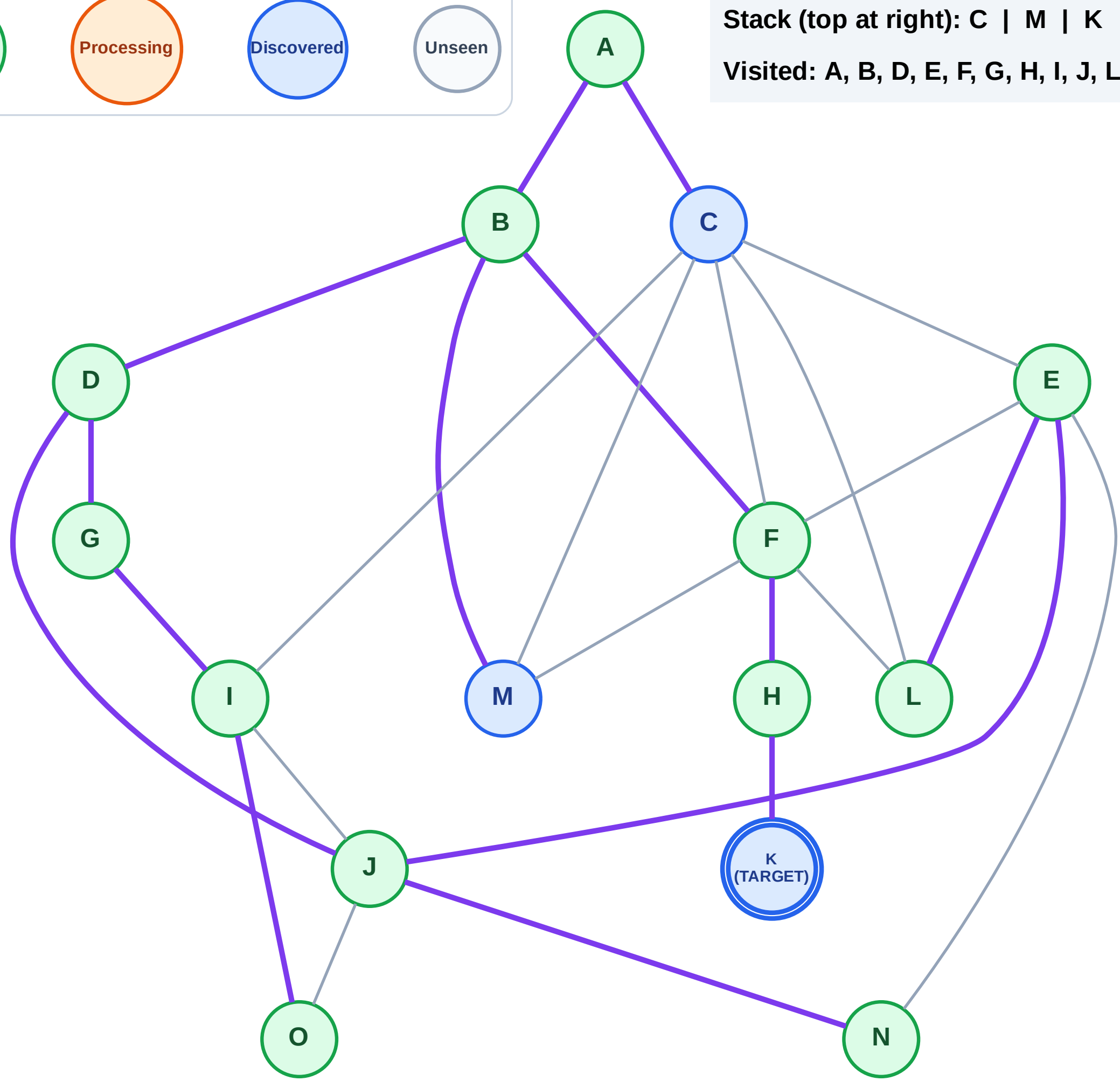
Complete

Processing

Discovered

Unseen

Stack (top at right): C | M | K  
Visited: A, B, D, E, F, G, H, I, J, L, N, O



# DFS Traversal

Step 26: Process node K

## Legend

Complete

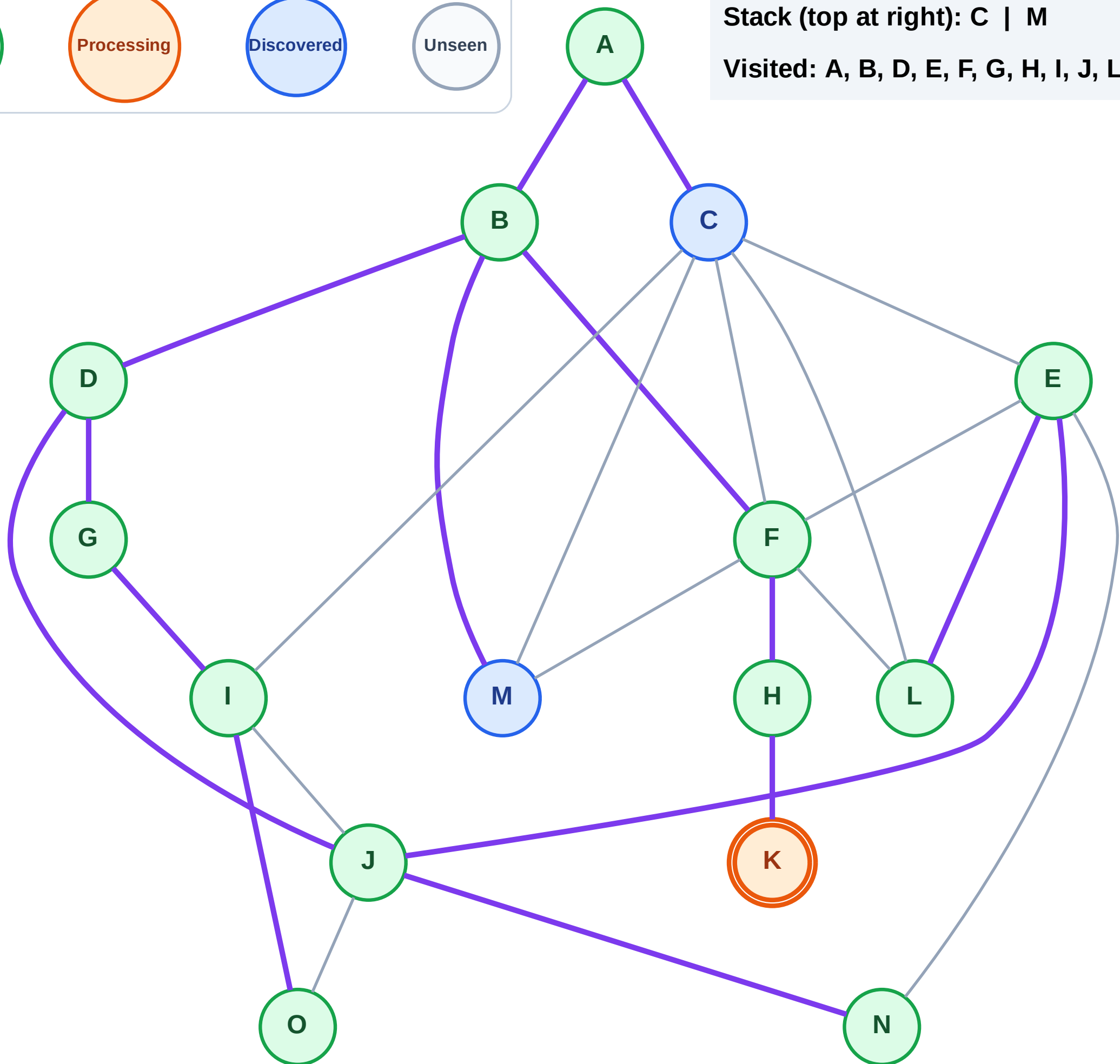
Processing

Discovered

Unseen

Stack (top at right): C | M

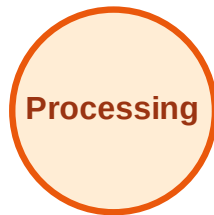
Visited: A, B, D, E, F, G, H, I, J, L, N, O



# DFS Traversal

Step 27: TARGET NODE K REACHED! Path (4 hops): A → B → F → H → K

## Legend



Stack (top at right): C | M

Visited: A, B, D, E, F, G, H, I, J, K, L, N, O

